Politically Induced Internal Trade Barriers in China *

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Abstract

Competition among subnational governments generates internal non-tariff trade barriers, fragmenting nominally integrated markets. In China, relative performance evaluation encourages politicians to capture local gains while limiting spillovers to rivals. Exploiting peer-turnover-induced variation in city party secretaries' age differentials (1996–2018), I show that stronger within-province competition reduces inter-city trade and firms' access to intermediate inputs while improving relative GDP performance. These barriers arise through resource allocation rather than explicit trade restrictions: politicians underinvest in connective motorways in counties bordering rival cities, redirect subsidies and land from tradable industries toward real estate, and devote less attention to internal trade in government work reports. Individually rational responses to career incentives thus produce collectively suboptimal market fragmentation: policies that improve politicians' relative standing need not benefit local tradable producers.

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1 Introduction

Trade drives prosperity, yet it is fundamentally constrained by political institutions. While standard benchmarks predict that falling transport costs should compel convergence (McCallum, 1995), evidence reveals persistent fragmentation even within national borders (Young, 2000; Poncet, 2005). A burgeoning literature demonstrates that, internationally, trade flows are endogenous to geopolitical alignment; strategic divergence and political risk act as implicit barriers that segment global markets (Fajgelbaum et al., 2020; Caldara and Iacoviello, 2022; Kleinman et al., 2024). Whether analogous political forces generate 'internal geopolitical distance' between subnational governments remains an open question.

This paper establishes that bureaucratic promotion tournaments function as a distinct source of such distance, driving domestic fragmentation through strategic resource misallocation. Because tariffs cannot be levied within a country, this distance takes the form of non-tariff barriers; the paper's main contribution is to show that promotion competition creates them through public-resource allocation. I examine how yardstick competition (Shleifer, 1985; Besley and Case, 1995)—originally conceptualized for democratic settings—operates in hierarchical authoritarian systems, where it generates internal geopolitical distance by transforming neighboring jurisdictions into rivals rather than partners in integration.[1] Under intense competition and budget constraints, this relative evaluation induces politicians to strategically reallocate public resources toward non-tradable sectors, particularly real estate, that generate rapid and locally appropriable GDP gains.[2] This reallocation away from tradable industries and trade-facilitating infrastructure creates non-tariff barriers to inter-jurisdictional trade, sacrificing aggregate economic efficiency

[1]The mechanism differs from canonical yardstick competition in one important respect: while Besley and Case (1995) emphasized voters comparing government performance, here upper-level governments conduct the comparison. Crucially, the distortions I document do not stem from "not correctly set incentives" but from a fundamental information constraint—upper-level governments cannot easily monitor cross-jurisdictional trade flows and spillovers, leading them to rationally evaluate officials on locally observable outcomes. This creates a systemic principal-agent problem where yardstick competition based on local performance metrics produces trade barriers as an equilibrium response.

[2]Politicians explicitly acknowledge this preference: "I sell land, I do real estate, today I build a park, tomorrow I build a center—it's all government investment. Once these investments go in, the KPI is completed and the GDP looks very pretty" (Voice of America Chinese, July 13, 2022, https://www.voachinese.com/a/6655247.html). Documented cases illustrate the pattern: Xiangtan's city party secretary launched 345 construction projects totaling 617 billion yuan (nearly five times annual fiscal revenue), explicitly stating his goal to "quickly show achievements" and secure vice-provincial promotion, resulting in 435 billion yuan debt and 33 unfinished projects (February 1, 2024, https://baijiahao.baidu.com/s?id=1789672752005565436&wfr=spider&for=pc). Similarly, Zhangzhou's city party secretary spent 2.11 billion yuan on six replica tourist stations and designed a 200-million-yuan opera house to create visible achievements "that attract attention" before superiors' visits; both projects were later abandoned (Central Commission for Discipline Inspection investigation, January 2024, https://news.sina.com.cn/c/2024-01-09/doc-inaawrkp5341897.shtml).

to maximize relative performance. The findings reveal a yet understudied drawback to certain forms of decentralized governance, contributing to our understanding of how institutional frameworks affect economic outcomes.[3]

China's institutional framework, which combines regional autonomy with centralized personnel control, provides an ideal setting for studying political economy and market integration. As documented by Maskin et al. (2000), Li and Zhou (2005), and Xu (2011), local politicians have autonomy in shaping economic policies similar to that in a democratic system, but their career prospects are not determined by voters. The promotion system operates within a strictly hierarchical structure, where politicians at each level are promoted by their immediate superiors.[4] In this promotion system, two key characteristics significantly influence evaluation: a politician's age and economic performance.[5][6] Consequently, politicians with similar characteristics face intense competition among their peers. When facing such competition under budget constraints, politicians strategically optimize their policy portfolios by reallocating scarce resources to maximize locally visible and appropriable economic gains within their evaluation period.

This evaluation system design reflects inherent information constraints in multi-tiered governance rather than policy mistakes. Upper-level governments cannot easily monitor or attribute bilateral trade flows between jurisdictions, while local GDP is readily observable through standardized reporting. Given limited administrative capacity, evaluation criteria rationally prioritize measurable local performance. When officials compete on such metrics, trade barriers emerge as equilibrium responses rather than policy failures. Section 2 details why this represents a second-best institutional solution.

I measure the intensity of yardstick competition with a novel proxy: the absolute difference between a city party secretary's start age and the average start age of the other secretaries in the same province.[7] Age is one of the two main predictors of promotion in

[3]The literature has identified other negative aspects of decentralized policy-making in multileveled government: suboptimal fiscal decisions (Oates and Schwab, 1988); tax policy distortions (Keen and Kotsogiannis, 2002); increased corruption (Fan et al., 2009); and populist tendencies (Acemoglu et al., 2013; Kishishita and Yamagishi, 2021).

[4]It is called the 'one-level-up' policy (Chen and Kung, 2019). For example, provincial-level organization departments, including the provincial governor, determine the appointments and promotions of city-level leaders.

[5]Several studies, e.g., Wang et al. (2020); Zeng and Zhou (2024), show that age is an important indicator for upper-level politicians when assessing local politicians' promotion potential. In the absence of electoral accountability, this age-based promotion system provides a standardized mechanism for screening and selecting politicians.

[6]Economic performance is measured by a jurisdiction's economic growth during a politician's tenure. As documented in Li and Zhou (2005), higher growth rates correlate with increased promotion probability.

[7]Section 3.1 details the construction. Politicians whose age is close to the provincial average fall within the prime promotion window and compete directly with many similarly placed peers, while those much older or younger face weaker pressure. The variation is plausibly exogenous because birth years cannot be manipulated, promotion age cutoffs are set institutionally, and the differential changes mainly through

China, and the differential relative to peers, rather than age itself, captures the tournament nature of the promotion system, in which politicians must "win" promotion from their peers.[8]

Controlling for politician and city fixed effects, I find that a one standard deviation increase in the age differential (weaker competition) raises total inter-city trade volumes by 2.6%, which corresponds to a tariff equivalent of about half a percent. The effect is largest for waterway trade (6.2%), where governments control port infrastructure, and 2.9% for motorway trade. Railway trade shows the same pattern with a smaller, statistically insignificant effect, as railway is governed at higher administrative levels over which city secretaries have less direct control, while airway trade shows no clear response.[9] For the same reason, railway is the natural outcome for competition among provincial leaders, and railway trade flows between provinces point the same way, with origin-province effects larger than destination effects. The fragmentation extends beyond domestic markets: export values rise by 3.1% when competition weakens, so the reallocation that restricts internal trade also undermines export performance.

I then trace how competition fragments markets. Competition improves relative performance: politicians facing intense competition outperform provincial peers by 0.6 percentage points of GDP relative to the provincial average (for a one standard deviation increase in competition), consistent with the tournament's relative-performance objective. They achieve this by redirecting public resources toward activities with rapid, locally appropriable returns and away from tradable activities whose benefits spill over to rivals. I document this reallocation across four policy margins. First, competition reduces investment in inter-city motorway networks that lower bilateral trade costs (Allen and Arkolakis, 2022), and the underinvestment concentrates in border counties where spillovers to rival cities are largest, with no effect in interior counties. Second, competition shifts industrial subsidies from tradable manufacturing toward real estate, providing the first empirical evidence for the strategic substitution framework of Becko and O'Connor (2024). Third, competition reallocates land from industrial and logistics uses toward residential development. Fourth, text analysis of annual government work reports shows that intense competition lowers the attention devoted to internal trade and raises the attention devoted

political turnover in other cities.

[8]Chinese leadership has historically emphasized age considerations in political appointments. A notable example is Deng Xiaoping's 干部队伍年轻化 (cadre rejuvenation) policy in the 1980s, which called for the promotion of cadres who were revolutionary, younger, more educated, and more technically specialized. This focus on age in political appointments continued under subsequent leaders, with Hu Jintao advocating for the promotion of experienced older politicians, and Xi Jinping further emphasizing age as a criterion in politician selection. https://news.ifeng.com/mainland/200702/0210_17_75079_1.shtml https://www.gov.cn/xinwen/2018-07/08/content_5304620.htm Retrieved on 31 January, 2025.

[9]Winston (1993) documents how different modes of transportation are subject to different regulations.

to real estate, direct evidence that the reallocation is deliberate.

Firm-level evidence explains why a secretary can tolerate the harm this allocation does to local tradable producers. Using inter-firm linkages identified from large language model (LLM) embeddings of firm descriptions and from historical provincial input-output (IO) tables, I find that intensified competition in one city reduces intermediate-input access, revenue, and labor productivity not only for tradable firms in that city but also for the firms connected to them elsewhere in the province. Because tradable producers are embedded in cross-city input-output linkages, the costs of withdrawing support from them do not stay at home: they propagate to firms in rival cities. In a tournament that rewards relative standing, a cost shared with rivals weighs far less on a secretary's position than its local size suggests, whereas the gains from real estate and other non-tradable activity remain locally appropriable. Politicians therefore favor activities whose gains stay local and withdraw from activities whose losses spread to competitors.

This paper relates to four strands of literature. First, it contributes to the literature on China's decentralized political economy. This literature argues that the M-form organization and decentralized governance strengthened local incentives and facilitated market-oriented reform (Montinola et al. 1995; Qian and Weingast 1997; Qian and Roland 1998; Maskin et al. 2000; Jin et al. 2005; Xu 2011). Maskin et al. (2000) emphasize that regional governments under the M-form structure are comparable, which allows yardstick competition among them. Xu (2011) describes China as a regionally decentralized authoritarian system, in which the center controls personnel and local governments hold substantial economic authority. Bai et al. (2020) argue that local governments use their broad discretion over land, infrastructure, and regulation to favor firms, and that competition across jurisdictions can constrain some forms of local predation. They leave open whether promotion is one reason local officials support firms. I distinguish two kinds of competition. Competition to attract businesses can discipline local governments. Competition among officials for promotion works differently, because the jurisdictions that the M-form framework treats as comparable are also trading partners. An investment that makes trade with neighboring cities easier, such as a motorway, also benefits those cities, which are the official's promotion rivals. Promotion competition therefore moves resources away from such infrastructure and from tradable industries, toward activities whose returns stay within the city, such as real estate.

Second, the paper contributes to the literature on political competition, yardstick competition, and growth. Yardstick competition can discipline politicians and improve policy performance (Besley and Case 1995), and stronger political competition induces more growth-oriented policies (Besley et al. 2010). In China, where political competition operates through promotion rather than elections, career concerns and relative performance

evaluation induce local officials to pursue growth (Li and Zhou 2005; Zhou 2007; Lu et al. 2022; Zeng and Zhou 2024) and shape policy choices in other domains (He et al. 2020; Jia 2024). Song and Xiong (2026) formalize this tournament in the Mandarin model of growth, in which regional governors evaluated on relative output overinvest in locally visible projects and behave short-termistically. My findings add a dynamic qualification to this growth-enhancing view and an inter-jurisdictional one to the Mandarin model. Stronger promotion competition raises a city's GDP relative to its provincial peers in the short run. It achieves this by shifting public investment away from activities whose returns depend on an integrated national market, because these activities also benefit rival jurisdictions. The same shift weakens the city's long-run development. Political competition can therefore improve short-run relative performance yet be dynamically inefficient.

Third, the paper speaks to the literature on local protectionism and internal market fragmentation in China (Young 2000; Bai et al. 2004; Poncet 2005; Bai and Liu 2019; Barwick et al. 2021) and to work quantifying the gains from lower domestic trade costs (Donaldson 2015). Bai et al. (2020) trace local protection to special deals: part of the deal granted to a favored firm is that its competitors are blocked from the local market. In their account, a city can block competitors only in its own market, which limits the damage. Most closely related, Fang et al. (2025) show that career-concerned officials discriminate against firms from politically competing regions. The barriers studied here require no such discrimination. Political competition changes the composition and spatial allocation of public resources themselves, raising inter-city trade costs without formal restrictions on non-local goods or explicit discrimination against outside firms. These policies serve a secretary's standing relative to provincial rivals, but they hurt the city's own tradable producers, whose input access, revenue, and labor productivity fall. The barriers are also not confined to the city's own market. Connected suppliers and partners in other cities of the same province lose inputs and revenue, while potential competitors show no response. The paper thus broadens the political economy of local protectionism from discrimination against non-local firms to barriers that need no discrimination and whose costs travel through production networks.

Finally, the paper contributes to the political economy of trade, in particular to the study of non-tariff barriers. Seminal work shows how political interests and institutions shape international trade policy and integration (Grossman and Helpman 1995; Dollar and Kraay 2003; Dutt and Mitra 2005; Levchenko 2007; Dutt and Traca 2010; Head et al. 2010; Antràs and i Miquel 2011), but less is known about how political incentives below the national level create non-tariff barriers within countries, where subnational governments control infrastructure, land, and industrial policy rather than tariffs. Studies of intra-national trade, including Jiang and Mei (2020) for China, show that political institutions affect

domestic integration but do not identify how competition among subnational politicians shapes the policy determinants of internal trade costs. I show that the career incentives of subnational politicians are a source of non-tariff internal trade barriers, providing a political micro-foundation for domestic trade frictions. A one standard deviation weakening of competition raises inter-city trade by 2.6%, a tariff equivalent of about half a percent. The effect is largest for waterway trade, where city governments control port infrastructure.

The paper is organized as follows: Section 2 describes the institutional background on yardstick political competition and internal trade in China; Sections 3 and 4 describe the data and identification strategy; Section 5 validates the competition measure and presents the main results on city trade; Section 6 examines the mechanism, namely relative economic performance and four policy channels, and then traces the consequences for firms and long-run development; Section 7 concludes.

2 Background

2.1 Politician's Age and Career Prospects

The institutional setup for political advancement in China's multi-level administrative system is key to understanding local policy-making. Promotions follow a top-down evaluation system under the 'one-level-up' policy (*xiaguanyiji*), where higher-level authorities evaluate subordinates using age, economic performance, political experience, and other political and organizational considerations. The literature documents age and relative economic performance as important dimensions of these evaluations: age is a key criterion in promotion decisions (Wang et al., 2020), and economic performance includes GDP growth (Jia et al., 2015) and infrastructure development (Wang et al., 2020).[10]

Formally, the appointment and promotion of city party secretaries run through the provincial Party Committee and its Organization Department under the 'one-level-up' system. The provincial Organization Department maintains cadre files, conducts evaluations, vets candidates, and proposes appointments of prefecture-level leaders, which the provincial Party Standing Committee then approves. The selection procedure follows the stages codified in the Party's Regulations on the Selection and Appointment of Leading Party and Government Cadres. When a vacancy arises, the Organization Department first solicits democratic recommendations from officials at and above the relevant rank, then conducts an inspection of shortlisted candidates that draws on cadre files, past assessment records, and structured interviews with superiors, colleagues, and subordinates. The Department

[10]While Chinese politicians may manipulate GDP growth for promotion, internal trade flows lack institutional salience and are not systematically monitored in cadre evaluations, making manipulation unlikely.

submits its proposal to the provincial Party Standing Committee, which deliberates and votes on the appointment, and successful candidates are publicly announced before taking office. The provincial leadership therefore acts as a single principal that hires, monitors, and promotes every prefecture-level secretary in its jurisdiction.

Evaluation is anchored in the annual cadre assessment (*kaohe*), which scores leaders against target-based criteria that include measurable economic indicators (GDP growth, investment, and fiscal revenue) as well as social stability and party-building responsibilities (Li and Zhou, 2005; Jia et al., 2015; Wang et al., 2020). Under the target responsibility system, provincial authorities assign each city a set of quantified 'hard' economic targets together with 'veto' items such as social stability, and assessment scores are recorded in secretaries' cadre files and consulted directly in appointment decisions (Li and Zhou, 2005). Because these criteria are standardized within a province and applied by a single principal, secretaries of the same rank in the same province are directly comparable, which is precisely the environment in which yardstick evaluation operates. For a city party secretary, promotion means elevation to a vice-provincial-rank position, typically vice governor, a seat on the provincial Party Standing Committee, or the leadership of a major provincial organ. The alternatives are lateral transfer to another city or a same-rank provincial department, retention in place, or movement toward retirement via ceremonial posts in the local people's congress or political consultative conference. Terms are nominally five years, but rotations driven by promotions, transfers, and retirements produce average tenures of two to three years. Rank matters for which tournament a secretary competes in: most prefecture-level cities carry director (*zhengtingji*) rank and their secretaries are appointed by the province, while provincial capitals and sub-provincial (*fushengji*) cities carry vice-provincial rank, and their secretaries, who typically sit on the provincial Party Standing Committee, compete in a different, higher-level pool. The estimation sample accordingly excludes vice-provincial-rank cities (Section 3).

Age is an important promotion factor because of institutional features. First, the combination of local decision-making power and central control creates a tournament-like promotion system where age becomes an easily observable measure of future potential (Li and Zhou, 2005; Xu, 2011; Wang et al., 2020; Fang et al., 2025; Zeng and Zhou, 2024).[11] Second, mandatory retirement ages–60 years for prefecture-level cities and 65 years for province-level cities–impose strict career limits.[12] In practice, formal retirement rules are reinforced by informal age ceilings for entering each rank, so that a secretary who has not reached vice-provincial rank by the mid-fifties is effectively out of contention for further

[11] Younger politicians may demonstrate higher ability by achieving the same rank as older peers in shorter time, while having more years for potential advancement.

[12] Politicians nearing these thresholds face reduced promotion chances and are often moved to ceremonial roles.

advancement. Without electoral accountability, age-based promotion offers a standardized method for comparing politicians across regions.

2.2 City Party Secretary and Within-province Competition

This study focuses on city party secretaries, who hold the highest political authority in city governance within the dual leadership structure alongside mayors. Party secretaries determine strategic priorities while mayors handle operations, highlighting the secretary's influential role in shaping local economic policies. City party secretaries face distinctive within-province competition. Their promotion opportunities are largely confined to provincial-level positions, creating vertical mobility constraints that differ from mayors, who enjoy more flexible career paths including potential promotion to secretary roles. This restriction intensifies competition among secretaries for limited higher-level positions–while provinces may contain over ten cities, available provincial positions are far fewer.[13]

Promotion data from 2004 to 2014 confirms this dynamic: about 18% of secretaries were promoted, and 98% of these promotions stayed within the province. In contrast, mayors enjoyed promotion rates exceeding 50% (Appendix Table H.1). Given that economic performance factors into evaluation, party secretaries provide valuable insights into how promotion incentives shape behavior through yardstick competition.

The institutional structure of China's promotion system generates competitive rather than cooperative incentives among similarly aged secretaries. Provincial-level positions are strictly limited–typically fewer than a dozen slots for provinces containing 10–20 cities–creating a zero-sum environment where one secretary's promotion reduces peers' chances. While factional ties, personal background, and career trajectories also shape appointments at every level of the hierarchy (Shih et al., 2012), relative economic performance benchmarked against provincial peers is one of the central criteria in the evaluation of city party secretaries (Li and Zhou, 2005; Wang et al., 2020). The "one-level-up" policy means secretaries compete for the attention of the same provincial organization department, which uses standardized age and performance criteria to rank candidates. These features ensure that age similarity intensifies rivalry rather than fostering collaboration. This study hypothesizes that secretaries adjust behavior based on age relative to provincial peers–those significantly younger or older face less intense competition, while those with

[13]While city administrative rank may correlate with promotion likelihood–with secretaries from provincial capitals or sub-provincial cities enjoying greater visibility–the inclusion of city fixed effects absorbs time-invariant differences in baseline promotion prospects across city tiers. The age differential measure captures relative competitive pressure among secretaries within the same province at the same time, conditional on this baseline heterogeneity.

similar ages compete most intensely for limited positions without age-based advantages. Appendix Figure G.1 shows the hypothesized non-linear relationship between politician age and trade volumes.

2.3 City Trade, Spillover, and Local Protectionism

In China's institutional context, trade volumes between cities via motorway and waterway transportation serve as important indicators of local protectionism, particularly across internal administrative boundaries. These modes, used primarily for shorter distances unlike airway and railway transportation, are particularly useful for studying protectionist behaviors from within-province competition. Railway, by contrast, carries mainly long-distance freight. In 2010, the average rail shipment traveled 759 km, against 177 km by road, so railway accounted for 29 percent of domestic ton-kilometers (excluding ocean shipping) despite a much smaller share of tonnage (China Statistical Yearbook 2011).

City party secretaries can implement protectionism through direct and indirect methods. Direct actions are most visible in the waterway sector, where local politicians exercise significant regulatory control over port infrastructure. Even in road transportation, secretaries can impose local tolls and administrative fees.[14]

Indirect methods are equally critical, driven by strong yardstick competition incentives. Competition motivates secretaries to strategically reduce subsidies to tradable sectors, preferring non-tradable sectors like real estate that generate locally-confined benefits without enhancing rival jurisdictions' performance. This strategic focus on minimizing positive spillovers is supported by evidence that trade activity correlates with beneficial economic outcomes in nearby cities (see Appendix Table H.2). Politicians possess additional policy instruments including land provision manipulation, road infrastructure development, and trade policy emphasis. These tools, explored in Section 6.2, collectively enable strategic management of inter-jurisdictional economic interactions for political advantage. Appendix Figure G.2 illustrates the proposed causal pathway.

3 Data

Appendix A provides further details on the data.

[14]Truck tolls in China differ widely across provinces and by charging method: some use weight, others axle or distance. Despite the nationwide ETC rollout in 2020, provincial autonomy means a loaded truck can still pay far more than an empty one, and similar routes can carry different costs across regions.

3.1 Yardstick Competition Measured by Politician Age Differential

This study utilizes biographical data for city party secretaries, including start age, education, and promotion history. The key measure to capture yardstick political competition intensity is the 'Politician Age Differential.'

I calculate this measure using Equation 1:

$$\text{Politician Age Differential}_{i,c,t} = \left|\text{Start Age}_{i,c} - \text{Average Start Age}_{p,t}\right| \tag{1}$$

where i, c, p, and t represent the politician, city, province, and year, respectively. $\text{Start Age}_{i,c}$ is the age of politician i when they began their term in city c. $\text{Average Start Age}_{p,t}$ is the average starting age of other city party secretaries within the same province during year t. Figure 1 illustrates how the replacement of a peer secretary mechanically shifts this measure for secretaries on either side of the peer norm.

3.2 City Trade Volumes and Export Value

The primary outcome variables are city-level trade volumes and export values sourced from the Chinese City Statistical Yearbook, spanning 1996 to 2018. Trade volumes encompass transportation across four modes: motorway, waterway, railway, and airway (measured in tons), while export values capture international trade flows (measured in USD). These indicators primarily reflect domestic trade patterns and regional commerce within China's borders. The analysis includes only cities with complete and consistent data coverage throughout the entire study period. Reflecting the rank distinction discussed in Section 2, the baseline estimation sample further excludes cities whose party secretary holds vice-provincial rank (provincial capitals and sub-provincial cities), since these secretaries compete in a different, higher-level tournament rather than the within-province tournament studied here; they enter the analysis only through the separate placebo test in Column (5) of Table 1. Appendix Figure G.3 maps average inter-province trade flows over the sample period.

Freight tonnage is a proxy for inter-regional trade, not a direct measure of it: it records the physical weight of goods moved, embedding transshipment, return trips, and intra-city logistics, a limitation long recognized in work on China's domestic integration (Poncet, 2003, 2005). Two considerations make it suitable nonetheless. First, since tonnage is the dependent variable, any wedge between measured volume and true trade acts as classical measurement error, widening standard errors without biasing the competition coefficient unless the error correlates with competition, which the fixed effects and the exogeneity of peer turnover rule out. Second, the freight measure closely tracks an independent,

input-output-based measure of inter-provincial trade (Appendix Figure G.4). By mode, road transport accounts for roughly three-quarters of domestic freight tonnage, and road and waterway are the modes over which city governments have the most discretion. Railway (about 9 percent of tonnage) is governed at higher administrative levels, where provincial governments coordinate freight with the national rail network, and mainly serves long-distance routes; airway freight is small and centrally regulated. I therefore treat motorway and waterway volumes as the main outcomes for competition among city secretaries. Railway trade, over which city secretaries have less direct control, shows the same pattern at the city level with a smaller and less precisely estimated effect, and it is the natural outcome for competition among provincial leaders; Section 5.5 reports both analyses. Railway is also the only mode for which freight between provinces is recorded by origin and destination province, in the Railway Statistical Yearbook (2000–2018). For road and waterway, the yearbooks report only each city's total freight, which does not show where goods go, so the analysis of bilateral trade between provinces uses railway (Appendix F.3).

3.3 Summary Statistics

Appendix Table H.3 presents summary statistics for the primary variables, covering 292 cities from 1996 to 2018.

The average Politician Age Differential is 2.934 years, with a standard deviation of 2.208 years and a range from 0 to 10 years. Politicians begin their tenure at an average age of 50.50 years, and the average start age of their provincial peers is 50.52 years.[15]

4 Empirical Strategy

4.1 Specification

To test whether age differential affects trade volumes, I estimate the following equation:

$$Y_{c,t} = \alpha_0 + \alpha_1 \text{Politician Age Differential}_{i,c,t} + X'_{i,c,t} + \delta_c + \sigma_i + \tau_t + \epsilon_{c,t} \tag{2}$$

where subscripts i, c, and t denote politician, city, and year, respectively. $Y_{c,t}$ is the logarithm of trade volumes (via road, waterway, train, airway, total, and export value) in city c and year t. The key independent variable, $\text{Politician Age Differential}_{i,c,t}$, measures the absolute difference between a city leader's age and the average age of other city leaders in the same

[15] Appendix Figure G.5 shows the distribution of the signed and absolute Politician Age Differential, and Appendix Figure G.6 shows distributions for the main trade outcome variables.

province, as defined in Equation 1. The vector $X'_{i,c,t}$ includes time-varying controls for city and politician characteristics (the average age of provincial peers and the politician's own age). I do not control for time-varying city GDP, which responds to competition (Table 2). δ_c denotes city fixed effects; σ_i denotes politician fixed effects; τ_t denotes year fixed effects. $\epsilon_{c,t}$ is the error term. Standard errors are clustered at the city level.

The coefficient of interest is α_1, which captures the effect of Politician Age Differential on trade volumes and export value. A positive coefficient would indicate that larger age differentials are associated with higher trade volumes, supporting the hypothesis that intense yardstick political competition (smaller age differentials) leads politicians to engage in local protectionism.

4.2 Identification Assumptions

The identification strategy assumes that, conditional on city, politician, and year fixed effects and time-varying controls, variation in Politician Age Differential is exogenous to unobserved determinants of city-level trade volumes.

From Trade Volumes to Trade Barriers Trade volume and trade barriers are conceptually distinct: higher barriers reduce volume, but low volume need not reflect high barriers. I address this by splitting the determinants of a city's trade volume into a persistent component (preferences, technology, geographic trade costs) and a variable component (local policy, business-cycle conditions, input prices). Identification rests on two assumptions.

Assumption 1 *City, politician, and year fixed effects absorb the persistent component, so identification comes only from the variable component.*

Assumption 2 *Yardstick competition, driven by peer secretary turnover in other cities, shifts the policy part of the variable component while remaining orthogonal to the rest.*

Under Assumptions 1 and 2, the competition-induced change in trade volume identifies the policy channel, even though volume levels do not themselves measure barrier levels; Section 6.2 documents the underlying policy instruments directly.

Plausibly Exogenous Variation Variation in age differential comes mainly from the staggered rotation of city party secretaries across the province and from politicians' start ages. When a peer politician elsewhere is replaced, the focal politician's relative age position shifts for reasons unrelated to conditions in the focal city, namely bureaucratic turnover elsewhere. These rotations follow institutional rules (terms up to five years, averaging

2-3 years) and are set by the provincial Organization Department based on province-wide personnel considerations largely orthogonal to local trade conditions, making the resulting variation plausibly exogenous (Jiang and Mei, 2020). Because the measure uses start ages rather than ages during the tenure, it also does not move with a politician's own short-term performance (Wang et al., 2020). Politician and city fixed effects absorb time-invariant differences that might otherwise correlate with both age differential and trade outcomes, guarding against strategic assignment or selection, so identification relies solely on within-politician variation in relative age position over time.

Balance Tests Appendix Figure G.7 shows no significant correlation between age differential and observable characteristics, including politician education, STEM background, and tenure, or city attributes such as population, GDP, and firm counts, once fixed effects are included. This supports the assumption that age differential does not proxy for other trade-related factors.

5 Results

This section proceeds in five steps. Section 5.1 shows what the Politician Age Differential captures: a secretary's career prospects depend on the ages of provincial peers, more close-in-age peers lower the chance of promotion, and the return to performance is largest near the peer average. Section 5.2 estimates the effect of competition on city trade. Section 5.3 traces the shape of the relationship and tests the symmetry that the absolute-value measure imposes, Section 5.4 examines its dynamics, and Section 5.5 summarizes the robustness checks.

5.1 What the Politician Age Differential Captures

To validate that age differential correlates with career prospects and within-province competition intensity, I analyze how age differential affects politician career outcomes using the following specification:

$$1(\text{Politician Career Prospect})_{i,c} = \sum_j \alpha_j \text{Signed Age Differential Bin}_{j,i,c} + X'_{i,c} + \delta_c + \sigma_t + \epsilon_{i,c} \tag{3}$$

The dependent variable captures two career outcomes: (1) Promotion, an indicator for whether politician i in city c is promoted following their tenure; and (2) Transfer/Retirement, an indicator for whether the politician is transferred or retired. Demotion serves as the reference category. The key explanatory variable represents discrete bins of the signed age

gap (politician's age minus provincial average), with the $[-1, 1]$ year range as the reference category; negative values indicate younger than peers, positive values indicate older. I use binned age differentials in Equation 3 to visualize the non-linear relationship across the age distribution in Figure 2; Appendix Tables H.4 and H.5 present regressions using a continuous measure. Controls include individual characteristics, city-level variables, connections to provincial politicians, city fixed effects, and tenure start year fixed effects. Standard errors are clustered at the city level.

Figure 2 presents the relationship between signed age differential bins and career prospects. The youngest politicians (relative to peers) exhibit the highest promotion probability, while the oldest have the lowest. Appendix Tables H.4 and H.5 present corresponding regressions: a one-year increase in signed age differential reduces promotion probability by 3.3 percentage points (Appendix Table H.4, column 4) and increases transfer/retirement probability by 3.9 percentage points (Appendix Table H.5, column 4). These effects are attenuated for a higher city tier, e.g., deputy-level cities, consistent with Wang et al. (2020).

Because the secretary's own age is held fixed, these estimates show that a secretary's prospects depend on the ages of the other secretaries in the province: the contest is relative, and provincial peers are the rivals. Promotion probability is highest for secretaries much younger than their peers, lowest for those much older, and intermediate near the peer average, where age no longer separates candidates and relative performance decides. Measured against a typical peer, promotion uncertainty is highest for secretaries near the peer average, whose promotion rate is close to the average (20 percent, against 18 percent overall), and lowest for the youngest and oldest secretaries, whose age gives them a clear advantage or disadvantage (Appendix Figure G.8). The absolute age differential measures distance from this zone. Two further results show that this is where competition is most intense.

The age differential is a continuous, cutoff-free measure of a politician's distance from the provincial peer norm, but two politicians with the same differential may face different numbers of close-in-age rivals. I therefore complement it with a measure of competitive density: the number of other city party secretaries in the same province whose start age falls within three years of the focal secretary's, a bandwidth motivated by Figure 2. For career outcomes I use the term-average count; for the annual trade analysis (Section 5.5) I use the annual count.

Figure 3 relates similar-age peer density to career outcomes, with one observation per secretary term. I regress indicators for promotion and for transfer or retirement at the end of the term on the term-average number of similar-age peers, controlling for the secretary's start age, education, and colleague and hometown ties to the provincial party secretary

and governor, and for the city's nighttime light density and population; standard errors are clustered by city. One additional similar-age peer lowers the probability of promotion by 0.9 percentage points and raises the probability of transfer or retirement by 1.0 percentage point. As the number of similar-age peers rises from zero to ten, the predicted probability of promotion falls from 0.22 to 0.13, and that of transfer or retirement rises from 0.76 to 0.86. These associations are consistent with tournament congestion: a denser set of close-in-age rivals worsens a secretary's career prospects.

Close-in-age competition should also change how strongly performance is rewarded, not only the level of promotion odds. Figure 4 relates promotion at the end of a tenure to growth performance, allowing the effect of performance to vary with the absolute age differential. For all four measures based on reported real GDP growth, the promotion payoff to better performance is largest for secretaries close to the peer average and declines as the age distance widens, consistent with stronger incentives among close-in-age rivals. The estimates are imprecise: the payoff at the 25th percentile of the differential exceeds that at the 75th percentile by 3.6 to 8.2 percentage points per standard deviation of performance, and none of these differences is significant at the 5 percent level.

5.2 Politician Age Differential and City Trade Volumes

Table 1 reports estimates of Equation 2, the effect of Politician Age Differential on city trade volumes. Panel A presents estimates for total city trade volumes across four progressively stringent specifications, followed by two placebo columns. Column (1) employs city fixed effects and year fixed effects as the baseline specification. Column (2) augments this with city and politician characteristics. Column (3) further adds politician fixed effects and replaces year fixed effects with province-year fixed effects to absorb province-specific time trends. Column (4) instead implements city-politician fixed effects together with year fixed effects to account for within-politician variation across different postings; I treat this as the paper's preferred specification throughout. The coefficient on Politician Age Differential remains positive and statistically significant across all four specifications. Under the preferred specification in Column (4), the estimated coefficient of 0.012 indicates that a one standard deviation increase in Politician Age Differential corresponds to a 2.6 percent increase in total trade volumes.[16]

The magnitude of this effect is economically substantial and comparable to findings in Li and Zhou (2005), and is consistent with evidence that promotion incentives near age thresholds raise reported GDP growth by about 3.4 percentage points (Zeng and Zhou,

[16]The standard deviation of Politician Age Differential is 2.208 years, yielding the calculation: $0.012 \times 2.208 = 0.026$.

2024), suggesting that political competition represents a significant factor in economic policy implementation.

Because these barriers are non-tariff, their size can be expressed as a tariff equivalent, the standard metric in the trade literature. In gravity models, a one percent increase in trade costs lowers trade by about θ percent, where θ is the trade elasticity. With θ between 4 and 5 (Simonovska and Waugh, 2014; Head and Mayer, 2014), the 2.6 percent response of total trade to a one standard deviation change in competition corresponds to an ad valorem tariff of about 0.5 to 0.65 percent; for waterway trade, the figure is about 1.2 to 1.6 percent.[17]

Panel B analyzes motorway trade volumes, where coefficients remain positive and significant across specifications, with a magnitude comparable to the total-trade effect (2.9 percent increase under Column 4). This finding aligns with evidence that highway-driven market access fosters trade integration (Faber, 2014).

Panel C examines waterway trade, revealing the largest effect with a 6.2 percent increase. This substantial impact reflects the high degree of government control over waterway and port infrastructure (Notteboom and Yang, 2017), which makes waterway systems particularly responsive to local officials' strategic decisions.

Panel D shows that international export value increases by 3.1 percent when domestic competition intensity decreases. While politicians may not be incentivized to restrict exports directly, tradable firms source many intermediate goods from other cities, so internal barriers make those inputs more expensive and harder to obtain. Combined with the diversion of subsidies and land away from tradable sectors, higher input costs leave these firms less competitive in global markets as well as at home (Section 6.3.1).

Column (5) provides a complementary placebo by restricting the sample to secretaries of provincial capitals and vice-provincial cities, who compete in a different, higher-level tournament and do not face the same within-province promotion competition as director-level secretaries (Section 2). Consistent with this institutional difference, the estimated coefficient for this subsample is small and statistically indistinguishable from zero across all four trade outcomes.

Column (6) provides a falsification test using city mayors, whose career incentives differ from those of party secretaries: a mayor's main advancement path leads to the secretary post, most often of the same city, so mayors do not compete in the same closed within-province tournament (Section 2).[18] When the same specification, with province-year in place of year fixed effects, is estimated with the age differential among city mayors, the coefficients are statistically indistinguishable from zero and economically negligible for

[17]The tariff equivalent equals the trade response divided by θ. The calculation treats freight responses as responses of trade values and is meant only to convey magnitudes.

[18]The promotion rate of mayors is around 50%, while that of party secretaries is approximately 18%.

every trade outcome. A horse-race specification that includes both the secretary's and the mayor's age differential confirms this pattern: the secretary's coefficient remains positive across all outcomes and statistically significant for total, motorway, and waterway trade, while the mayor's coefficient is small and statistically insignificant throughout (Appendix Table H.6).

5.3 Shape and Symmetry

To trace out the shape of this relationship rather than impose linearity, I re-estimate the trade regression using discrete bins of the signed age differential in place of the continuous absolute-value measure:

$$\ln(Y_{c,t}) = \sum_{\substack{j=1 \\ j\neq 4}}^{7} \beta_j \, \text{Signed Age Differential Bin}_{j,c,t} + \eta_{c\times i} + \tau_t + \epsilon_{c,t} \tag{4}$$

where $Y_{c,t}$ is the trade outcome for city c in year t, and the seven bins are defined as in Equation 3, with the Peer Average bin ($j = 4$, the $[-1, 1]$ year range) omitted as the reference category. $\eta_{c\times i}$ and τ_t denote city-politician and year fixed effects, respectively. Standard errors are clustered at the city level.

This non-parametric specification traces the pattern across the full distribution of the age differential (Figure 5): trade volumes are lowest when a politician's age is close to the provincial average and rise as the differential widens in either direction. The pattern holds on both sides of the distribution, ruling out the possibility that the baseline result is driven by seniority alone; politicians who are much younger than their peers, not only those who are much older, also generate higher trade volumes. The U-shape is most pronounced for waterway trade and comparatively muted for motorway trade, mirroring the panel-by-panel ranking in Table 1. A formal test supports the absolute-value measure used in the baseline: when the effect is allowed to differ on each side of the peer average and to bend on each side, the restriction to a linear and symmetric relationship cannot be rejected for any outcome ($p = 0.75$, 0.37, 0.98, and 0.30 for total, motorway, waterway, and export trade).[19]

The absolute-value construction also embeds a symmetry restriction: a given distance from the peer norm is assumed to matter equally for secretaries above and below it. The hypothesis itself requires only that competition weakens as a secretary moves away from the peer average in either direction, not that the two sides move trade by the same amount.

[19] I add separate quadratic terms in the years younger and the years older than the peer average to the two-slope specification; the absolute-value measure is the special case with equal slopes and no curvature. The quadratic terms are small and insignificant, and cubic terms do not improve the fit.

Appendix Table H.7 tests this restriction directly by decomposing the differential into years younger and years older than the peer average. The coefficients are positive on both sides for every trade outcome, and tests of equality yield p-values of 0.240, 0.212, 0.491, and 0.869 for total, motorway, waterway, and export trade, respectively, so the restriction is not rejected, although I do not take a strong stance on whether the two sides affect trade by exactly the same amount.

5.4 Dynamics: Peer Arrivals and Tenure

To provide dynamic evidence and distinguish the age-similarity channel from a generic turnover shock, I compare the arrival of a similar-age peer secretary (within three years of the focal secretary's start age) with the arrival of a dissimilar-age peer in another city of the same province. For each group $g \in \{S, D\}$, I estimate the following event-study specification separately:

$$Y_{c,t} = \sum_{\substack{k=-4 \\ k\neq 0}}^{4} \beta_k^g D_{i,c,t}^g(k) + X'_{i,c,t}\gamma + \eta_{c\times i} + \tau_t + \epsilon_{c,p,t}, \qquad g \in \{S, D\}, \tag{5}$$

where $D_{i,c,t}^g(k)$ indicates that city c is observed k years relative to the arrival of a peer of type g: S denotes an incoming secretary whose age is within three years of the focal secretary's age, and D denotes an incoming secretary outside that band. The specification includes four leads and four lags, omits the arrival year ($k = 0$), and includes the same controls and city-politician and year fixed effects used in the preferred specification. Standard errors are clustered at the city level. The lead coefficients provide a diagnostic for pre-trends: a competition response should follow, rather than precede, the arrival of a similar-age rival.

Figure 6 reports the results. The pre-arrival estimates do not display a systematic decline in trade for the similar-age series.[20] After arrival, the similar-age series shifts downward, whereas the dissimilar-age series shows no comparable decline. Because dissimilar-age arrivals are also turnover events, this contrast is consistent with an age-similarity channel rather than a generic province-wide turnover effect.

The effect weakens when I use an alternative, wider similar-age band of five years (Appendix Figure G.9), as the wider band admits less comparable peers and weakens rivalry

[20] Joint tests that the four pre-arrival coefficients are zero do not reject for total, motorway, or waterway trade, either for similar-age arrivals ($p = 0.55$, 0.28, and 0.34) or for dissimilar-age arrivals ($p = 0.11$, 0.19, and 0.92). Export value may show a pre-trend after similar-age arrivals: the test rejects at the 10 percent level ($p = 0.09$, against $p = 0.83$ after dissimilar-age arrivals), driven by the coefficient four years before arrival. However, the post-arrival decline after similar-age arrivals is much larger than the placebo response after dissimilar-age arrivals (average post-arrival coefficients of -0.034 and -0.012).

intensity, confirming that the three-year result reflects genuine age-based competition rather than the bandwidth choice.

Splitting the dissimilar-age arrivals into peers more than three years younger and more than three years older than the focal secretary gives a symmetric picture (Appendix Figure G.10). After either type of arrival, the average post-arrival coefficient is small and insignificant for every outcome, ranging from -0.022 to 0.007, and tests that younger and older arrivals have the same average post-arrival effect do not reject (p between 0.46 and 0.74). The decline in trade is therefore specific to similar-age arrivals, and younger and older newcomers affect trade alike.

A further dynamic pattern is consistent with career incentives that strengthen as the promotion decision approaches: the effect is positive in every year of the secretary's tenure and rises modestly at its end (Appendix Figure G.11).

5.5 Robustness Checks

The baseline estimates survive a battery of checks, reported in the Appendix. First, cross-provincial age differentials, which should not respond to within-province competition, show no effect (Appendix Table H.8). Second, the response is strongest for the transport modes that city governments control most directly: railway trade, which is governed at higher administrative levels, shows the same pattern with a coefficient close to the motorway estimate but not statistically significant, airway trade shows no clear response (Appendix Table H.9), and a stacked regression with mode-specific coefficients and mode-interacted fixed effects cannot reject equality across modes (Appendix Table H.10). Third, alternative sets of fixed effects and alternative clustering (Appendix Table H.11) and an instrumental variable strategy (Appendix Table H.12; first-stage relationship in Appendix Figure G.12) give similar estimates. Fourth, because leadership turnover may itself induce changes in local policy (Shen et al., 2022), I exclude city-years in which the focal city's own secretary turns over, so that the variation comes only from turnover in other cities; the estimates remain positive and are larger for total and motorway trade, consistent with the turnover year being a transition year in which trade reflects both the outgoing and the incoming secretary's policies (Appendix Table H.13). Fifth, the estimates are robust to controlling for lagged trade volume and historical economic conditions (Appendix Table H.14), and railway infrastructure, which is centrally planned and outside local politicians' control, shows no corresponding response to competition (Appendix Table H.15). Sixth, an alternative measure of competition, the annual number of similar-age peers, reproduces the result with the mirrored sign: one additional similar-age peer is associated with lower total, motorway, and waterway trade, while the export estimate is negative but imprecise

(Appendix Table H.16).[21] Alternative competition measures based on tenure, education, or working experience show no comparable effects, confirming that age-based competition drives the results (Appendix F.2). Seventh, provincial-level analysis using inter-provincial railway trade, the mode over which provincial leaders have more influence (Appendix F.3), yields consistent findings, with the trade-reducing effects originating primarily from competition intensity in the exporting region. Finally, the effect is smaller for regions that rely more on trade and manufacturing sectors (Appendix F.5), shows no pre-existing trend prior to a peer turnover (Appendix F.6), and is not an artifact of mechanical tenure effects (Appendix F.7).

6 Mechanisms and Additional Results

Having documented that yardstick competition reduces inter-city trade, I now connect the political objective to policy choices, firm outcomes, and long-run development. First, relative GDP performance shows that competitive politicians improve their standing against provincial peers. Second, the policy evidence tests one allocation mechanism: competition favors activities whose benefits remain concentrated locally over those benefiting rivals. Third, firm-level results trace the consequences for input sourcing, revenue, and productivity within cities and across business networks. Fourth, the long-run analysis examines development beyond the immediate performance comparison. The sequence links what politicians seek, how they allocate resources, who bears the costs, and how those costs persist.

6.1 Mechanism: Competition and Relative Economic Performance

Table 2 presents results on the relationship between Politician Age Differential and contemporaneous GDP performance relative to provincial peers. The analysis employs city-politician fixed effects and year fixed effects to isolate within-politician variation across postings and control for common temporal shocks. Columns (1) and (2) measure the GDP gap in hundreds of millions of RMB between a city and the provincial average, Columns (3) and (4) express this gap as a percentage deviation from the provincial mean, and Columns (5) and (6) use next-year GDP. Odd-numbered columns use Politician Age Differential and even-numbered columns use the number of similar-age peers.

The results reveal a significant negative relationship between Politician Age Differential and relative performance. A one standard deviation increase in Politician Age Differential

[21] The pattern extends to complementary outcomes. More similar-age peers predict stronger short-run GDP performance relative to provincial peers (Table 2). In provincial bilateral trade, both origin and destination counts are negative (Appendix Table H.17).

is associated with a 15.4 hundred million RMB decrease in GDP relative to the provincial average (Column 1), and a 0.6 percentage point reduction in relative GDP performance (Column 3).[22]

The negative coefficients indicate that politicians facing stronger competitive pressure–those with smaller age differentials from their provincial peers–achieve better short-term GDP performance relative to competitors. This finding is consistent with tournament theory predictions (Lazear and Rosen, 1981; Li and Zhou, 2005) and with the Mandarin model of growth, in which relative evaluation drives governors to boost measured regional output (Song and Xiong, 2026).

Integrating these findings with the trade results in Table 1 reveals the logic of yardstick competition. Politicians facing intense competition simultaneously improve GDP standing relative to provincial peers and raise trade barriers against them. The tournament rewards this relative standing, not the profitability of local firms as a whole. A city can gain ground through stronger own performance, weaker peer performance, or both. With fiscal resources and land quotas limited, consider two equally costly projects generating the same local output gain. If one also raises output in competing cities, it delivers a smaller improvement in the secretary's relative standing, even when its total economic return is higher. The political choice therefore depends on who shares the benefits. The policy evidence follows this logic: secretaries withdraw support from tradable subsidies, industrial land, and cross-border roads whose returns reach rivals, redirecting resources toward locally confined activity such as real estate (Section 6.2). The first shift lowers spillovers to peers; the second supports locally appropriable output. Productive local firms can consequently lose support even as the allocation improves the secretary's competitive position.

This logic also distinguishes the mechanism from producer-oriented protectionism, in which local firms gain at outsiders' expense. Here the city's own tradable firms lose access to inputs and markets, and connected firms in other cities lose with them (Section 6.3.1), even though the allocation improves the secretary's relative position. Better relative performance, weaker local industry, and lower aggregate efficiency therefore move together.

6.2 Policy Channels Behind the Non-tariff Trade Barriers

The reduced-form results show that stronger yardstick competition raises internal trade barriers, but not through which instruments. Because tariffs cannot be levied within

[22]Calculations: $6.953 \times 2.208 = 15.4$ hundred million RMB; $0.270 \times 2.208 = 0.6$ percentage points, where 2.208 years is the standard deviation of Politician Age Differential.

a country, these are non-tariff barriers.[23] I identify four policy channels behind these barriers: underinvestment in motorways in counties bordering rival cities (Table 3), a shift of industrial subsidies from tradable sectors toward real estate (Table 4), a similar shift in land supply (Table 5), and less attention to internal trade in government work reports (Table 6). None restricts goods or firms from other regions directly; together they raise the cost of trading across jurisdictions through the allocation of public resources and the policy priorities behind it. Across all four, politicians facing intense competition do not cut spending indiscriminately; instead, they strategically withhold investments most likely to spill over to rival jurisdictions while sustaining spillover-free alternatives.

6.2.1 Endogenous Trade Costs: Competition and Intercity Motorway Network Development

Motorway construction is the canonical high-spillover policy: it connects multiple jurisdictions, lowers bilateral trade costs, and re-routes economic activity beyond investing city boundaries. Spatial trade models show that improving transport links lowers generalized trade costs across many origin-destination pairs, with benefits diffusing well beyond the directly treated links,[24] and in China expressways lower ad-valorem transport costs by roughly 20 percent relative to regular roads and raise aggregate exports (Fan et al., 2023). When competitive pressure is acute, incumbents tilt spending toward low-spillover, place-based amenities with geographically contained benefits; when it weakens, the political return to high-spillover projects rises, making cross-jurisdictional roads more attractive (Bougheas et al., 1999; Limao and Venables, 2001).

I test whether competition depresses connective infrastructure investment using a GIS panel that combines political data with geospatial transport networks (Davis et al., 2025).[25] Rather than estimate a uniform average effect, I exploit spatial variation in spillover exposure across counties, since the theory predicts where underinvestment occurs, not that it occurs everywhere. I classify each county as *border* or *interior* depending on whether it adjoins a rival city's counties within the same province. Motorway investment in border counties mechanically lowers bilateral trade costs to rivals and generates substantial cross-jurisdictional spillovers through general equilibrium adjustments (Allen and Arkolakis, 2014); investment in interior counties, buffered from rival boundaries, yields mostly local

[23] Non-tariff measures are policy measures other than ordinary customs tariffs that can affect the quantities or prices of traded goods; subsidies form a separate chapter of the international classification (UNCTAD, International Classification of Non-Tariff Measures, 2019 version).

[24] Allen and Arkolakis (2014) estimate the U.S. Interstate Highway System raised welfare by about 3.5 percent; Allen and Arkolakis (2022) similarly show network improvements reshape flows and welfare far beyond treated links.

[25] The Davis et al. (2025) dataset provides annual GIS records for highways, motorways, and rail, with routes and access points detailed enough to construct consistent connectivity measures over time.

returns with little leakage. If politicians concentrate underinvestment where the spillover-to-local-cost ratio is highest, the effect of Politician Age Differential on motorway provision should be concentrated in border counties and weak or absent in interior ones.

Table 3 reports interaction-term regressions across five motorway network measures. The interaction between Politician Age Differential and the border-county indicator is positive and statistically significant across all five outcomes, while the uninteracted (interior-county) effect is small in magnitude and statistically indistinguishable from zero. This pattern indicates that competitive underinvestment in motorway infrastructure is not a uniform cut to spending but concentrates precisely in border locations, exactly where the spillover-based mechanism predicts it should. Appendix Table H.18 in Appendix F.8 further decomposes the border-county effect by administrative class: the response is concentrated in national expressways, the largest and most spillover-intensive projects, whose implementation (land expropriation, matching funds, and construction pacing) runs through local governments even though their corridors are centrally designated. Appendix Table H.19 shows the effect of motorway development on trade volume and exports.

A natural question is how an instrument that accumulates over multi-year horizons can be reconciled with annual variation in trade flows. The answer is that while the motorway stock itself moves slowly, the margins through which it affects trade costs are adjusted year by year. Construction is incremental: local governments decide annually which segments to fund, accelerate, or delay, so even a persistent infrastructure strategy translates into yearly changes in effective network capacity (Donaldson, 2018; Allen and Arkolakis, 2014). Moreover, trade responds to effective rather than physical trade costs, which local governments can raise without a formal process, for example through charges at road blocks, border controls on the movement of goods, and discriminatory fees and certification requirements on goods from other regions (Young, 2000; Poncet, 2005). The stock of motorway capacity thus pins down the medium-run level of trade costs a city carries, while annual choices over construction pace and network operation generate the year-to-year responses documented in Table 1.

6.2.2 Production Costs and Productivity: Industrial Policy Spending

Industrial policy, particularly subsidies, is a central tool of Chinese economic governance (Aghion et al., 2015). Recent work on geoeconomics argues that peacetime industrial policy substitutes for credible conflict-time trade punishment: governments unable to threaten future trade taxes rely more heavily on industrial policy to build strategic capacity today (Becko and O'Connor, 2024). Applied to the subgovernmental context, where local politicians lack direct trade instruments such as tariffs, this logic implies that political

incentives should instead show up in the composition of subsidies across tradable and non-tradable sectors. Politicians facing intense yardstick competition may shift subsidies away from tradable sectors, whose productivity gains could spill over to rival jurisdictions, and toward non-tradable sectors, whose benefits stay local.

To test this mechanism, I construct a city-sector panel and estimate:

$$\begin{aligned}\text{Log(Subsidy)}_{s,c,t} = \beta_0 + \beta_1 \text{ Politician Age Differential}_{c,t} \times \text{Sector}_s \\ + X'_{s,c,t} + \delta_{c,s} + \sigma_i + \tau_{s,t} + \gamma_{c,t} + \epsilon_{c,s,t},\end{aligned} \tag{6}$$

where $\text{Log(Subsidy)}_{s,c,t}$ is the logarithm of subsidies received by sector s in city c during year t. Sector_s equals 1 if the sector is tradable (manufacturing and wholesale following (Brandt et al., 2017)), 0 otherwise. Controls include city-sector characteristics and politician characteristics; fixed effects are at the city-sector, politician, sector-year, and city-year levels. Standard errors are clustered at the city-sector level.

Table 4 shows that the interaction between Politician Age Differential and the tradable-sector indicator is positive and significant across specifications. In the full model (Column 3), the coefficient of 0.087 implies that a one-standard-deviation increase in age differential (weaker yardstick competition) is associated with a 27.3 percent increase in subsidies to tradable sectors relative to non-tradables.[26] Consistent with the strategic-substitution logic in Becko and O'Connor (2024), when direct trade-policy instruments are unavailable at the local level, politicians facing less competitive pressure deploy industrial policy more aggressively toward tradables despite cross-jurisdiction spillovers. By contrast, the interaction with real-estate sectors in the full model is negative (-0.121, Column 6) and statistically significant, suggesting that a one–standard-deviation increase in age differential (weaker yardstick competition) is associated with a 34.0 percent decrease in subsidies to real-estate sectors relative to other sectors.

Relative to other sectors, subsidies to tradable sectors show no significant change before a similar-age peer arrives and fall after the arrival, whereas dissimilar-age arrivals show no comparable decline (Appendix Figure G.13).

These magnitudes, on the order of 50 percent shifts in relative subsidy allocations, represent economically meaningful policy distortions, consistent with evidence on local political incentives in China (Chen and Kung, 2019; Wang et al., 2020; Lu et al., 2022).

6.2.3 Agglomeration and Capacity: Land Allocation

Land allocation is a first-order policy lever for local politicians, shaping both urban form and the sectoral composition of economic activity. Land-use planning and the timing,

[26] Using $(e^{0.087} - 1) \times 100 = 9.1\%$, then $2.98 \times 0.091 = 0.273$.

quantity, and composition of land released to the market thus become instruments for influencing observable outcomes (GDP growth, investment, fiscal take) (Brueckner, 2000). Given the post-1994 fiscal architecture, land-lease revenues ("land finance") provide a large, discretionary funding source for local governments, further heightening the political salience of land supply decisions (Cai et al., 2013; Fang et al., 2019).

To examine these mechanisms, I re-estimate Equation 2 at the city level, rotating the outcome variable across distinct measures of land provision and value. Specifically, I compile city-year aggregates for (i) manufacturing/tradable industrial land, (ii) real estate land (residential/commercial), and (iii) other non-tradable sector land. Industrial land transactions are drawn from the Ministry of Natural Resources' public registry,[27] which reports universe-level transactions from 2000 to 2018; I classify parcels by designated end use following standard administrative categories used in urban land markets (see, e.g., Cai et al., 2013). For each category, I study both physical provision (area released) and contemporaneous market value (total transaction value), recognizing that quantities and prices capture different margins of political intervention.

Table 5 shows how the intensity of yardstick competition maps into land allocation across sectors. Consistent with the mechanism, weaker competition (larger age differential) induces a reallocation toward tradables and away from non-tradables.

A one standard deviation increase in Politician Age Differential raises industrial land provision by 12.4% and the total value of industrial land sold by 20.1% (Columns 1 and 2), and lowers real estate land provision by 13.9% and its value by 32.5% (Columns 3 and 4); land provision for other non-tradable uses falls by 14.2% and its value by 23.8% (Columns 5 and 6). Politicians facing less immediate benchmarking pressure are more willing to invest in slower-maturing, high-spillover industrial capacity (Duranton and Puga, 2004), whereas intense competition pushes them toward projects with rapid, locally appropriable gains in measured GDP and fiscal revenues.

Industrial land and related infrastructure generate spatial spillovers through supply chain linkages, labor pooling, and knowledge externalities that extend beyond jurisdictional borders (Duranton and Puga, 2004; Greenstone et al., 2010; Combes et al., 2011), whereas real estate and other non-tradables deliver benefits that are highly visible, quickly realized, and geographically confined. Consistent with this reallocation, weaker competition is also associated with more new firm registrations in tradable sectors and fewer in real estate (Appendix Table H.20); see Appendix F.9 for details.

[27] China's Ministry of National Land and Resources website https://www.landchina.com.

6.2.4 Stated Policy Priorities: Trade and Real Estate Attention in Government Work Reports

The three instruments above operate through infrastructure, budgets, and land quotas. A complementary question is whether the underlying reallocation is visible in politicians' own stated priorities. Annual government work reports are the canonical statement of a city government's policy agenda: delivered at the start of each year, they set out the leadership's priorities for the year ahead. Because the reports are drafted under the direct control of the city leadership and revised every year, stated priorities can respond to competitive pressure at the same annual frequency as the competition measure itself, making this the fastest-moving of the four channels.[28]

To measure policy attention, I construct length-weighted indices of trade-related and real-estate-related policy mentions from the full text of each city's annual report, weighting the frequency of topic-related mentions by the length of the corresponding policy text. Appendix Figure G.14 reports the keyword composition underlying the two indices. I then estimate specifications analogous to Equation 2, with the attention indices as outcomes.

Table 6 reports the results. Columns (1) to (3) examine trade-policy mentions and Columns (4) to (6) real-estate mentions, progressively adding politician characteristics and city-politician fixed effects. In the most demanding specification, a larger Politician Age Differential (weaker competition) significantly raises attention to internal trade (0.302, Column 3) and significantly lowers attention to real estate (-0.181, Column 6). This textual evidence complements the budgetary and allocative margins: the reallocation documented in subsidies, infrastructure, and land shows up in what politicians say they will do, indicating a deliberate shift in policy priorities rather than an incidental by-product of other decisions.

6.3 Competition and Firm Supply Chain Disruption

In this section, I examine the effect of competition on firm outcomes, especially those connected via business networks.

[28] Government work reports are widely used in the literature on Chinese local governance to measure the targets and priorities of city leaderships. Li et al. (2019) extract GDP growth targets from provincial and prefectural work reports to study target setting in promotion tournaments, and Campante et al. (2023) track the frequency of stability-related language in work reports to show that local governments reallocate attention toward public security. I follow this approach and treat the allocation of text across policy topics as a revealed measure of the leadership's stated priorities.

6.3.1 Local and Spillover Effects of Politician Age Differential on Firms

Yardstick competition can affect economic activity not only through a city's own trade flows but also through the inter-city linkages in which firms are embedded: intense competition may lead politicians to underinvest in policies that generate beneficial network spillovers. I therefore hypothesize that stronger competition (a smaller age differential) worsens the performance of firms reliant on inter-regional network ties. To test this, I construct a firm-level panel from the Annual Survey of Industrial Firms (ASIF), spanning 2000–2007 following Brandt et al. (2017), combined with inter-firm business network connections.

Business networks are identified in two complementary ways. The first approach builds on Breitung and Müller (2025): firm pairs are linked when the similarity of their business profiles, which combine industry classification, business activities, and financial characteristics, is at least 0.95; Appendix B details the construction, embedding model, and threshold choice, Appendix Table H.21 provides example matched pairs, and Appendix Figure G.15 visualizes the resulting networks. The second approach uses province-specific Input-Output (IO) tables to measure sectoral connectedness within each province (Appendix C).

The empirical strategy estimates the impact of yardstick political competition faced by the politician in firm i's city on the economic outcomes of the connected firm j located in a different city. I estimate the following equation:

$$Y_{j,s,c_j,t} = \beta_0 + \beta_1 \text{Politician Age Differential}_{i,c_i,t} + \mathbf{X}'_{j,s,c_j,t} + \delta_{pair} + \sigma_i + \tau_{s,t} + \gamma_{c_j,t} + \epsilon_{j,s,c_j,t} \tag{7}$$

where $Y_{j,s,c_j,t}$ represents firm outcomes (e.g., log intermediate inputs, log revenue) for the connected firm j in sector s and city c_j at year t. The key independent variable is Politician Age Differential$_{i,c_i,t}$ for the politician governing firm i's city c_i at year t. Note that c_i and c_j denote distinct cities: c_i is the focal city where political competition is measured, and c_j is the connected city where firm j's outcomes are observed. $\mathbf{X}'$ includes time-varying controls for firm j. The specification incorporates firm-pair fixed effects (δ_{pair}) to control for time-invariant characteristics of each specific firm linkage, politician fixed effects (σ_i) for the politician in city c_i, sector-year fixed effects ($\tau_{s,t}$), and connected-city-year fixed effects ($\gamma_{c_j,t}$). Standard errors are clustered at the city level. The coefficient of interest, β_1, captures the effect of yardstick political competition in city c_i on its network partners in city c_j.

Table 7 presents the results. Columns (1)-(4) report direct effects on the focal firm i:[29]

[29] For direct effects, I estimate an analogous specification in which firm i is both the focal and outcome unit; the firm-pair fixed effect then collapses to a firm fixed effect. Outcomes are intermediate inputs, profit, revenue, and revenue per worker, all in logs.

a larger Politician Age Differential (weaker competition) is associated with significantly better performance across all four outcomes. A one standard deviation increase raises intermediate good inputs by 5.0% and revenue by 3.5%, with profits and revenue per worker of similar magnitude (Table 7).[30] Columns (5)-(8) show that the spillover effects on connected firms j, estimated from Equation (7), are qualitatively identical and slightly attenuated, as expected for effects transmitted through the network. The estimates are robust to replacing the text-based network with pre-sample (1997) provincial input-output linkages (Appendix Table H.22). Together, the direct and spillover estimates indicate that intense competition generates negative externalities (trade frictions and underinvestment in shared infrastructure) that harm firms within the jurisdiction and propagate to their network partners elsewhere in the province. This propagation is also what makes the harm to local tradable producers politically tolerable: because part of the cost travels through input-output linkages to firms in rival cities, it weighs less on a secretary's relative standing than its local size suggests, while the gains from non-tradable activity stay local.

A natural objection is that tournament competition should *help* local firms: career-concerned officials favor them through preferential procurement and implicit barriers against outsiders (Fang et al., 2025; Barwick et al., 2021). The ownership split shows which firms bear the costs of this allocation response, alongside preferential support for selected firms. The split (Appendix Table H.23) shows a clear distributional pattern: the adverse competition effect is borne almost entirely by private firms, while for state-owned enterprises the SOE interaction offsets the main effect to a near-zero net, for both focal and connected firms. State-owned firms thus appear more insulated from fragmentation costs. This accommodates special deals, whose beneficiaries can also be privately owned (Bai et al., 2020). The average effect is negative because private, trade-reliant firms dominate the sample.

Firm-level losses and short-run relative GDP gains are consistent (Section 6.1). The GDP outcome benchmarks the whole local economy against peers; the ASIF outcomes measure industrial producers' own performance. Under competitive pressure, leaders channel resources toward non-tradables, particularly real estate, which accounts for roughly a quarter of Chinese GDP and is directly influenced through land conveyance (Rogoff and Yang, 2022). A decomposition of city GDP points the same way: weaker competition raises the tradable share and lowers the non-tradable share, although these imprecise estimates serve as directional corroboration (Appendix Table H.24).

A placebo test using connected firms located in bordering cities across provincial lines shows no effect (Appendix Table H.25), consistent with the within-province tournament

[30]The standard deviation of Politician Age Differential is 2.192 in the focal-firm sample and 1.986 in the connected-firm sample.

mechanism rather than broader regional shocks or unobserved confounders.

I further examine firm heterogeneity in competition effects and find that the effect is concentrated among downstream firms, consistent with these firms being more embedded in and reliant upon upstream network connections for inputs. The effect is also concentrated among supplier/partner relationships rather than potential competitors, consistent with the mechanism operating through disruption of input-sourcing linkages. Appendix F.10 reports these results.

6.3.2 Economic Costs of Yardstick Competition on Firms

To gauge magnitudes, I translate the firm-level estimates into an illustrative cost in value added. In the baseline specification (Panel A of Table 7), a one standard deviation increase in yardstick competition lowers the revenue of industrial firms by 5.0% and their intermediate inputs by 5.9%. Because intermediate inputs make up about three quarters of revenue, their value added falls by about 2.4% (Appendix Table H.26). The secondary sector produces about 46% of city GDP over the period of the firm sample. If every firm in this sector shared this loss, GDP would fall by about 1.1%, or approximately $13 billion in 2000 USD (Appendix Table H.27). This is a cost to the secondary sector, not an overall GDP effect: it assumes that every firm in the sector shares the loss of the industrial firms in the data, and it leaves out the expansion of real estate that competition also induces. Appendix E details the calculation.

6.4 Persistent Consequences of Yardstick Competition

This section examines whether sustained yardstick competition yields persistent growth advantages or ultimately undermines development.

To examine long-term consequences, I estimate the following cross-sectional specification:

$$\Delta Y_{c,1996-2018} = \alpha_0 + \alpha_1 \overline{\text{Politician Age Differential}}_{c,1996-2018} + X_c' + \delta_p + \epsilon_c \tag{8}$$

where $\Delta Y_{c,1996-2018}$ represents the change in economic outcomes for city c between 1996 and 2018. The key variable, $\overline{\text{Politician Age Differential}}_{c,1996-2018}$, captures a city's cumulative exposure to competitive pressure over more than two decades by averaging the age differential between each party secretary and their provincial peers. The vector X_c' includes baseline city characteristics, and δ_p denotes province fixed effects. Standard errors are bootstrapped (500 times) at the province level.

Table 8 contrasts with the short-run relative-performance advantage: cities with higher

average age differentials, indicating lower competitive intensity and less protectionism, exhibit significantly better long-term outcomes. A one standard deviation increase in the average age differential is associated with 2.25% higher annual GDP growth, 2.14% higher lighting growth, 2.82% higher population growth, and 2.89% higher trade growth, with particularly strong effects on population and trade. These patterns suggest that cities whose party secretaries faced less intense competition, and consequently erected fewer trade barriers, achieved stronger market integration, which attracted labor and capital, fostered agglomeration economies, and enabled sustained development (Donaldson, 2018; Faber, 2014).

6.4.1 Reconciling Short-Term Gains with Long-Term Costs

Tables 2 and 8 reveal a fundamental tension between short-run relative GDP gains and weaker long-run development. The resolution lies in the distinction between effort and instruments. Competitive pressure induces greater effort, consistent with tournament theory (Lazear and Rosen, 1981; Li and Zhou, 2005), but the protectionist instruments deployed create persistent institutional distortions. Trade barriers become entrenched through vested interests and bureaucratic inertia (Acemoglu et al., 2001), and the accumulated stock of protectionist policies erected by successive administrations ultimately outweighs the transient benefits of enhanced effort. The multitask agency problem reinforces this mechanism (Holmstrom and Milgrom, 1991): competitive politicians rationally prioritize policies that boost short-term, measurable GDP even when market fragmentation undermines long-term growth. Song and Xiong (2026) derive the same short-termism from the governors' tournament; the evidence here shows that one of its instruments is the erection of internal trade barriers, whose costs fall on the integrated market rather than on the jurisdiction that erects them.

7 Conclusion

This paper establishes that internal non-tariff trade barriers are endogenous to the design of subnational political institutions. Tariffs cannot be levied within a country, but promotion competition raises the cost of trading across jurisdictions through the allocation of infrastructure, subsidies, and land. I demonstrate that yardstick competition – often praised for disciplining local politicians – can generate perverse incentives that systematically impede economic integration. When career advancement is tied to relative performance and economic activities generate spillovers to rivals, rational politicians have an incentive to fragment markets.

To test this argument, I exploit exogenous variation in yardstick competition intensity in China, proxied by relative age differentials among city-level party secretaries within provinces. I find that a one standard deviation increase in this age differential (indicating lower competition) is associated with a 2.6% increase in inter-regional trade volume.

My analysis uncovers specific policy levers used by politicians under competitive pressure: underinvestment in trade-enhancing infrastructure near jurisdictional borders, misallocation of industrial policy spending, biased land provision, and a shift in stated policy priorities away from internal trade. These distortions propagate through production networks, lowering input use, revenue, and productivity for firms inside and outside the jurisdiction. They are also not transitory; regions subject to more intense yardstick competition exhibit lower long-term economic development, evidenced by slower growth in GDP, nighttime lights, population, and trade.

The policy implications extend beyond China's context. Any system combining subnational autonomy with relative performance evaluation risks creating incentives for local protectionism. Because these barriers work through public spending, land, and infrastructure rather than tariffs, trade agreements and tariff policy cannot remove them.

This research demonstrates that the political architecture of decentralization, not just its degree, fundamentally shapes economic outcomes. As countries devolve authority to subnational governments, addressing these institutional design challenges becomes critical for realizing the benefits of both political decentralization and economic integration.

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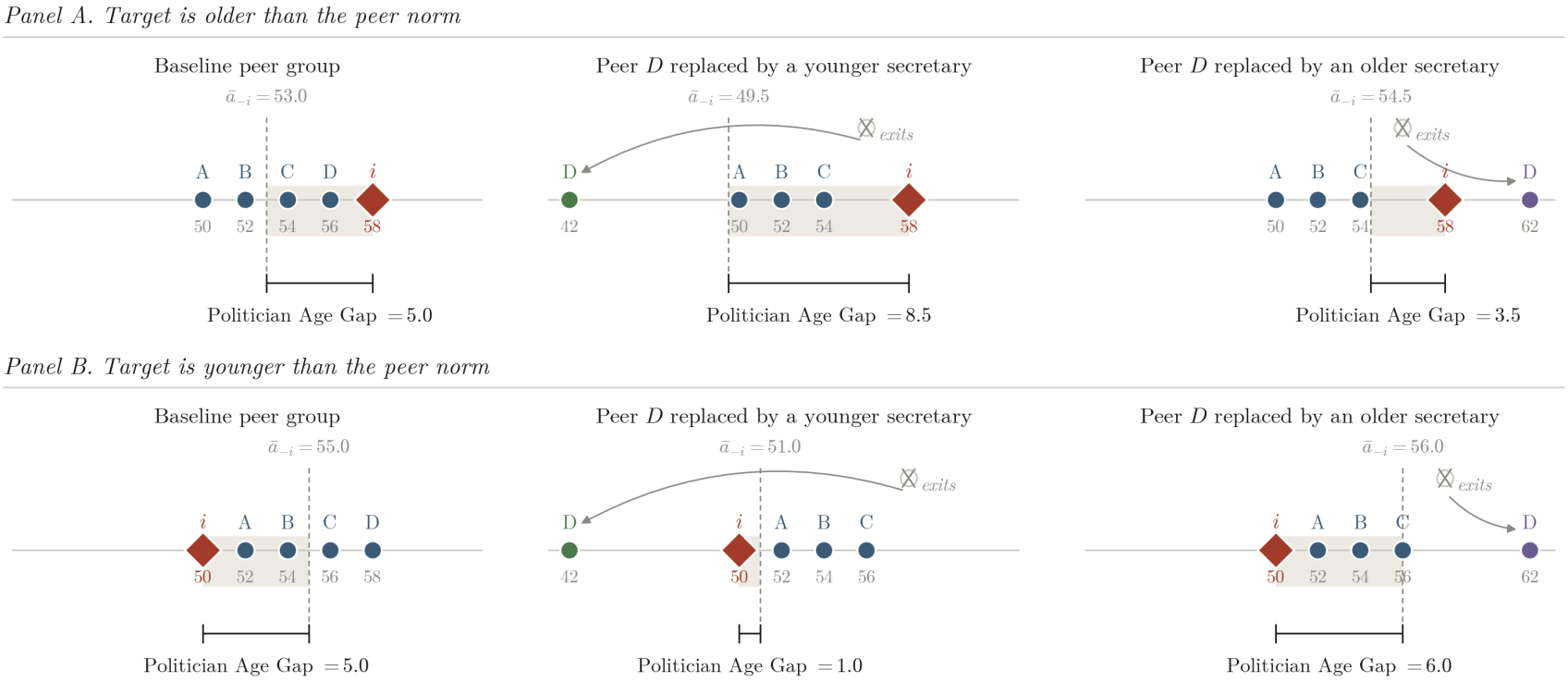
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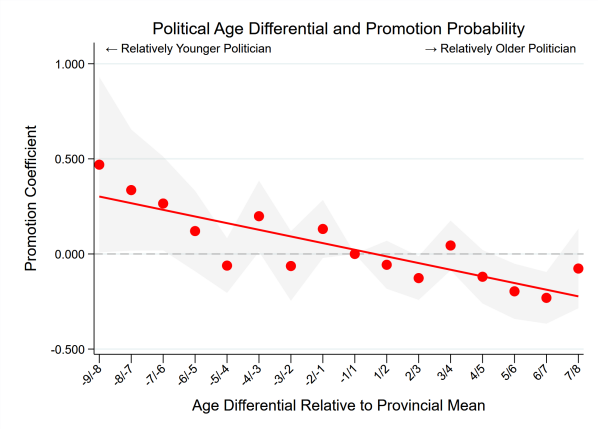
Figures

Figure 1: Construction of the Politician Age Differential Under Peer Turnover

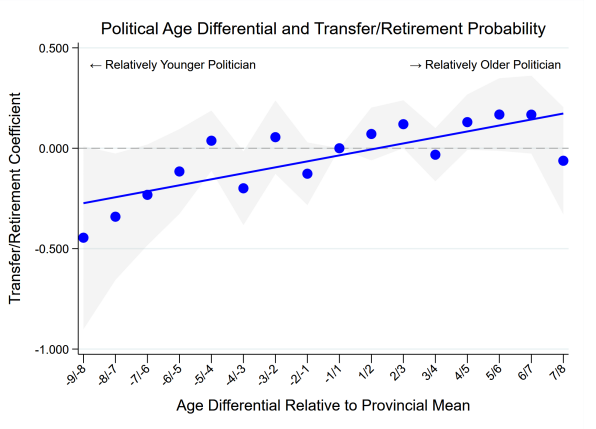


Notes: Politicians *A*–*D* denote the peers of secretary *i*, the other city party secretaries in the same province and year. The Politician Age Gap shown in each panel is the Politician Age Differential of Equation 1, $|a_i - \bar{a}_{-i}|$, the absolute value of the difference between *i*'s own start age and the mean start age of *i*'s peers. Panel A considers a target politician who is older than the peer norm at baseline; Panel B considers a target politician who is younger than the peer norm at baseline. Within each panel, the first column shows the baseline peer group, the second column shows the effect of replacing peer *D* with a younger secretary, and the third column shows the effect of replacing peer *D* with an older secretary. Reading down each column, the same replacement event moves the Politician Age Gap in opposite directions for the two targets, since each begins on a different side of the peer norm; this illustrates that the measure is symmetric in distance from the peer norm, not in the direction of peer turnover.

Figure 2: Signed Politician Age Differential and Career Prospects (Controlling Politician's Age)



(a) Promotion after the Tenure



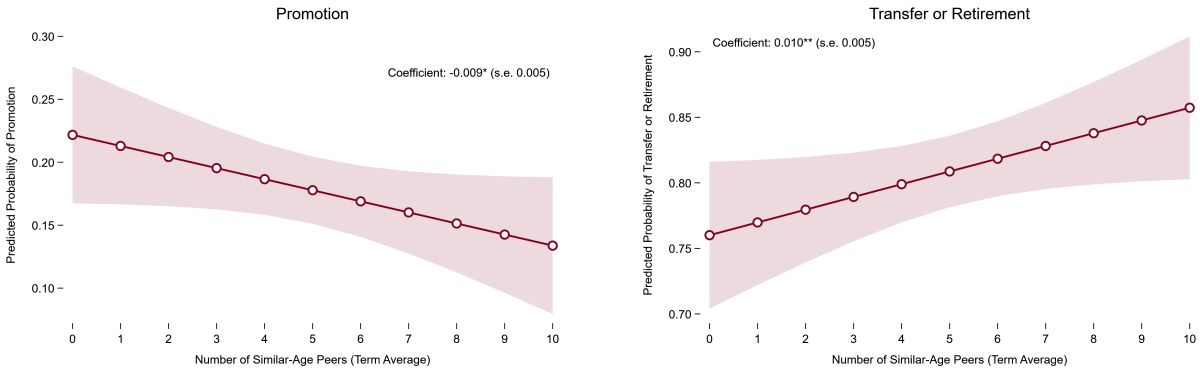
(b) Transfer/Retirement after the Tenure

Notes: This figure shows how the signed age differential affects career prospects. The signed age differential is calculated as a politician's age minus the mean age of other party secretaries in the same province during the term; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions instead use the absolute value of this gap, which captures competition intensity regardless of direction. The x-axis shows binned age differentials, with [-1,1] as the reference category. The analysis uses politician term-level data from 2000 to 2014. Controls include city characteristics, politician characteristics, politician's provincial connections, city fixed effects, and tenure start year fixed effects. Standard errors clustered at the city level. 95% confidence intervals shown.

Figure 3: Similar-Age Peer Density and Politician Career Outcomes

(a) Promotion

(b) Transfer or Retirement

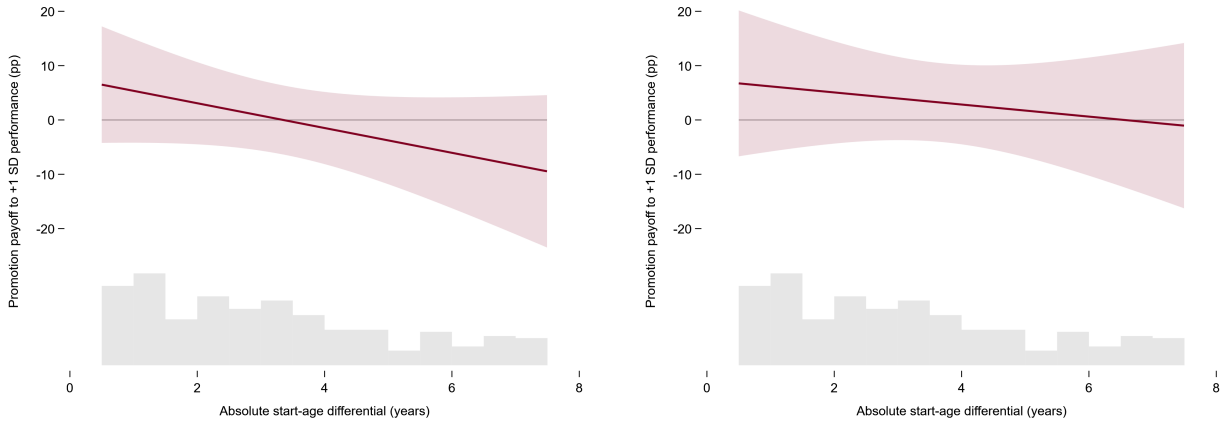


Notes: Each observation is a city party secretary's term (661 terms). The dependent variable is an indicator equal to one if the secretary was promoted at the end of the term in Panel (a), and an indicator equal to one if the secretary was transferred to a non-promotion position or retired in Panel (b). The horizontal axis is the number of similar-age peers, the term average of the number of other city party secretaries in the same province whose start age is within three years of the focal secretary's start age, shown over its 5th to 95th percentiles. Each regression controls for the secretary's start age and term averages of education, the city's nighttime light density and population, and colleague and hometown ties to the provincial party secretary and governor. Each panel plots the predicted probability with the other regressors at their sample means; the printed coefficient is the slope on the number of similar-age peers. Shaded areas are 95% confidence intervals based on standard errors clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4: Promotion Payoff to Growth Performance by Age Distance to Provincial Peers

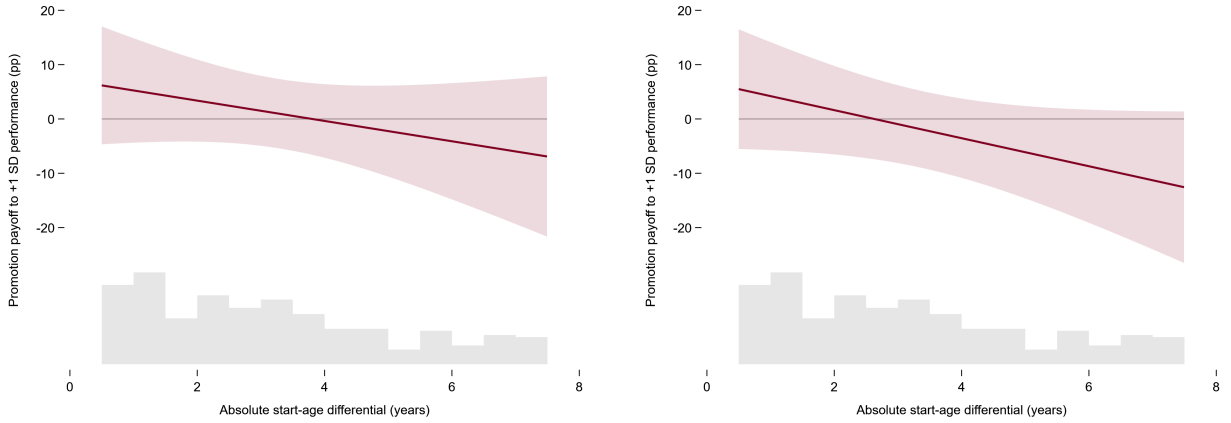
(a) Real GDP Growth Relative to Peer Mean

(c) Provincial Rank of Real GDP Growth



(b) Real GDP Growth Relative to Peer Median

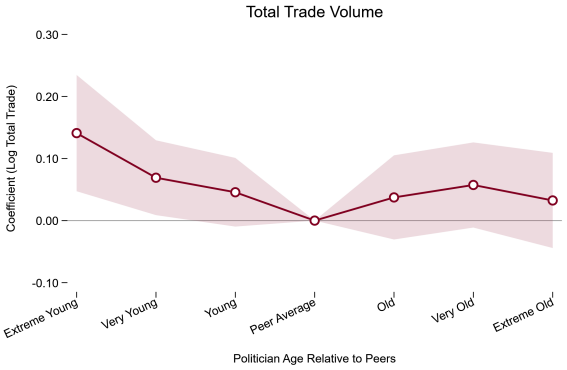
(d) Own Real GDP Growth



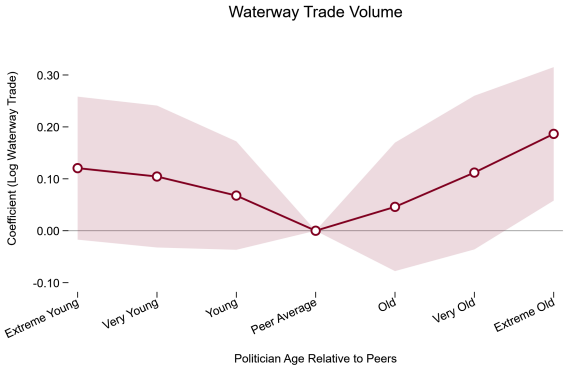
Notes: Each panel plots the estimated effect of a one standard deviation increase in growth performance on the probability of promotion at the end of a city party secretary's tenure (percentage points), as a function of the absolute start-age differential from provincial peers. Estimates come from a linear probability model with one observation per city-secretary tenure, regressing a promotion indicator on the performance measure, the absolute age differential, and their interaction, controlling for start age, education, log population, the number of provincial peers, and ties to the provincial party secretary and governor, with city and tenure-start-year fixed effects. Performance is the tenure average of annual reported real GDP growth minus the leave-one-out mean (panel a) or median (panel b) of other cities in the same province, the within-province percentile rank of growth (panel c), and own growth (panel d). Shaded areas are 95% confidence intervals based on standard errors clustered at the city level. Gray bars show the distribution of the age differential; the horizontal range covers its 5th to 95th percentiles.

Figure 5: The U-shaped Relationship between Politician Age Differential and Trade Volumes

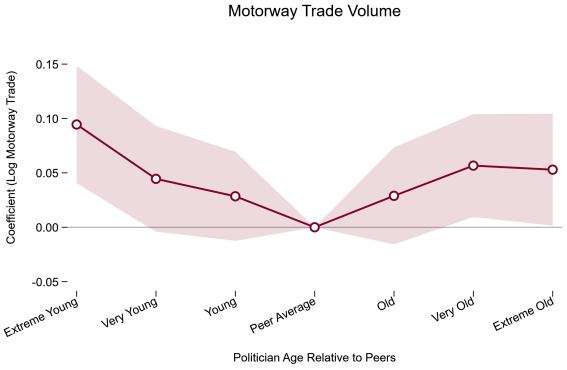
(a) Total Trade Volume



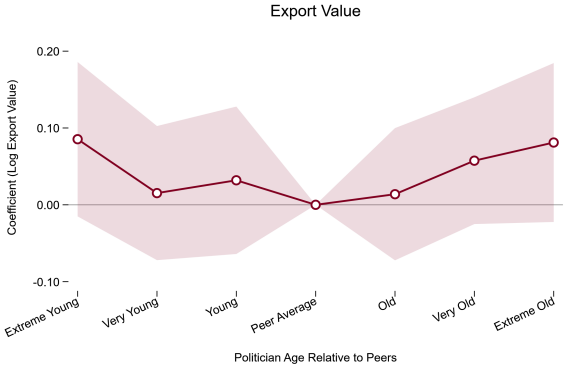
(c) Waterway Trade Volume



(b) Motorway Trade Volume



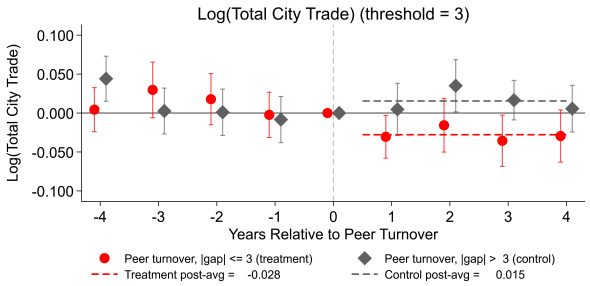
(d) Export Value (USD)



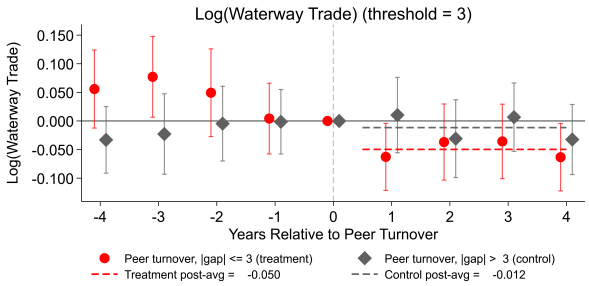
Notes: These figures show the estimated effects of Politician Age Differential on different trade volumes across different age categories. The analysis uses city-level panel data from 1996 to 2018. The x-axis shows seven categories of age differential relative to the provincial peer average: 'Extreme Young', 'Very Young', 'Young', 'Average' (reference category), 'Old', 'Very Old', and 'Extreme Old'. The y-axis shows the estimated coefficients from a regression of log trade volume on age differential category indicators. I control for city and politician characteristics, city-politician fixed effects, and year fixed effects. Standard errors are clustered at the city level. 95% confidence intervals are shown.

Figure 6: Event Study: Similar- versus Dissimilar-Age Peer Arrivals

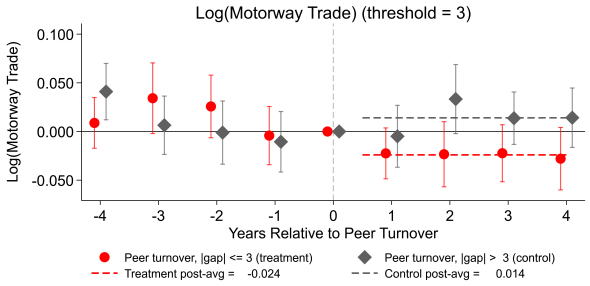
(a) Total Trade Volume



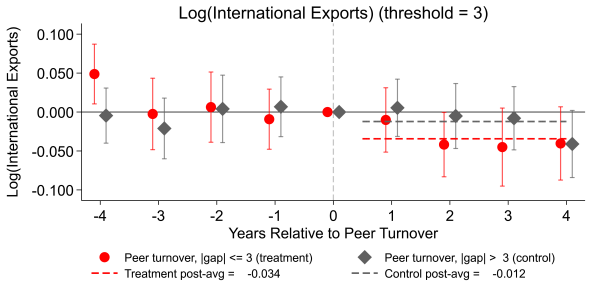
(c) Waterway Trade Volume



(b) Motorway Trade Volume



(d) Export Value



Notes: These figures present event-study estimates of the effect of a within-province peer secretary's arrival on city trade outcomes. The similar-age series (red circles) consists of arrivals for which the incoming peer secretary's age is within three years of the focal secretary's start age; the dissimilar-age series (gray diamonds) consists of arrivals outside that band. The x-axis shows years relative to the peer arrival, and the y-axis shows the estimated coefficients. The horizontal dashed lines report the average post-arrival coefficient for each group. All specifications include city-politician and year fixed effects and the baseline controls. Standard errors are clustered at the city level. Vertical bars show 90% confidence intervals.

Tables

Table 1: Effects of Politician Age Differential on City Trade

	City Party Secretary (Within-province Competition)				Placebo: Vice-Provincial City Secretary	Placebo: City Mayor
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Total Trade Volume						
Politician Age Differential	0.018***	0.017**	0.012***	0.012***	0.005	-0.000
	(0.007)	(0.007)	(0.004)	(0.004)	(0.007)	(0.006)
Observations	4,624	4,624	4,559	4,624	657	5,107
R-squared	0.656	0.657	0.927	0.871	0.971	0.937
Panel B: Motorway Trade Volume						
Politician Age Differential	0.023***	0.023***	0.015***	0.013***	0.007	0.004
	(0.006)	(0.006)	(0.004)	(0.003)	(0.007)	(0.004)
Observations	4,480	4,480	4,414	4,480	621	4,858
R-squared	0.801	0.801	0.941	0.927	0.958	0.954
Panel C: Waterway Trade Volume						
Politician Age Differential	0.015	0.016	0.030***	0.028***	0.000	-0.003
	(0.015)	(0.015)	(0.010)	(0.008)	(0.014)	(0.008)
Observations	2,530	2,530	2,461	2,530	328	2,708
R-squared	0.864	0.864	0.972	0.964	0.976	0.976
Panel D: Export Value (USD)						
Politician Age Differential	0.016*	0.014*	0.008	0.014**	0.006	-0.007
	(0.008)	(0.008)	(0.007)	(0.007)	(0.005)	(0.004)
Observations	3,404	3,404	3,384	3,404	500	3,754
R-squared	0.924	0.924	0.980	0.971	0.986	0.988
City Characteristics	–	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	–	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	–	–	–
Politician FE	–	–	Yes	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	–
Province-Year FE	–	–	Yes	–	–	Yes
City-Politician FE	–	–	–	Yes	Yes	Yes

Notes: This table examines how politician age differentials between party secretaries (within the same province) affect city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variable Politician Age Differential measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Columns (5) and (6) report placebo tests: column (5) restricts the sample to provincial capitals and vice-provincial cities, and column (6) analyzes the city mayor's age differential, neither of which features within-province competition of the same kind. Panel A shows effects on total trade volume, Panel B on motorway trade, Panel C on waterway trade, and Panel D on the value of international exports (USD). All trade volumes are logged and measured at 10,000 tons. Column (1) includes city and year fixed effects; column (2) adds city and politician characteristics; column (3) adds politician fixed effects and replaces year fixed effects with province-year fixed effects, which drops singleton province-years; columns (4) and (5) use city-politician and year fixed effects; column (6) uses city-politician and province-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Competition Mechanism: Effect of Politician Age Differential on Short-Term GDP Performance Among Competitors

	GDP Gap Relative to Provincial Average (100M RMB)		GDP as % Deviation From Provincial Average		Next-Year GDP (100M RMB)	
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	-6.953*		-0.270*		-6.696	
	(4.135)		(0.157)		(4.431)	
# Similar Age Peers		9.933**		0.275**		9.772**
		(4.470)		(0.120)		(4.917)
Observations	4,967	4,967	4,967	4,967	4,967	4,967
R-squared	0.972	0.972	0.989	0.989	0.980	0.980
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the relationship between within-province politician competition and short-term GDP performance relative to provincial peers. The analysis uses city-level panel data from 1996 to 2018. Politician Age Differential measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. # Similar Age Peers is the count of within-province peer secretaries whose age similar to the focal secretary. Columns (1) and (2) measure the GDP gap in hundreds of millions of RMB between a city and the provincial average. Columns (3) and (4) express this gap as a percentage deviation from the provincial mean. Columns (5) and (6) use next-year GDP (100M RMB) as the outcome. All specifications include city characteristics, politician characteristics, and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Spatial Heterogeneity: Trade Infrastructure Under-investment under Competition

VARIABLES	Log(Motorway Sum Length) (1)	Log(Motor Exit/ Entry Count) (2)	Log(Motorway Link Count) (3)	Log(Motorway Node Count) (4)	Log(Motorway Betweenness) (5)
Politician Age Differential	-0.063	-0.005	-0.008	-0.005	-0.045
	(0.055)	(0.006)	(0.008)	(0.006)	(0.031)
Politician Age Differential × Border Region	0.114**	0.013*	0.016*	0.013*	0.106***
	(0.057)	(0.007)	(0.008)	(0.007)	(0.039)
Observations	38,094	38,094	38,094	38,094	38,094
R-squared	0.925	0.949	0.947	0.949	0.924
County Characteristics	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes
County-Politician FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the differential effects of politician age differential on motorway infrastructure investment between border and interior counties through interaction terms. Border counties are defined as counties that share geographic boundaries with counties belonging to rival cities within the same province, where cross-jurisdictional spillovers from infrastructure investment are most salient. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. The interaction term 'Politician Age Differential × Border Region' captures the additional effect of yardstick competition in border counties of the politician's jurisdiction. Dependent variables include log-transformed measures of motorway sum length, motor exit/entry count, motorway link count, motorway node count, and motorway betweenness centrality. All specifications include county characteristics, politician characteristics, county-politician fixed effects and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of Politician Age Differential on Industrial Policy Spending: Tradable vs. Real Estate

	City-Sector Log(Mean Subsidies)					
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential × Tradable Sector	0.043*	0.076***	0.087***			
	(0.024)	(0.027)	(0.030)			
Politician Age Differential × Real Estate Sector				-0.066	-0.101*	-0.121**
				(0.057)	(0.058)	(0.056)
Observations	9,739	9,739	9,739	9,739	9,739	9,739
R-squared	0.644	0.707	0.743	0.644	0.707	0.743
City-Sector Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	Yes	Yes	–
Politician FE	–	Yes	Yes	–	Yes	Yes
Sector-Year FE	–	Yes	Yes	–	Yes	Yes
City-Year FE	–	–	Yes	–	–	Yes

Notes: This table examines how age differentials between party secretaries (within the same province) differently affect tradable and real estate sectors. The analysis uses city-sector panel data from 2005 to 2017. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. The dependent variable is the log of city-sector subsidies. 'Tradable' is a dummy variable for sectors producing tradable goods, while 'Real Estate' is a dummy for the real estate sector. Controls include city-sector fixed effects, city-sector characteristics, politician fixed effects, sector-year fixed effects, and city-year fixed effects. Standard errors (in parentheses) are clustered at the city-sector level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effects of Politician Age Differential on Land Provision and Value: Tradable vs. Real Estate

	Manufacturing/Tradable Land		Real Estate Land		Other Non-tradable Land (Commercial)	
VARIABLES	Log(Industrial Land Provision) (1)	Log(Total Value of Industrial Land Sold) (2)	Log(Real Estate Land Provision) (3)	Log(Total Value of Real Estate Land Sold) (4)	Log(Other Sector Land Provision) (5)	Log(Total Value of Other Sector Land Sold) (6)
Politician Age Differential	0.040*	0.065*	-0.045*	-0.105**	-0.046	-0.077*
	(0.022)	(0.039)	(0.026)	(0.052)	(0.028)	(0.047)
Observations	4,155	4,155	4,155	4,155	4,155	4,155
R-squared	0.955	0.956	0.896	0.904	0.864	0.881
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how age differentials between party secretaries affect land provision and usage across different categories. The analysis uses city-level panel data from 2000 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Columns (1)-(2) show effects on different aspects of industrial land allocation: total provision and total value of land sold. Columns (3)-(4) show effects on real estate land provision and value, while Columns (5)-(6) show effects on land provision of other non-tradable sectors. All dependent variables are in logarithmic form. All specifications include controls, city-politician fixed effects, and province-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effect of Politician Age Differential on Policy Attention: Government Report

	Trade Policy Mentions			Real Estate Policy Mentions		
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	0.335*	0.350**	0.302**	-0.180*	-0.196**	-0.181**
	(0.170)	(0.173)	(0.153)	(0.096)	(0.097)	(0.084)
Observations	2,679	2,636	2,574	2,679	2,636	2,574
R-squared	0.782	0.784	0.780	0.678	0.680	0.672
City Characteristics	–	Yes	Yes	–	Yes	Yes
Politician Characteristics	–	Yes	Yes	–	Yes	Yes
City FE	Yes	Yes	–	Yes	Yes	–
Politician FE	Yes	Yes	–	Yes	Yes	–
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	–	–	Yes	–	–	Yes

Notes: This table examines how the politician age differential affects the policy attention of city governments, measured by length-weighted policy mentions in city policy documents, where the frequency of topic-related mentions is weighted by the length of the corresponding policy text. Columns (1) to (3) report trade-related mentions, and Columns (4) to (6) report real estate mentions. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Columns (1) and (4) include city, politician, and year fixed effects; Columns (2) and (5) add city and politician characteristics; Columns (3) and (6) replace the city and politician fixed effects with city-politician fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Effect of Politician Age Differential on Firm Network and Firm Outcomes

	Firm$_i$ Located in City$_i$ Direct Effect on Focal Firms				Firm$_j$ Connected to Firm$_i$ in Business Network Indirect/Spillover Effect on Firms with a 95% Similarity Score (LLM)			
VARIABLES	Log(Intermediate Good Inputs) (1)	Log(Profits) (2)	Log(Revenue) (3)	Log(Revenue per Worker) (4)	Log(Intermediate Good Inputs) (5)	Log(Profits) (6)	Log(Revenue) (7)	Log(Revenue per Worker) (8)
Panel A: Baseline Specification								
Politician Age Differential in City$_i$	0.027***	0.030*	0.023***	0.023***	0.020**	0.057**	0.022**	0.026***
	(0.007)	(0.016)	(0.007)	(0.007)	(0.009)	(0.028)	(0.009)	(0.009)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.824	0.731	0.879	0.846	0.790	0.691	0.845	0.819
Panel B: With Firm Controls								
Politician Age Differential in City$_i$	0.028***	0.031*	0.024***	0.024***	0.026***	0.065**	0.029***	0.028***
	(0.008)	(0.016)	(0.007)	(0.007)	(0.009)	(0.028)	(0.010)	(0.009)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.844	0.744	0.902	0.850	0.818	0.708	0.878	0.825
Panel C: With Firm Controls, Sector-Year Fixed Effects, and City$_j$-Year Fixed Effects								
Politician Age Differential in City$_i$	0.023***	0.011	0.016***	0.016***	0.008***	0.015*	0.005**	0.008***
	(0.005)	(0.013)	(0.005)	(0.006)	(0.003)	(0.009)	(0.002)	(0.003)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.870	0.760	0.911	0.864	0.870	0.777	0.906	0.862
Firm Pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how age differentials between party secretaries affect firm-level outcomes and business network connections. The analysis uses firm-level panel data from 2000 to 2007, where Columns (1)-(4) show effects on the focal firms (Firm$_i$), and Columns (5)-(8) show effects on firms connected to the focal firms through business networks (Firm$_j$). The key independent variable 'Politician Age Differential in City$_i$' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Panel A shows baseline results, Panel B adds firm-level controls, and Panel C further includes sector-year and connected city-year fixed effects. All specifications include firm pair fixed effects, politician fixed effects, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Effects of Politician Age Differential on Long-Term Economic Outcomes (1996 to 2018)

VARIABLES	ΔLog(City GDP) (1)	ΔLog(Night-time Lighting) (2)	ΔLog(City Population) (3)	Log(Average Trade Volumes) (4)
Avg. Politician Age Differential$_{1996-2018}$	0.234***	0.223***	0.294**	0.301***
	(0.061)	(0.050)	(0.122)	(0.072)
Observations	153	288	139	283
R-squared	0.424	0.637	0.308	0.560
Province FE	Yes	Yes	Yes	Yes
Baseline Characteristics	Yes	Yes	Yes	Yes

Notes: This table examines the long-term impacts of politician age differentials on city development. The analysis uses cross-sectional city-level data comparing outcomes between 1996 and 2018, averaging the politician age differential over the full window. The key independent variable is the average absolute age differential between a city's party secretary and those in other cities within the same province over the entire period. The dependent variables are log-differences (ΔLog) between 1996 and 2018: city GDP (Column 1), nighttime lighting intensity (Column 2), city population (Column 3), and average trade volumes over the period (Column 4). All specifications include province fixed effects and baseline city controls. Standard errors are bootstrapped at the province level by 500 times. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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A Data Description

A.1 Provincial Bilateral Trade Flows

To validate the baseline city-level findings and examine the underlying mechanism – namely, how local politicians strategically influence inter-regional trade – I employ provincial bilateral trade flow data spanning 2000 to 2018. These data, obtained from the Railway Statistical Yearbook, capture trade volumes between province pairs transported via railway networks.

A.2 City-sector Subsidies and Firm-level Data

City-sector government subsidies data are drawn from the China Stock Market & Accounting Research (CSMAR) Database. Firm-level data, including employment, revenue, and intermediate inputs, come from the Annual Survey of Industrial Firms (ASIF). The ASIF provides comprehensive coverage of industrial enterprises exceeding a revenue threshold (typically 5 million RMB), accounting for over 95% of China's total industrial output.

A.3 Land Provision Data

Land provision data is obtained from China's Ministry of National Land and Resources website.[31] The dataset contains comprehensive land transaction information including city, purpose, area, and price. I track keywords to identify industrial/residential land provision from purpose descriptions. During 2000-2018, there are over 2.5 million land transactions, collapsed into city-year panel data.

A.4 Motorway Network Data

Motorway network data are sourced from the comprehensive GIS database of China's transport network compiled by Davis et al. (2025). This database provides granular, annual GIS records of national highways, motorways, and railways with detailed spatial coordinates and network characteristics. The original data are aggregated to the city level for analysis. Key variables include motorway length, network nodes, network links, and betweenness centrality metrics that capture each city's position within the broader transportation network.

[31] The data can be obtained from www.landchina.com

A.5 Firm Entry Data

This study utilizes comprehensive firm registration data from China's State Administration for Industry and Commerce (SAIC), including all registered companies. The dataset provides registered addresses, registration dates, operational status, and industry classifications. I developed a county-industry-year panel tracking new firm entries from 1996-2018, containing approximately 5.8 million observations.

B Business Network Construction

To analyze the propagation of politically induced trade barriers through inter-firm relationships, this study constructs a comprehensive business network using China's Annual Survey of Industrial Firms (ASIF) dataset covering 2000-2007. The methodology follows Breitung and Müller (2025) for network identification.

The network construction proceeds in three stages. First, firm-level data is extracted for enterprises meeting the revenue threshold (typically 5 million RMB), representing over 95% of China's industrial output. Key attributes include industry subcategory, primary business activities, geographical location, and financial metrics (output value, sales, exports, assets, liabilities, revenue, profit, and intermediate goods usage). Firms with missing values in critical fields are excluded.

Second, composite feature vectors are engineered for each firm by combining textual and quantitative characteristics. Textual descriptions are transformed into embeddings using the paraphrase-multilingual-MiniLM-L12-v2 SentenceTransformer model, while financial variables are standardized to zero mean and unit variance. These components are concatenated to form unified feature vectors.

Third, business connections are established using cosine similarity between feature vectors. Network linkages require: (1) cosine similarity ≥ 0.95 to capture only the strongest relationships, (2) firms located in the same province but different cities to identify inter-city connections relevant to internal trade barriers, and (3) unique pair recording to avoid redundancy. On the threshold choice: Breitung and Müller (2025) link firms whose similarity lies in the top 1 percent of all pairs to identify global firm networks; this study instead uses a fixed cosine similarity threshold of 0.95 to capture the closer business relationships that exist among firms within the same province, where geographic proximity makes moderately similar firms far more likely to be genuinely connected.

The matched firm pairs from 2000 are extended across the 2000-2007 period, incorporating time-varying financial data when both firms remain active and satisfy geographical constraints. All industry subcategories are grouped into 45 sectors using K-Means clus-

tering on text embeddings, with dummy variables indicating sector membership and same-sector linkages.

This methodology yields a dynamic panel dataset of inter-city firm relationships within provinces, providing the empirical foundation for analyzing how political competition and trade frictions propagate through interconnected firms across jurisdictions. Figure G.16 visualizes the intensity of connections among the top 15 sectors, while Figure G.15 displays network structures for selected provinces.

C Construction of Sectoral Connectedness (IO Table)

To capture structural interdependencies at the sectoral level, I construct a measure of connectedness using province-specific Input-Output (IO) tables. I specifically use the 1997 IO tables, which predate the main sample period, to mitigate potential endogeneity concerns arising from contemporaneous shocks that might jointly affect firm performance and sectoral linkages.

While the similarity-based measure identifies latent similarities in business scope, the IO approach explicitly maps vertical supply chain linkages between industries. Let Z_{sk}^{p} denote the intermediate input flows from upstream sector s to downstream sector k within province p, and let X_{k}^{p} represent the total inputs of sector k. The direct input coefficient (technical coefficient) a_{sk}^{p} is calculated as:

$$a_{sk}^{p} = \frac{Z_{sk}^{p}}{X_{k}^{p}}. \tag{9}$$

This coefficient represents the value of inputs from sector s required per unit of total inputs in sector k, thereby proxying the intensity of the direct supply chain linkage.

To distinguish significant economic linkages from negligible transactions, I define sectoral connectedness based on the distribution of these coefficients within the IO table. Let M^{p} denote the median value of all nonzero technical coefficients in province p. The binary connectedness indicator C_{sk}^{p} is defined as:

$$C_{sk}^{p} = \mathbf{1}\left(a_{sk}^{p} > M^{p}\right), \tag{10}$$

where $\mathbf{1}(\cdot)$ is the indicator function. Sectors s and k are thus considered *connected* only if their direct input-output dependency exceeds the median linkage intensity observed in the 1997 provincial economy. This threshold approach ensures that the measure captures salient structural relationships while filtering out minor or incidental cross-sector transactions.

D Firm Pair Classification Procedure: Potential Competitors/Suppliers

This appendix details the classification of high-similarity firm pairs into "potential competitors" and "potential suppliers/partners" for the heterogeneity analysis in Appendix F.10.

D.1 Conceptual Framework

Following Chen (1996), I identify competitors based on two dimensions: market commonality and resource similarity. Direct competitors are firms that operate in the same market segment and possess comparable strategic endowments. Bergen and Peteraf (2002) emphasize that true horizontal competitors should exhibit similarity in both product offerings and operational characteristics. In contrast, vertically related firms may have overlapping product descriptions but differ in their financial profiles due to different positions in the value chain (Fresard et al., 2020).

D.2 Classification Criteria

Among the firm pairs already linked by the similarity of their business profiles (Section 6.3.1), I classify pairs as **potential competitors** if they satisfy two additional conditions:

1. **Same main business**: The descriptions of the two firms' primary business activities are closely similar (text similarity of at least 0.85), indicating direct market overlap.

2. **Similar financial status**: Firms have comparable operating profit margins (within 5 percentage points) and asset turnover ratios (within 0.5) in the same year, indicating similar business models and scale of operations.

Firm pairs that do not meet both conditions are classified as **potential suppliers/partners**. This residual category captures vertical relationships where firms have related products but occupy different positions in the supply chain, typically reflected in different main activities, profitability, and asset utilization patterns.

D.3 Why One Similarity Metric Identifies Both Relationship Types

The similarity score and the classification criteria play distinct roles, and it is the combination, not the similarity score alone, that separates horizontal from vertical relationships. A high similarity of business profiles establishes only that two firms are *economically related*:

firms whose industry descriptions, business activities, and financial characteristics overlap are plausibly connected, but the nature of the connection is ambiguous, since both a rival producing the same good and a supplier producing its key intermediate will describe closely related products (Hoberg and Phillips, 2016; Fresard et al., 2020). The second-stage criteria then resolve this ambiguity with a separate rule applied to each pair every year: firms with closely similar primary business activities and comparable margins and asset turnover occupy the same market position and are classified as horizontal competitors, while related-product pairs with divergent activities or financial profiles are consistent with different positions along the value chain and are classified as vertical suppliers/partners. The similarity metric is therefore a *relatedness filter* applied to all pairs, and the competitor/supplier distinction rests entirely on the market-overlap and financial-profile criteria, so no single measure is asked to proxy for two different relationship types at once. The IO-table-based connectedness measure (Appendix C), which identifies vertical linkages from input-output coefficients without any text input, provides an independent check: results using this purely vertical measure closely track the supplier/partner classification (Table H.22).

E Details of the Value-Added Cost Calculation

This appendix details the value-added cost reported in Section 6.3.2 (Tables H.26 and H.27).

From revenue and input losses to value added. The starting point is the firm-level effect of competition in the baseline specification of Table 7 (Panel A). A one standard deviation decrease in Politician Age Differential lowers revenue by 5.0% and intermediate inputs by 5.9%.[32] Value added equals revenue minus intermediate inputs. If intermediate inputs make up a share s of revenue, the percentage change in value added is

$$g_{VA} = \frac{g_R - s\, g_M}{1 - s}, \tag{11}$$

where g_R and g_M are the percentage changes in revenue and intermediate inputs. Intermediate inputs make up about three quarters of revenue in the firm sample ($s \approx 0.75$), so value added falls by about 2.4% (Table H.26).

Reallocation. The market fragmentation underlying these input losses also reallocates production from efficient producers toward less efficient local producers, since firms producing intermediate inputs are, on average, more productive than the local non-tradable

[32]The estimated coefficients are 0.023 for revenue (Column (3)) and 0.027 for intermediate inputs (Column (1)), each multiplied by 2.192, the standard deviation of Politician Age Differential in the firm sample.

firms that benefit from protection.[33] The calculation below captures only the direct loss in value added and therefore understates the total cost.

Aggregation. The secondary sector produces 46% of city GDP on average over 2000–2007, the period of the firm sample. If every firm in this sector shared the value-added loss of the industrial firms in Table 7, GDP would fall by about 1.1% (Table H.27). This is a cost to the secondary sector rather than an economy-wide effect: it leaves out the expansion of real estate that competition also induces.

F Additional Results

F.1 Instrumental Variable Estimates

To address potential concerns about endogeneity and reverse causality between politician age differentials and regional trade outcomes, I employ instrumental variable estimation using the one-year lagged value of Politician Age Differential. While the variation in Politician Age Differential is plausibly exogenous to unobservable characteristics affecting regional trade volume, the IV approach provides additional robustness by exploiting predetermined variation in political competition.

I instrument Politician Age Differential using its one-year lagged value. This approach exploits the predetermined nature of lagged political competition while maintaining relevance to current competitive dynamics. The lagged instrument is valid under the assumption that past realizations of politician age differentials affect current economic outcomes only through their influence on current political competition, rather than through direct channels.

The relevance condition is satisfied because political competition exhibits persistence over time–age differentials that existed in the previous period continue to influence current competitive dynamics through ongoing career trajectories, established networks, and accumulated political capital. However, the one-year lag ensures that the instrument is predetermined with respect to current-period shocks that might simultaneously affect both political competition and economic outcomes.

The exclusion restriction requires that lagged politician age differentials do not directly affect current trade outcomes except through their impact on current political competition:

$$E[\epsilon_{i,c,t}|Z_{i,p,t-1}, X_{i,c,t}, \alpha_c, \gamma_i, \delta_t] = 0 \tag{12}$$

where $Z_{i,p,t-1}$ represents the one-year lagged Politician Age Differential, $\epsilon_{i,c,t}$ is the error

[33] The relationship between trade participation and productivity is well established (Melitz, 2003; Bøler et al., 2015), as is the efficiency cost of misallocation (Hsieh and Klenow, 2009).

term in the structural equation, $X_{i,c,t}$ denotes control variables, and α_c, γ_i, and δ_t represent city, politician, and year fixed effects, respectively.

Table H.12 presents the second-stage results from the instrumental variable estimation, which confirm the baseline findings of a positive relationship between Politician Age Differential and internal trade volumes. The IV estimates are positive and statistically significant for total trade volume (0.021, significant at the 5 percent level), motorway trade volume (0.022, significant at the 5 percent level), and waterway trade volume (0.040, significant at the 10 percent level). The coefficient for export value is close to zero and imprecisely estimated, so the IV estimates are not informative about exports.

The first-stage F-statistics range from 277 to 363 across specifications, far above conventional thresholds for weak instrument concerns, indicating strong relevance of the lagged instrument. Figure G.12 illustrates the first-stage relationship, showing a clear positive correlation between the one-year lagged Politician Age Differential and its current value.

These IV results provide further evidence that the positive relationship between politician age differentials and internal trade reflects a causal effect of political competition rather than spurious correlation, reinforcing confidence in the robustness of the main findings.

F.2 Alternative Measures of Yardstick Competition

To assess the robustness of the main findings, this section re-estimates the baseline specification using alternative measures of yardstick competition. While the primary hypothesis centers on age, competition could plausibly occur along other dimensions such as tenure, education, or experience. The results of these estimations are presented in Table H.28.

First, I examine alternative specifications of the age-based competition measure. Panel B restricts the sample to politicians aged 57 or younger, isolating the group with the strongest promotion incentives. The results are nearly identical to the baseline in Panel A, confirming that the findings are not driven by politicians near retirement who might have different motivations. Panel C replaces the start-of-term age with the politician's current age in constructing the differential. The estimates remain positive and statistically significant for motorway trade, waterway trade, and exports, though the coefficient for total trade volume is smaller and imprecisely estimated, consistent with the start-of-term age being the more relevant benchmark in the promotion tournament.

Next, I test whether competition based on characteristics other than age produces similar effects. Panels D, E, and F replace the age differential with differentials in tenure, education level, and total working years, respectively. Across all trade categories, these alternative measures of competition are consistently statistically insignificant, with the

sole exception of a marginally significant coefficient for waterway trade in Panel F. This suggests that the observed effect is specific to the institutional importance of **age** in the promotion tournament, rather than being a general proxy for differences in experience or human capital.

Finally, Panel G measures competition intensity by counting the number of peers within a close range (±3 years) of a politician's working experience. A higher count signifies a more crowded competitive field, so the expected sign is negative. Consistent with this, a greater number of similarly experienced peers is associated with significantly lower waterway trade volumes, while the remaining coefficients are small and insignificant. Taken together, these robustness checks reinforce the conclusion that yardstick competition in this context is uniquely and robustly tied to politicians' relative age.

F.3 Bilateral Trade Flows and Provincial Yardstick Competition

To validate that the yardstick competition mechanism operates beyond the city level and through bilateral trade relationships, I examine inter-provincial trade flows using province-pair panel data. This analysis addresses two objectives: first, demonstrating that the competition effects persist when using bilateral trade flows rather than aggregate city volumes; second, showing that the mechanism operates across different levels of government where yardstick competition exists.

A note on the choice of transport mode is warranted. In the city-level analysis, road and waterway freight are the main outcomes because city-level officials' jurisdiction is concentrated on local road and waterway infrastructure. Railway is governed at higher administrative levels and primarily handles long-distance inter-provincial routes; at the city level it shows the same pattern as road trade, with a smaller and less precisely estimated effect (Appendix Table H.9). At the provincial level, railway freight becomes the appropriate outcome variable precisely because it captures inter-provincial trade and falls within the sphere of influence of provincial leaders. Provincial leaders possess greater authority over railway logistics within their jurisdictions—including freight scheduling, terminal access, and coordination with national rail networks. Thus, the shift from road/waterway freight at the city level to railway freight at the provincial level reflects the change in spatial scale: the politically salient transport mode depends on which infrastructure falls within the relevant leader's sphere of influence.

At the provincial level, party secretaries compete nationally for central government positions, creating a different but analogous tournament structure to the within-province competition among city leaders. I construct the provincial-level age differential by comparing each provincial party secretary's age to the average age of all other provincial

party secretaries nationally. This measure captures the intensity of yardstick competition among provincial leaders, where those closer to the national average age face more intense competition for limited central government positions.

The empirical specification examines bilateral trade flows between province pairs:

$$\text{Log(Bilateral Trade)}_{ij,t} = \alpha_0 + \alpha_1 \text{Age Diff}_{i,t} + \alpha_2 \text{Age Diff}_{j,t} + \delta_{ij} + \tau_t + \gamma_{i,pol} + \gamma_{j,pol} + \epsilon_{ij,t} \tag{13}$$

where i and j index origin and destination provinces, δ_{ij} represents province-pair fixed effects controlling for time-invariant bilateral trade relationships, τ_t denotes year fixed effects, and $\gamma_{i,pol}$, $\gamma_{j,pol}$ are province-politician fixed effects capturing individual leader characteristics.

Table H.17 presents the results using bilateral provincial trade data spanning approximately 900 province-pairs per year. The findings demonstrate that yardstick competition affects trade flows at the provincial level, with origin province competition showing stronger effects than destination competition. A larger age differential in the origin province (indicating weaker competitive pressure) is associated with significantly higher bilateral trade volumes, with coefficients of 0.021-0.022 that are statistically significant at the 5% level.

The asymmetric effects between origin and destination provinces align with the mechanism. Politicians have greater control over policies affecting outbound trade flows from their jurisdiction—through industrial policies, regulatory approaches, and inter-provincial coordination—than over inbound flows, which depend primarily on destination province policies.

Notably, the destination province age differential shows weaker and statistically insignificant effects (0.011-0.012), consistent with the notion that trade-restricting policies primarily operate through origin-side mechanisms. When examining only origin effects (Column 3) or destination effects (Column 4) separately, the origin competition measure remains robust while destination effects remain insignificant.

These results provide two important validations of the main findings. First, they confirm that the competition mechanism operates through bilateral trade relationships rather than being an artifact of aggregate city-level trade measurement. Second, they demonstrate that yardstick competition effects extend beyond the specific institutional structure of city-provincial promotion tournaments to operate across different levels of government where similar competitive dynamics exist.

The consistency between city-level and provincial-level findings strengthens the interpretation that career-driven yardstick competition systematically influences politicians' willingness to implement policies that facilitate or restrict inter-jurisdictional economic integration, regardless of the specific administrative level at which this competition occurs.

F.4 Controlling for Past Trade Volume and Economic Growth

To ensure the main findings are not driven by pre-existing economic trends that might correlate with both political competition and current trade volumes, I conduct a robustness check by controlling for lagged economic indicators. The main specification is re-estimated, including lagged values of nighttime lighting (as a proxy for historical economic conditions) and the specific trade volume being analyzed. The specifications in Table H.14 use both a one-year lag and a three-year average for these controls, while maintaining the full set of fixed effects.

The results confirm the robustness of the primary finding. The coefficient for the Politician Age Differential remains positive and statistically significant across all three freight trade modes in both the lagged and three-year average specifications, ranging from 0.008 for total trade with lagged controls to 0.024 for waterway trade with three-year average controls. For exports, the coefficient is positive and significant with lagged controls (Column 7) and remains positive, though imprecisely estimated, with the three-year average controls in the smaller sample of Column (8). As expected, the lagged trade volumes themselves are strong predictors of current trade, indicating a high degree of persistence. The stability of the main coefficient after including these dynamic controls provides further confidence that the estimates are not biased by unobserved, slowly evolving city-specific trends.

F.5 Heterogeneous Effects by Initial Economic Conditions

A key question in political economy is how a region's initial economic conditions shape the impact of distortions caused by competition. The mechanism suggests that politicians' incentives to manipulate trade under yardstick competition may vary with local market integration and industrial development. To test this implication, the baseline model is extended by interacting the Politician Age Differential with six measures of city characteristics: an indicator for the eastern region (more developed region), the initial trade share, firm tax revenue, the manufacturing value-added share, industrial employment, and the number of industrial firms. Table H.29 reports the results. The estimated coefficient on the Politician Age Differential remains positive in every specification and is statistically significant at conventional levels, ranging from 0.016 to 0.056. Thus, the baseline relationship between a larger age differential (weaker political competition) and trade is robust across these alternative dimensions of initial economic conditions.

The interaction coefficients are negative in all six specifications, indicating that the positive association between the Politician Age Differential and trade weakens in more developed or economically integrated cities. The interaction with the eastern-region

indicator is −0.012 (Column (1)). Given the main coefficient of 0.016, the implied effect for eastern cities is approximately 0.004, roughly a quarter of the effect for inland cities, so the distortion associated with political competition is largely, though not entirely, eliminated in the developed coastal region. The interaction with the initial trade share is −0.048, against a main coefficient of 0.033 (Column (2)). Taking the point estimates at face value, the effect declines with initial trade exposure and reaches zero at an initial trade share of approximately 0.69 (0.033/0.048); in cities above this threshold, political competition no longer distorts trade. The interaction with firm tax revenue is negative but small and imprecisely estimated (−0.003, Column (3)), offering little indication of heterogeneity along this dimension. The interactions with the manufacturing value-added share (−0.001), industrial employment (−0.007), and the number of industrial firms (−0.006) are likewise negative (Columns (4)–(6)). The corresponding turning points are a manufacturing value-added share of approximately 43 percent (0.043/0.001), manufacturing employment of approximately 4.9 (0.034/0.007), and approximately 9.3 for the number of industrial firms (0.056/0.006), in the units reported in Table H.29. Beyond these levels of market integration and industrial development, the estimates imply that the politically induced trade distortion disappears entirely. Taken together, these patterns indicate that political competition suppresses trade primarily in less developed and less industrialized cities, and that this distortion attenuates, and ultimately vanishes, once market integration and industrial depth are sufficiently advanced.

From an institutional standpoint, these findings suggest that politicians' strategic responses are conditioned by their local economic environment. In eastern, trade-oriented, and industrially developed cities, firms are more deeply embedded in interregional markets and production networks. Policies that disrupt trade can therefore impose larger costs by damaging supply chains, reducing firm performance, and weakening the absolute economic outcomes used to evaluate local officials. These economic constraints counteract the incentives generated by relative political evaluation. Consequently, although weaker political competition remains positively associated with trade across the specifications, the relationship is substantially attenuated in cities with stronger initial trade linkages and deeper industrial structures.

F.6 Pre-trend in Trade Volume

To further support the causal interpretation of the relationship between Politician Age Differential and trade volumes, I conduct a placebo test to examine the possibility of pre-existing trends. Specifically, I test whether future levels of political competition (measured by the average age differential in a later period) are correlated with past changes in city

trade volumes. If cities that were set to experience lower or higher competition intensity in the future already exhibited different trade growth trajectories before those competition levels were realized, it would cast doubt on the identification strategy, suggesting potential reverse causality or omitted variables driving both competition and trade patterns.

For this test, I regress the change in city trade volumes (total, motorway, waterway, and export) between 2000 and 2005 on the average Politician Age Differential observed during the subsequent 2006 to 2010 period. The regression includes province fixed effects and baseline city controls from 2000.

The results are presented in Figure G.17. I find no statistically significant relationship between the average Politician Age Differential during 2006-2010 and the change in trade volumes during the earlier 2000-2005 period. The estimated coefficients are close to zero and statistically insignificant across all trade categories examined.

This lack of correlation between future competition intensity and past trade dynamics suggests that the main findings are unlikely to be driven by pre-existing trends. It provides additional evidence supporting the interpretation that changes in yardstick political competition, proxied by the Politician Age Differential, have a causal impact on inter-regional trade volumes, rather than merely reflecting pre-determined city-specific trade trajectories.

F.7 Politician Age Differential and Tenure Years

One alternative explanation for the effect of the Politician Age Differential on trade volumes is that it merely captures a politician's time in office or accumulated experience rather than the intensity of yardstick competition. Politicians with longer tenures might mechanically exhibit particular age differentials while independently influencing trade. Two pieces of descriptive evidence and one formal test speak against this interpretation.

Figure G.18 documents how the differential behaves over a secretary's term. The measure exhibits no mechanical relationship with tenure: its average level remains flat at roughly three years of age distance from provincial peers at every tenure year, and the average *signed* year-over-year change, reported below the line, never exceeds two percent of the level. At the same time, the measure is far from static. The average *absolute* year-over-year change is 0.53 to 0.58 years, roughly one-fifth of the level, in every year of a term. Because a secretary's own start-of-term age is fixed within a term, all of this movement originates in the turnover of peer secretaries in the same province, which is precisely the variation that the city-politician fixed-effects design exploits. The measure thus generates ample identifying variation within terms, none of which tracks the focal politician's time in office.

Figure G.19 plots the same two objects by calendar year and delivers two further

messages. First, the identifying variation is present throughout the sample period: the average absolute within-term change is sizable in every year from 1996 to 2018, and its peaks follow the political appointment calendar. The reshuffle waves after the Party Congresses (most visibly 1998, 2007, and 2012 to 2013) rearrange provincial age structures and generate bursts of variation in the measure. The estimates therefore do not rely on any particular episode or era. Second, the average level of the differential declines steadily, from about 3.8 years in the late 1990s to about 2.1 years by 2018: city secretaries have become increasingly similar in age to their provincial peers, consistent with the progressive standardization of age-based appointment norms documented in Section 2. Through the lens of the tournament framework, this age compression implies that the average intensity of yardstick competition has been *rising* over the sample period. Because this aggregate trend is common to all cities, it is absorbed by the year fixed effects and does not contribute to the within-city estimates. It does, however, underscore that the tournament pressure this paper studies is a growing, not a fading, feature of the system.

To test formally whether the estimated effect reflects tenure or experience rather than competition, I decompose the age differential by the *tenure proximity* of provincial peers. For each city-year, peer secretaries are split into three disjoint bins according to the absolute gap between the focal and peer secretary's tenure year: near-tenure peers (a gap of at most two years), mid-tenure peers (a gap of three to five years), and far-tenure peers (a gap of six or more years). The age differential computed within each bin then enters the baseline specification jointly. If the baseline coefficient merely proxied for the focal politician's tenure duration, accumulated experience, or incumbency status, there would be no reason for the effect to depend on *which* peers the differential is measured against. The results are presented in Table H.30.

The effect is concentrated among near-tenure peers: the age differential computed over peers within a two-year tenure gap is positive and significant for total trade (0.008), motorway trade (0.009), and waterway trade (0.030), whereas the differentials computed over mid- and far-tenure peers are small and generally insignificant. This pattern is exactly what the tournament interpretation predicts, since secretaries compete most intensely with peers at a comparable stage of their terms, who will be evaluated by the provincial leadership over the same horizon. It is difficult to reconcile with tenure- or experience-based alternatives, which imply no differential sensitivity to peers' tenure proximity. Together with the descriptive evidence in Figures G.18 and G.19, these results indicate that the yardstick-competition effect is distinct from the influence of tenure duration, accumulated experience, or incumbency status.

F.8 Heterogeneity by Motorway Administrative Class: Evidence from the National Expressway Network

This section decomposes the border-county result in Table 3 by motorway class: national expressways (G-prefix), provincial expressways (S-prefix), and urban motorways. National expressways stand out on the two dimensions that matter for the incumbent's calculus. They are the largest projects a locality hosts, absorbing the most land, financing, and administrative effort per segment, and they form the trunk network carrying long-distance inter-city trade, so their benefits leak most heavily to rival jurisdictions. Provincial segments serve within-region connectivity, and urban motorways yield almost entirely local returns. The spillover mechanism therefore predicts underinvestment should concentrate in the national network.

Although national expressways are centrally planned, route designation and implementation are distinct stages. The center fixes the corridor map, but local governments conduct land expropriation, arrange matching funds, issue permits, and control construction pacing; the larger the segment, the more such margins for delay. Central designation thus determines whether a link exists, while local implementation determines when it is completed. This division also aids identification: corridor placement is plausibly exogenous to local conditions, so any effect of competition on G-class capacity reflects protectionist foot-dragging on centrally designated projects.

Table H.18 reports the border-county interaction by different class. The effect is concentrated in national motorways (column 2), with no significant response for provincial or urban motorways (columns 1 and 3). Underinvestment thus concentrates in the largest, most spillover-intensive projects, operating through the local implementation of central plans.

F.9 New Firm Entry: Tradable vs Real Estate

Building upon established evidence that political incentives shape policy outcomes under yardstick competition, I further examine how such competition influences local economic structure through its effects on new firm entry.[34] I hypothesize that yardstick competition generates heterogeneous effects on firm creation contingent upon sectoral characteristics, particularly tradability and potential for generating inter-jurisdictional spillovers that may affect competing politicians' performance metrics.

Using the SAIC firm registration data previously described, this dataset covers the

[34]While firm entry represents a critical driver of economic dynamism, politicians can strategically manipulate entry patterns to enhance their relative performance within promotion tournaments, effectively constituting a form of industrial policy.

universe of all firms that were ever registered in China. The data contains detailed information on firms' registered address, date of registration, current state of existence, and industry (3-digit). I thus construct a county-industry-year panel data, measuring the number of firm entries. Overall, I have obtained about 5.8 million county-industry-year observations from the data period 1996 to 2018. I estimate the following equation:

$$Y_{s,c,t} = \alpha_0 + \alpha_1 \text{Politician Age Differential}_{c,t} \times \text{Sector}_s + \delta_{s,c} + \sigma_i + \gamma_{c,t} + \epsilon_{s,c,t} \quad (14)$$

where $Y_{s,c,t}$ denotes the logarithm of the number of firm registrations plus one in 3-digit sector s, county c, and year t. Politician Age Differential$_{c,t} \times$ Sector$_s$ denotes the interaction terms between Politician Age Differential and tradable/real estate sector indicators. The specification includes county-sector fixed effects ($\delta_{s,c}$), politician fixed effects (σ_i), and county-year fixed effects ($\gamma_{c,t}$). Note that the main effect of Politician Age Differential is absorbed by the county-year fixed effects, while the main effect of Sector is absorbed by the county-sector fixed effects. Standard errors are clustered at the city level.

Table H.20 reports the results. I find that weaker yardstick competition increases the number of new firm entries within tradable sectors. Specifically, the interaction term between Politician Age Differential and an indicator for tradable sectors is positive and significant across all specifications in Columns (1)-(3). A one standard deviation increase in Politician Age Differential would lead to a 1.7 percent increase in tradable firm entries.[35] This finding is congruent with the study's central argument that diminished competitive pressures incentivize politicians to adopt policies more conducive to activities with potential inter-jurisdictional spillovers, such as fostering tradable industries through favorable subsidy allocations or infrastructure investments, thereby creating a more attractive environment for entrepreneurial endeavors in these domains.

In contrast, the analysis reveals an inverse relationship concerning the real estate sector. The interaction term between Politician Age Differential and real estate sectors yields a negative and statistically significant coefficient across specifications in Columns (4)-(6). The results suggest that a one standard deviation decrease in Politician Age Differential (stronger yardstick competition) would lead to a 2.1 percent increase in real estate firm entries.[36] This observation aligns with the mechanism in which politicians facing intense rivalry strategically reallocate resources towards non-tradable sectors that offer primarily localized benefits, such as real estate, which generate economic outcomes confined to their own jurisdiction without benefiting competing politicians.

[35] Using the coefficient 0.0077 from the most complete specification in Column(3) and assuming a standard deviation of 2.2 for Politician Age Differential: 0.0077 $\times$ 2.2 = 0.017 or 1.7 percent.

[36] Using the coefficient -0.0094 from Column(6): 0.0094 $\times$ 2.2 = 0.021 or 2.1 percent.

These patterns of firm entry, conditional on the interplay between yardstick competition intensity and sectoral characteristics, reveal a further channel through which the career concerns of local politicians can shape the contours of local industrial structure. An environment characterized by attenuated political competition appears to foster greater dynamism in tradable sectors, whereas heightened competition may, ceteris paribus, make entry into non-tradable sectors like real estate relatively more prevalent as politicians prioritize geographically circumscribed economic outcomes.

F.10 Firm-Level Heterogeneity: Supplier/Competitor Relationships and Supply-Chain Position

This subsection extends the firm-level network evidence in Section 6.3.1 along two additional dimensions: the type of relationship connecting firm pairs, and each firm's position in the supply chain.

F.10.1 Disruption of Supply Chain: Competitors versus Suppliers/Partners

The preceding analysis established that the intensity of yardstick political competition generates spillover effects impacting firms connected through business networks. This section examines these network effects more deeply by distinguishing between different types of inter-firm relationships. Specifically, I investigate whether the impact of competition in city c_i on a connected firm j varies depending on whether firm j is primarily a potential competitor or a potential supplier/partner to firms in city c_i. The underlying hypothesis is that the mechanisms of local protectionism induced by intense competition might disproportionately affect vertical supply chain linkages compared to horizontal competitive relationships.

To explore this heterogeneity, I categorize the firm pairs identified through network analysis (as described in Section 6.3.1) into two groups: potential competitors (characterized by high similarity in products and business activities) and potential suppliers/partners (representing likely vertical or collaborative links).[37] I then re-estimate the core specification from Equation 7 separately for each group, examining the effect of the Politician Age Differential in city c_i on the performance outcomes of firm j. The results are presented in Table H.31.

[37]For potential competitors, if firms have very similar products and business activities and comparable firm scale (based on firm assets, revenue, profits and other financial status) but are located in different cities, they are classified as potential competitors operating in the same market segment. I use LLM to match their business activities and product information, which helps identify firms with highly comparable industry classifications, product offerings, and target customer bases, making them direct rivals in their respective local markets. Detailed classification can be found in Appendix D.

Panel A of Table H.31 displays the estimation results for firm pairs classified as potential suppliers or partners. In this group, a larger Politician Age Differential in city c_i (indicating weaker yardstick competition) is associated with statistically significant improvements in the performance of the connected firm j. Specifically, the coefficients are positive and significant for intermediate good inputs, revenue, and revenue per worker. The coefficient for Log(Profits) is positive but does not reach statistical significance. These findings imply that when yardstick competition is less intense in city c_i, its potential suppliers or partners in city c_j experience tangible benefits, notably in terms of access to intermediate inputs and overall revenue generation.

Conversely, Panel B presents the results for firm pairs identified as potential competitors. The coefficients associated with the Politician Age Differential are consistently small and statistically insignificant across all four outcome variables: Log(Intermediate Good Inputs), Log(Profits), Log(Revenue), and Log(Revenue per Worker). This lack of significance suggests that the intensity of political competition faced by the politician in city c_i does not translate into discernible positive or negative performance impacts for its potential competitor firms located in city c_j, at least through the network channels captured here.

Comparing the results across Panel A and Panel B reveals a clear distinction. The economic consequences of yardstick political competition, as transmitted through inter-city firm networks, appear to primarily manifest in relationships involving suppliers and partners, rather than direct competitors. Intense competition (smaller age differential) seems to impede economic activity within these vertical or collaborative chains more substantially than it affects parallel competitors. This suggests that the strategic trade barriers or reduced inter-regional cooperation resulting from political pressures may be particularly detrimental to the functioning of supply networks and partnerships connecting different jurisdictions. This finding refines our understanding of the market fragmentation mechanism, indicating that politically induced frictions disproportionately disrupt established inter-firm dependencies along the value chain.

F.10.2 Heterogeneity by Firm Supply-Chain Position: Upstream vs. Downstream

This subsection examines how yardstick competition affects firm performance differentially across the value chain. I estimate the direct effects specification (as described in Section 6.3.1) at the firm level, focusing on firm i's own performance across four key metrics: intermediate inputs, revenue, revenue per worker, and profit (all in logarithmic form). A positive coefficient indicates that weaker competition (larger age differential) correlates with improved firm outcomes.

I group firms by their position in the supply chain, using the industry subcategory recorded for each firm. Natural Resources, which consists almost entirely of coal mining

and quarrying, sits at the top of the supply chain: these firms extract local deposits and buy comparatively few intermediate inputs. The remaining groups are downstream of their suppliers and transform purchased intermediate inputs into goods. Processing covers firms that process purchased materials, mainly textile dyeing and finishing, grain, feed, and meat processing, and metal casting, forging, and rolling; Textiles covers spinning, weaving, and garments; Products covers finished goods such as knitwear and leather, paper, plastic, ceramic, and metal products; and Manufacturing is the broadest group, spanning building materials, garments, food and beverages, chemicals, machinery, and vehicles. Figure G.20 reports the estimated coefficients for each group, together with the Full Sample and the Tradable sector (all groups except Natural Resources), across the four outcomes (Panels a–d).

In the Full Sample and the Tradable sector, weaker yardstick competition is associated with statistically significant increases in intermediate inputs, revenue, and revenue per worker; the profit estimates are positive but imprecise (Panel b).

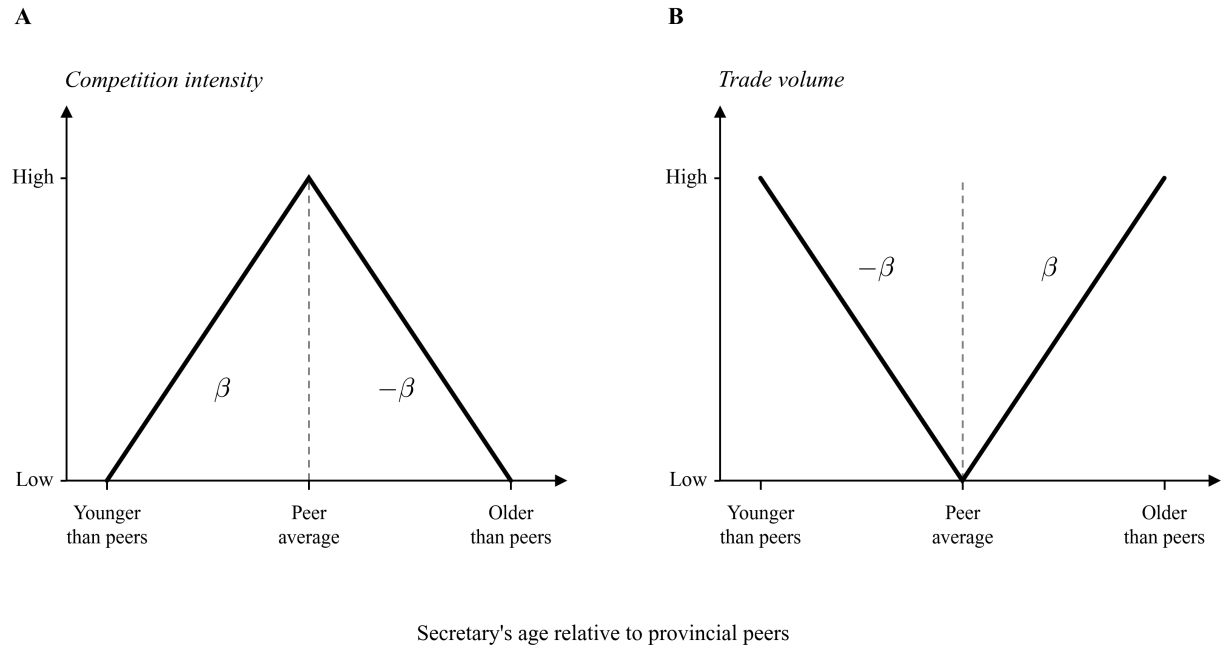
The heterogeneity follows supply-chain position. Natural Resource firms, the upstream group, show no statistically significant response on any outcome, and their point estimates for revenue and revenue per worker are the smallest in the figure. The four downstream groups respond. Each shows a positive and statistically significant revenue gain when competition weakens (Panel c), and intermediate inputs rise in every downstream group (Panel a), significantly so except for Products, whose confidence interval just includes zero. The largest input response is among Processing firms, whose production consists of processing materials bought from suppliers. Revenue per worker rises significantly for Textiles and Manufacturing (Panel d), and the profit estimates are positive but imprecise in every group (Panel b).

This pattern matches the mechanism in Section 6.3.1. Downstream firms depend on intermediate inputs, many of them sourced from suppliers in other cities. Political frictions between rival cities raise delivered input costs and make supply less reliable, so firms that rely on purchased inputs cut input use and output when competition intensifies and expand both when it weakens. Upstream resource extraction draws on local deposits and few purchased inputs, so it has little exposure to these frictions and its response is muted.

The results in Figure G.20 therefore locate the firm-level costs of yardstick competition among downstream firms that rely on intermediate inputs, while upstream resource firms remain largely insulated. Together with the supplier/partner evidence above, they indicate that politically induced frictions operate mainly through the sourcing of intermediate inputs across jurisdictions.

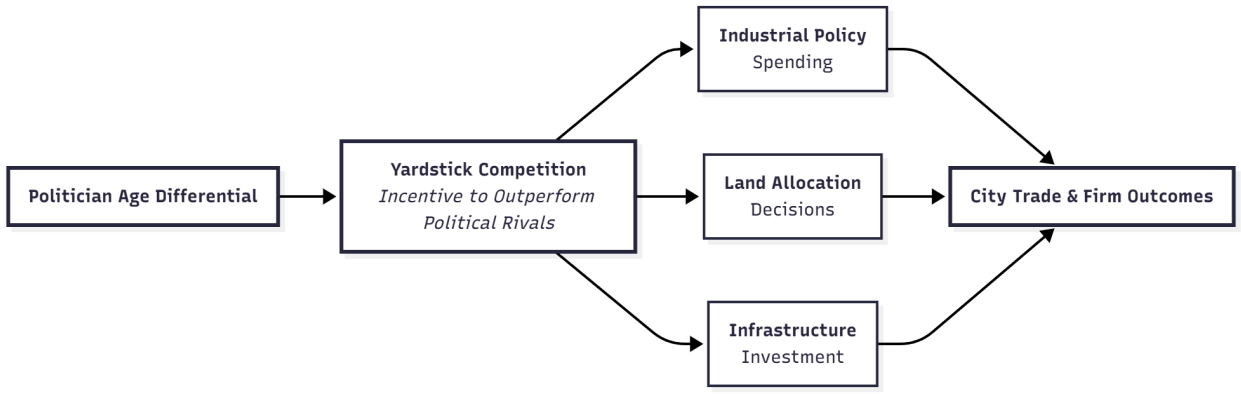
G Figures

Figure G.1: Hypothesized Relationship between Politician Age Differential and Trade Volumes



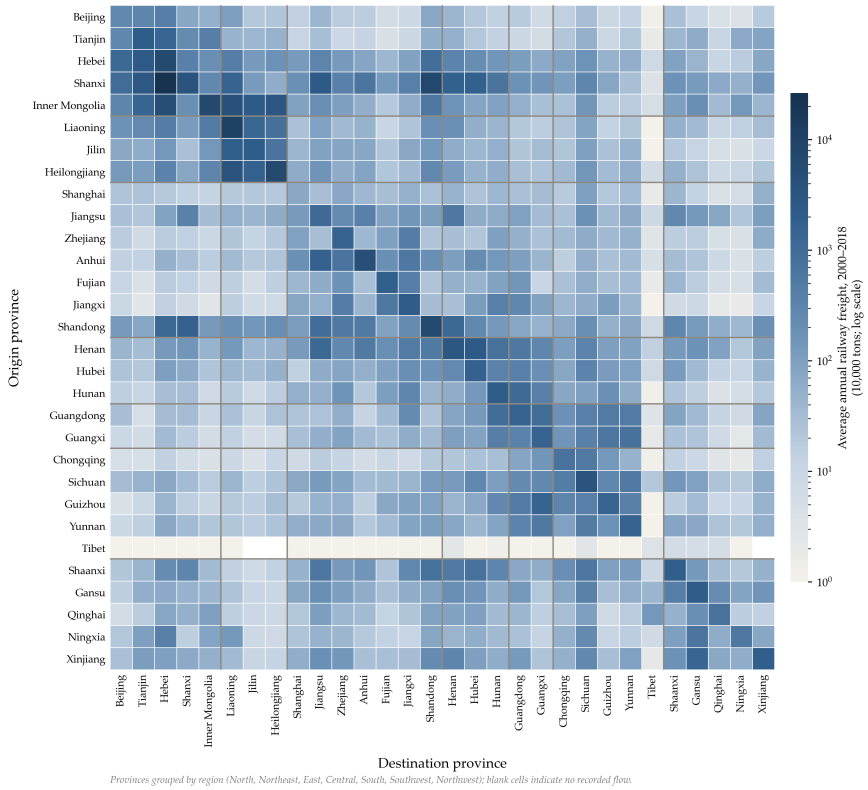
Notes: This figure illustrates the hypothesized relationship between politicians' age differential, competition intensity, and city trade volumes.

Figure G.2: Causal Chains from Politician Age Differential to Trade Outcomes



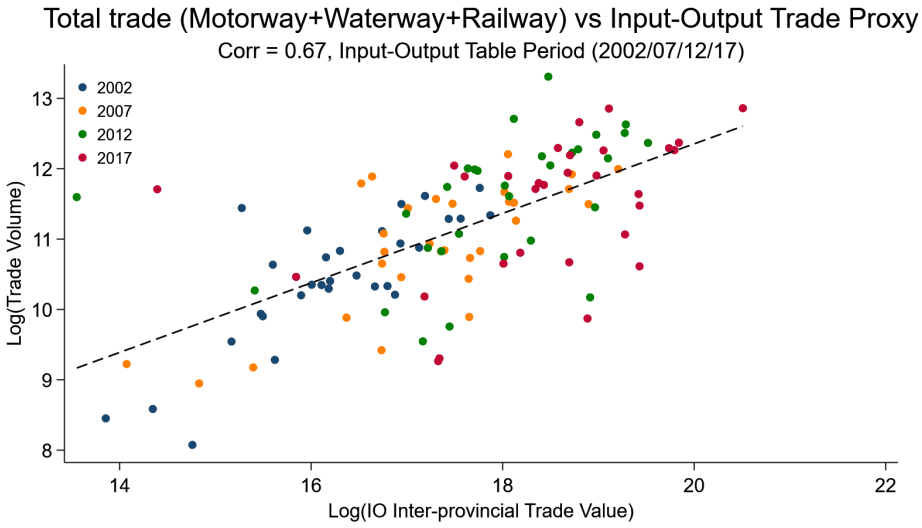
Notes: This figure illustrates the flow chart of causal chains from politician age differential to city and firm trade outcomes.

Figure G.3: Inter-province Trade Flows

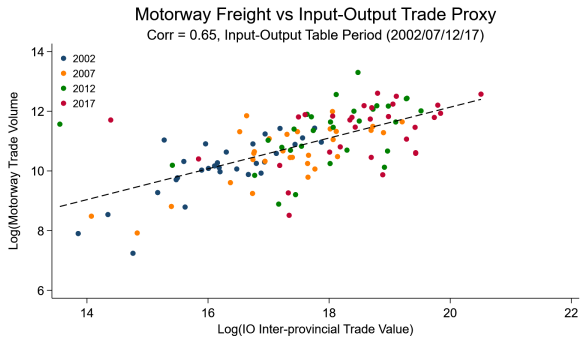


Notes: This figure illustrate the average inter-province trade flows from 2000 to 2018.

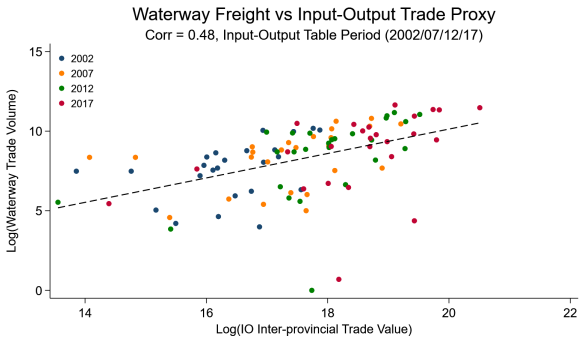
Figure G.4: Validation of the Freight-Based Trade Measure against Input-Output Inter-provincial Trade



(a) Total Trade Volume (Motorway+Waterway+Railway)



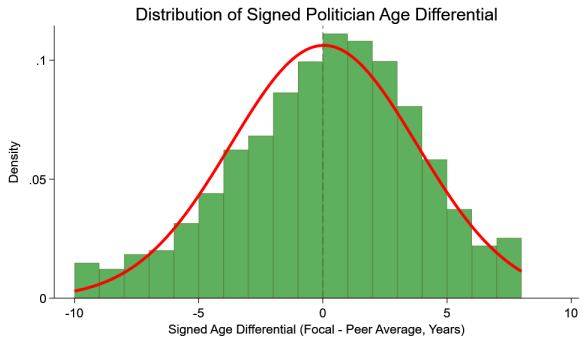
(b) Motorway Trade Volume



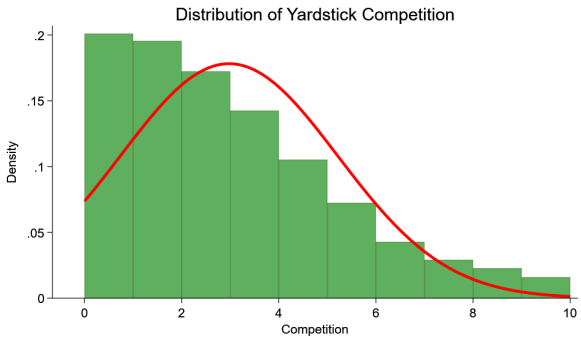
(c) Waterway Trade Volume

Notes: This figure validates the freight-based trade measures against an independent, accounts-based trade proxy constructed from China's 30-province multi-regional input-output (MRIO) tables. For each benchmark year (2002, 2007, 2012, and 2017), inter-provincial trade is computed by summing each province's tradable-sector shipments to and from all other provinces. City-level freight volumes from the City Statistical Yearbook are aggregated to the province level. Each panel plots the log freight measure against the log IO inter-provincial trade value, pooling the four benchmark years; the dashed line is the fitted linear relationship. The pooled log-log correlations are 0.67 for total trade, 0.65 for motorway freight, and 0.48 for waterway freight. Because the IO flows are denominated in value while freight is denominated in tonnage, the two measures are not expected to coincide in levels; the tight co-movement indicates that the freight measure is a valid volume proxy for inter-provincial trade.

Figure G.5: Distribution of Politician Age Differential



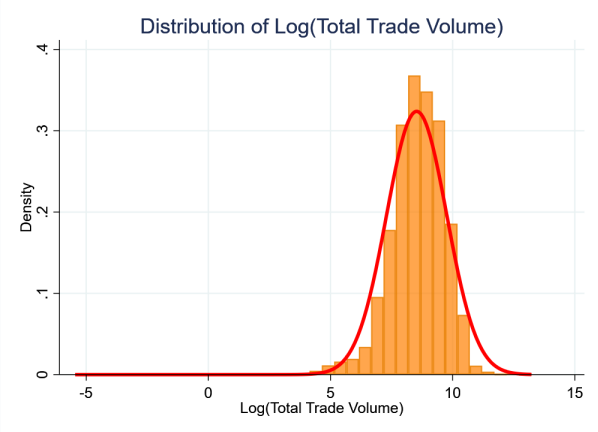
(a) Signed Politician Age Differential



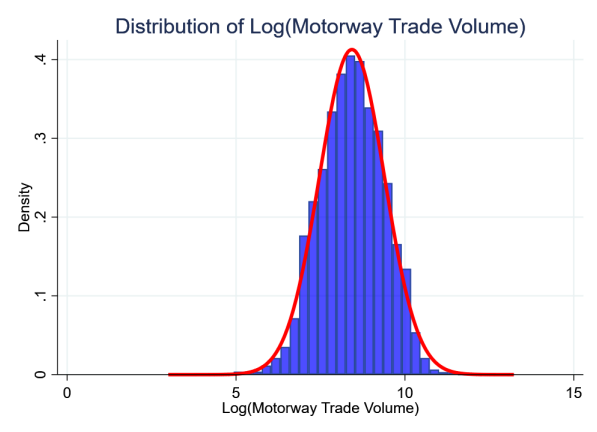
(b) Politician Age Differential

Notes: This figure presents frequency distributions of key variables in the analysis. Panel (a) displays the signed Politician Age Differential. Panel (b) illustrates Politician Age Differential. All distributions include normal curve overlays in red for reference.

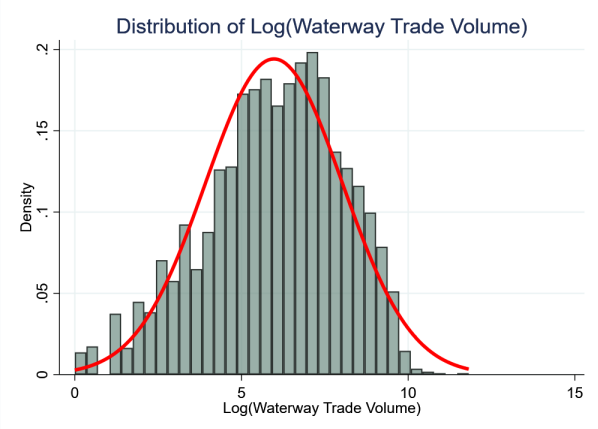
Figure G.6: Distribution of Trade Indicators



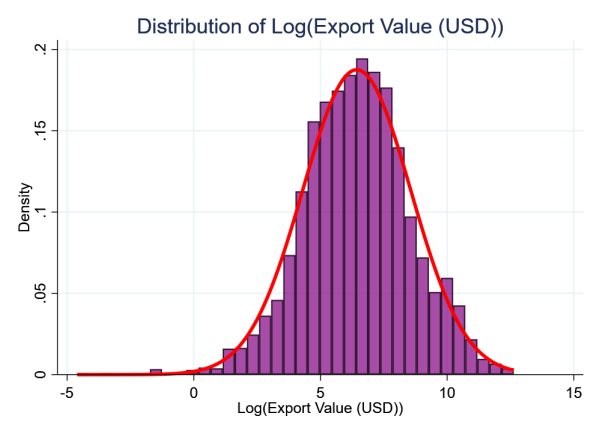
(a) Log (Total Trade Volume)



(b) Log (Motorway Trade Volume)



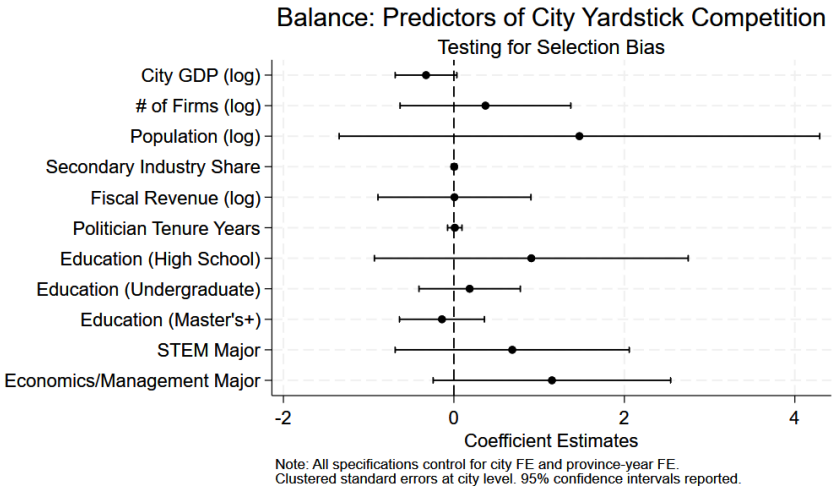
(c) Log (Waterway Trade Volume)



(d) Log (Export Value USD)

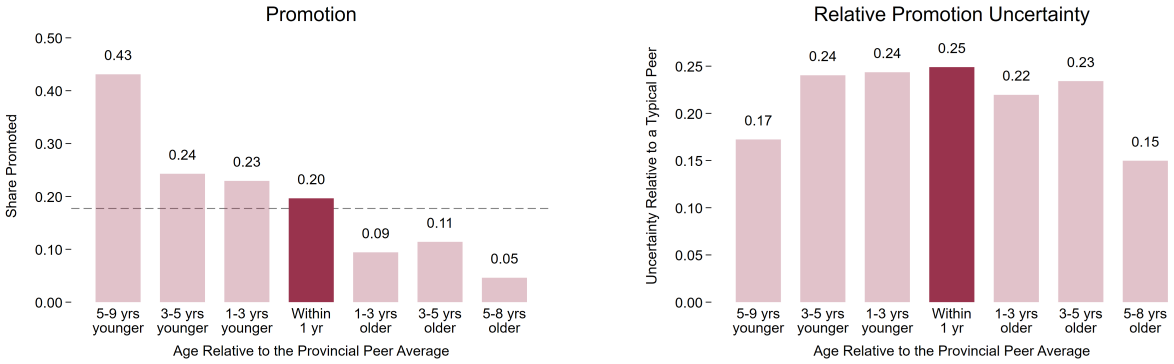
Notes: This figure presents frequency distributions of key variables in the analysis. Panel (a) displays the log-transformed total trade volume. Panels (b), (c), and (d) present infrastructure-related trade measures: log-transformed motorway trade volume, waterway trade volume, and export value in USD, respectively. All distributions include normal curve overlays in red for reference.

Figure G.7: Balance Test: Relationship between Politician Age Differential and Politician and City Characteristics



Notes: This figure presents a balance test examining the relationship between Politician Age Differential and various politician and city characteristics. The lack of statistically significant relationships suggests that age differential is plausibly exogenous to these characteristics.

Figure G.8: Promotion Rates and Relative Promotion Uncertainty by Age Relative to Provincial Peers



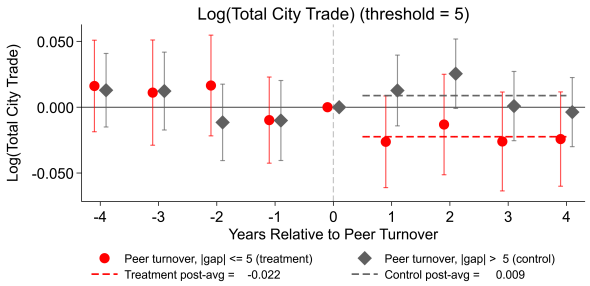
(a) Share Promoted

(b) Relative Promotion Uncertainty

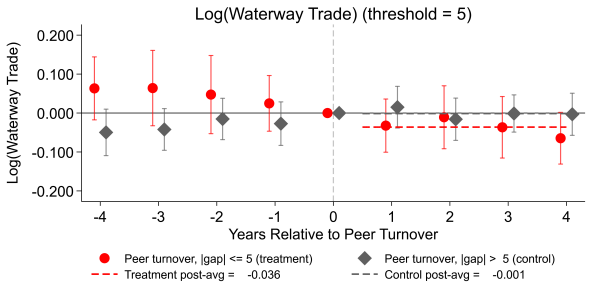
Notes: Each bar is a group of city party secretary terms (666 terms) by the signed age differential, the secretary's start age minus the average start age of the other city party secretaries in the same province; the darker bar is the group within one year of the peer average. Panel (a) reports the share promoted, p, the number of terms in the group that ended in promotion divided by the number of terms in the group. The dashed line is the average share promoted across all terms, $\bar{p} = 0.177$. Panel (b) reports relative promotion uncertainty, $q(1-q)$, where $q = o/(o+\bar{o})$ is the chance that a secretary in the group finishes ahead of a typical peer. Here $o = p/(1-p)$ is the group's odds of promotion and $\bar{o} = \bar{p}/(1-\bar{p}) = 0.215$ is the odds of a typical peer, whose chance of promotion is the average $\bar{p}$. A group with the average share promoted has no age advantage or disadvantage against a typical peer ($q = 0.5$), where relative uncertainty reaches its maximum of 0.25.

Figure G.9: Robustness: Similar- versus Dissimilar-Age Peer Arrivals (Five-Year Threshold)

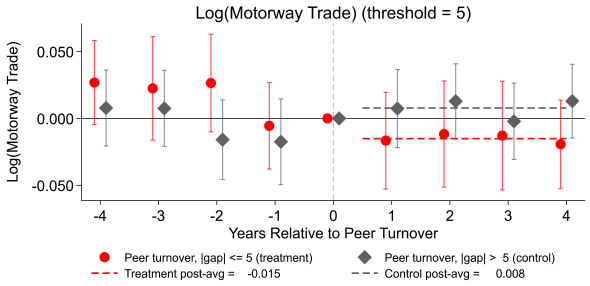
(a) Total Trade Volume



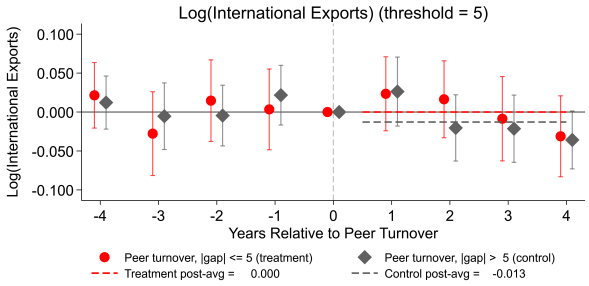
(c) Waterway Trade Volume



(b) Motorway Trade Volume



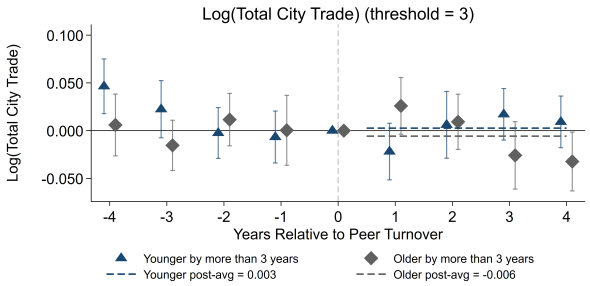
(d) Export Value



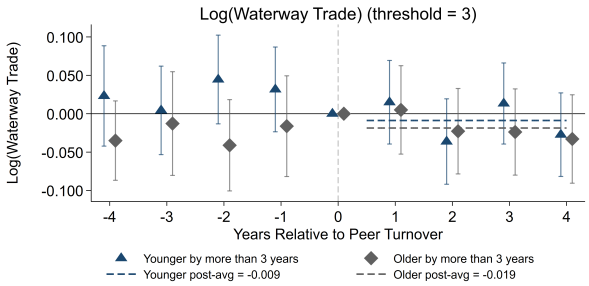
Notes: These figures report an alternative event study using a five-year age-gap threshold. The similar-age series (red circles) consists of arrivals for which the incoming peer secretary's age is within five years of the focal secretary's start age; the dissimilar-age series (gray diamonds) consists of arrivals outside that band. The x-axis shows years relative to the peer arrival, and the y-axis shows the estimated coefficients. The horizontal dashed lines report the average post-arrival coefficient for each group. All specifications include city-politician and year fixed effects and the baseline controls. Standard errors are clustered at the city level. Vertical bars show 90% confidence intervals.

Figure G.10: Dissimilar-Age Peer Arrivals: Younger versus Older Peers

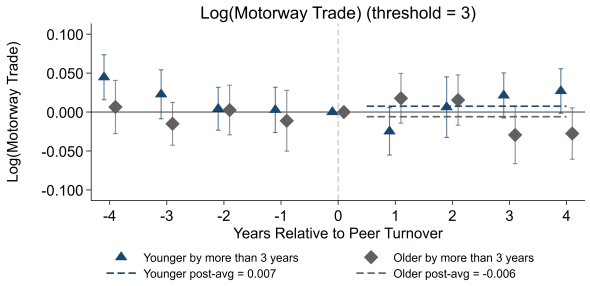
(a) Total Trade Volume



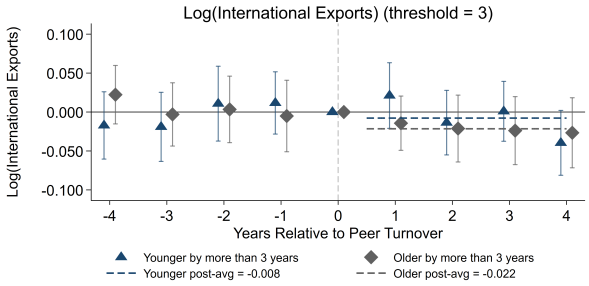
(c) Waterway Trade Volume



(b) Motorway Trade Volume



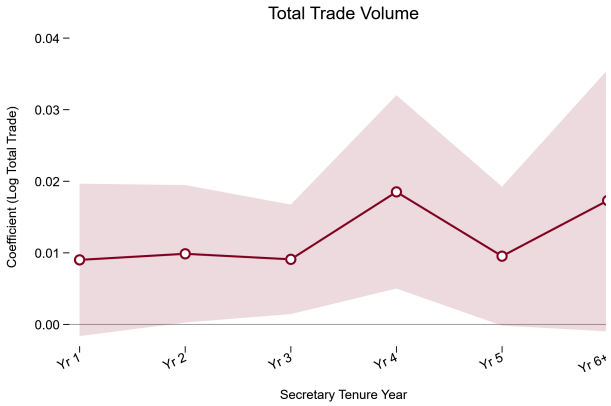
(d) Export Value



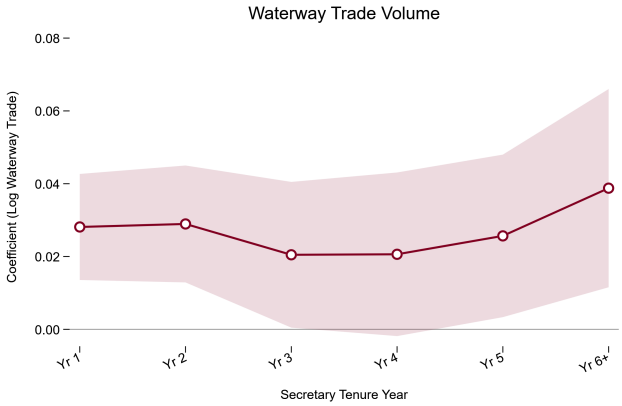
Notes: These figures split the dissimilar-age arrivals of Figure 6 by the direction of the age gap. The younger series (blue triangles) consists of arrivals for which the incoming peer secretary is more than three years younger than the focal secretary's start age; the older series (gray diamonds) consists of arrivals for which the incoming peer secretary is more than three years older. The two series are estimated separately with the specification of Figure 6, and the arrival year is the reference period. The x-axis shows years relative to the peer arrival, and the y-axis shows the estimated coefficients. The horizontal dashed lines report the average post-arrival coefficient for each group. All specifications include city-politician and year fixed effects and the baseline controls. Standard errors are clustered at the city level. Vertical bars show 90% confidence intervals. In a regression that includes both series, tests that the average post-arrival coefficients are equal for younger and older arrivals do not reject for any outcome ($p = 0.60$, 0.46, 0.74, and 0.67 for total, motorway, waterway, and export trade).

Figure G.11: Evolving Effects of Politician Age Differential on Trade Volumes by Secretary's Own Tenure Year

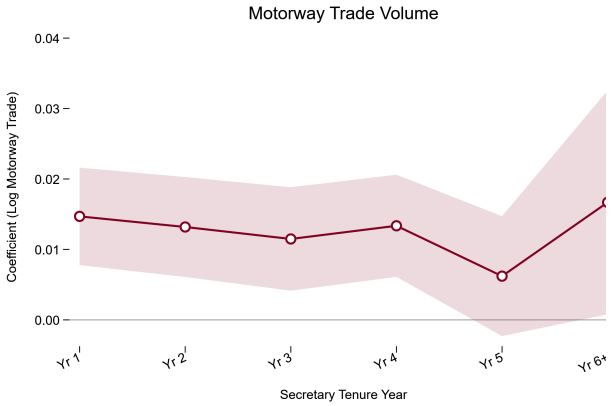
(a) Total Trade Volume



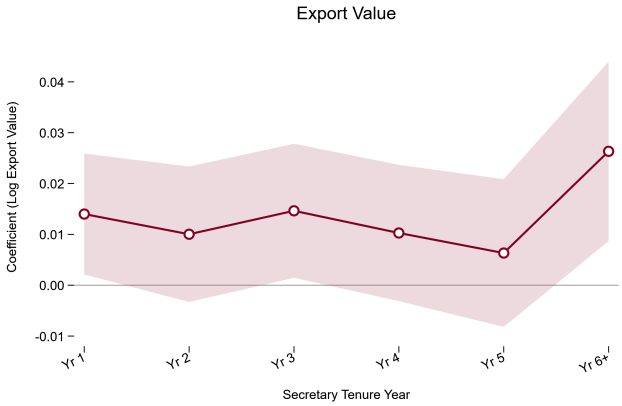
(c) Waterway Trade Volume



(b) Motorway Trade Volume

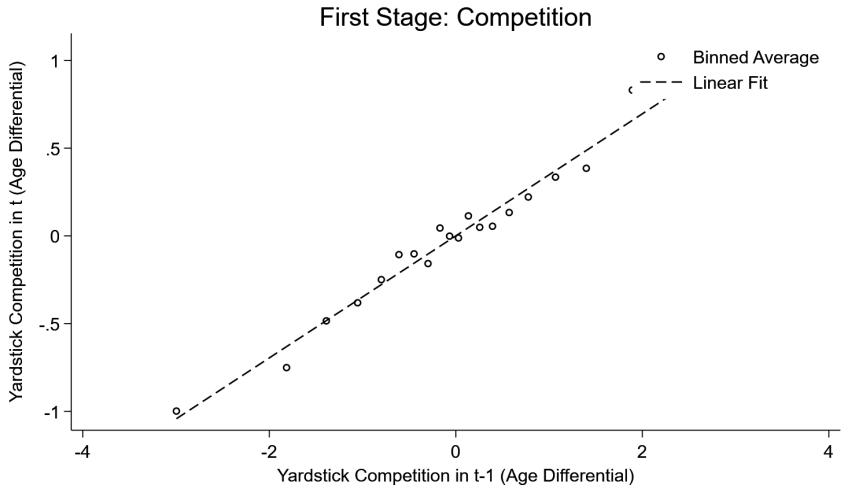


(d) Export Value (USD)



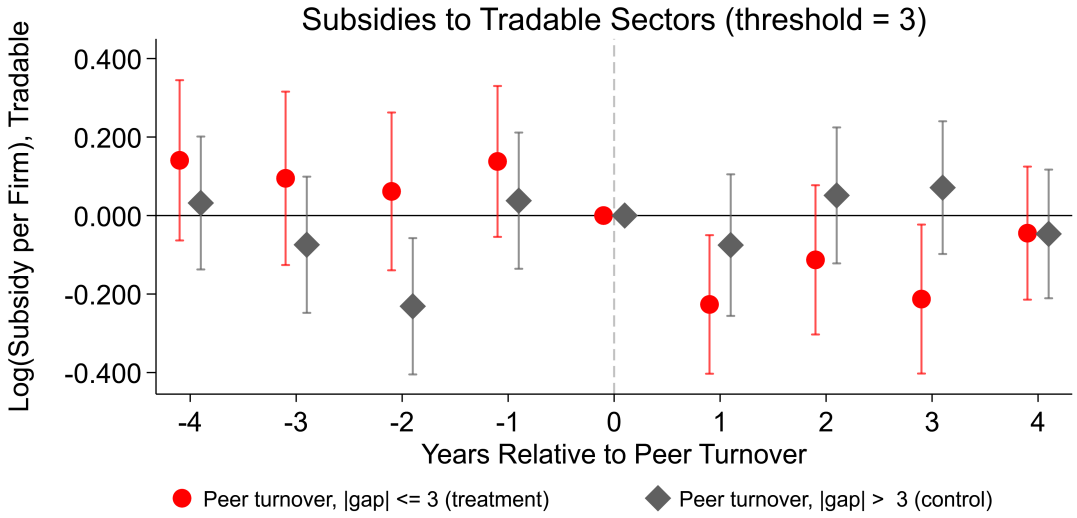
Notes: These figures plot the estimated effect of Politician Age Differential on different trade outcomes by the city party secretary's tenure year. The analysis uses city-level panel data from 1996 to 2018. The x-axis shows the secretary's tenure year, from Year 1 through Year 6 and above. The y-axis shows the estimated coefficients from a regression of the log trade outcome on Politician Age Differential interacted with tenure-year indicators. I control for city and politician characteristics, city-politician fixed effects, and year fixed effects. Standard errors are clustered at the city level.

Figure G.12: First Stage Relationship: One-Year Lagged Politician Age Differential



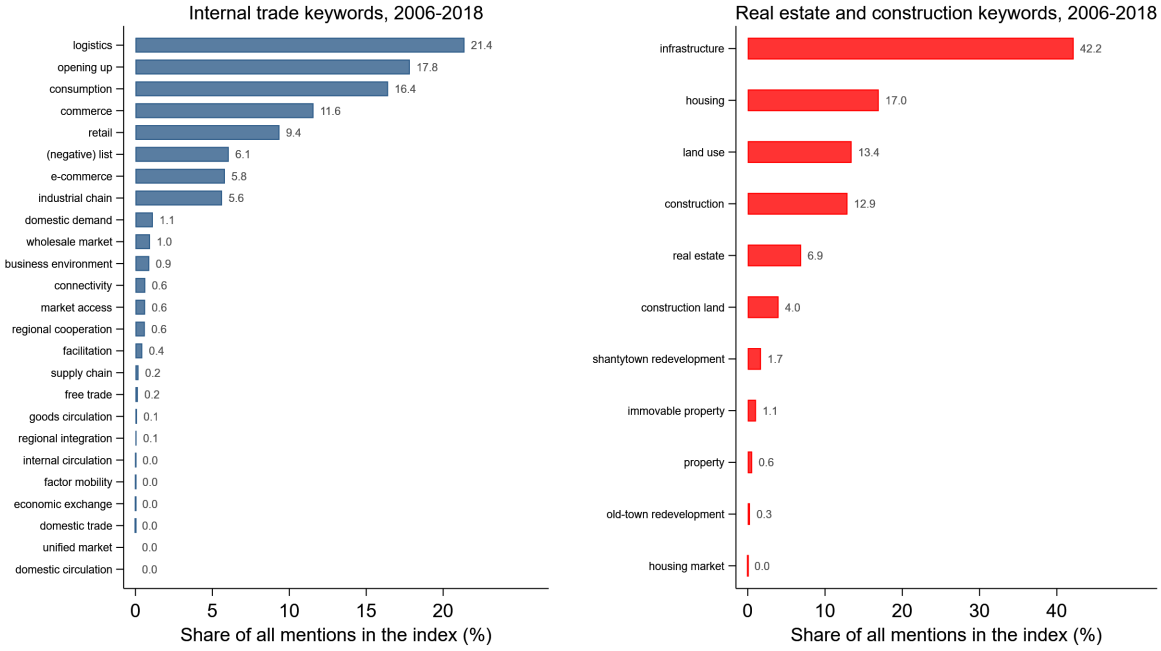
Notes: This figure displays the first-stage relationship underlying the instrumental variable estimates in Table H.12, which instrument the Politician Age Differential with its one-year lag. Both the endogenous variable and the instrument are residualized on politician characteristics and city-politician and year fixed effects (Frisch-Waugh-Lovell), using the first-stage estimation sample. Observations are grouped into twenty equal-sized bins of the residualized instrument; each circle plots the within-bin mean of the residualized endogenous variable against the within-bin mean of the residualized instrument, and the dashed line is a linear fit through the bin means weighted by bin size. The strong positive relationship corresponds to first-stage F-statistics between 277 and 363 across specifications, well above conventional weak-instrument thresholds.

Figure G.13: Event Study: Subsidies to Tradable Sectors around Similar- and Dissimilar-Age Peer Arrivals



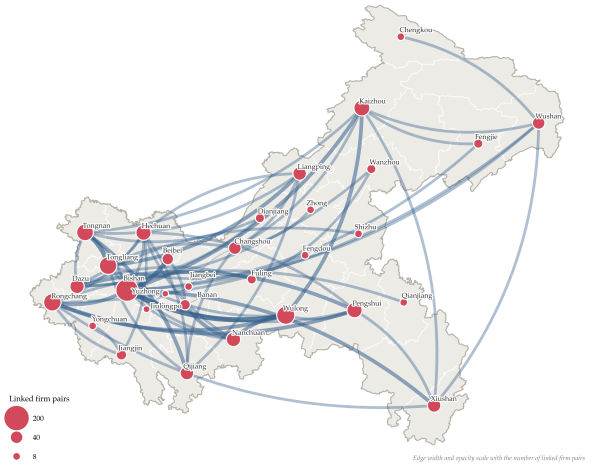
Notes: This figure reports an event study of subsidies around peer arrivals, using the city-sector panel of Table 4. The outcome is the logarithm of average subsidies per listed firm in a city-sector-year. The coefficients are the interactions between indicators for years relative to a peer arrival and the tradable-sector indicator, so they trace subsidies to tradable sectors relative to other sectors in the same city and year. The similar-age series (red circles) consists of arrivals for which the incoming peer secretary's age is within three years of the focal secretary's start age; the dissimilar-age series (gray diamonds) consists of arrivals outside that band. The two series are estimated separately, and the arrival year is the reference period. All specifications include city-sector, politician, sector-year, and city-year fixed effects. Standard errors are clustered at the city-sector level. Vertical bars show 90 percent confidence intervals. Joint tests that the four pre-arrival coefficients are zero do not reject for either series ($p = 0.68$ for similar-age arrivals and $p = 0.14$ for dissimilar-age arrivals).

Figure G.14: Keyword Composition of the Internal Trade and Real Estate Policy Indices

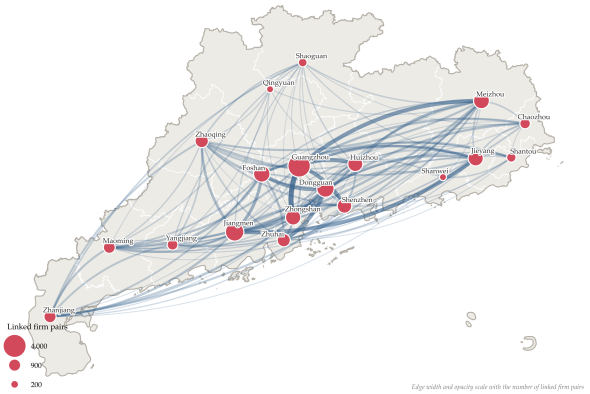


Notes: This figure reports the composition of keywords underlying the internal trade (panel a) and real estate/construction (panel b) policy attention indices. Keywords are extracted from the text of annual government work reports issued by regional governments in China, with mentions counted across the full text of each report. Each bar reports a keyword's share of total mentions within its category, so that shares sum to 100% within each panel. Keyword-to-category assignment is mutually exclusive. English labels are translations of the original Chinese keywords.

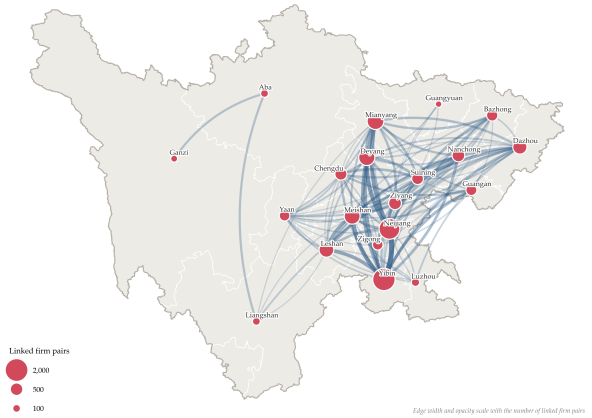
Figure G.15: Firm Networks in Four Selected Provinces and Municipality



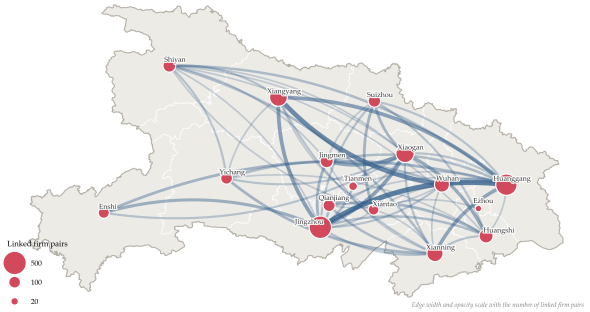
(a) Firm Network for Chongqing Municipality



(b) Firm Network for Guangdong Province



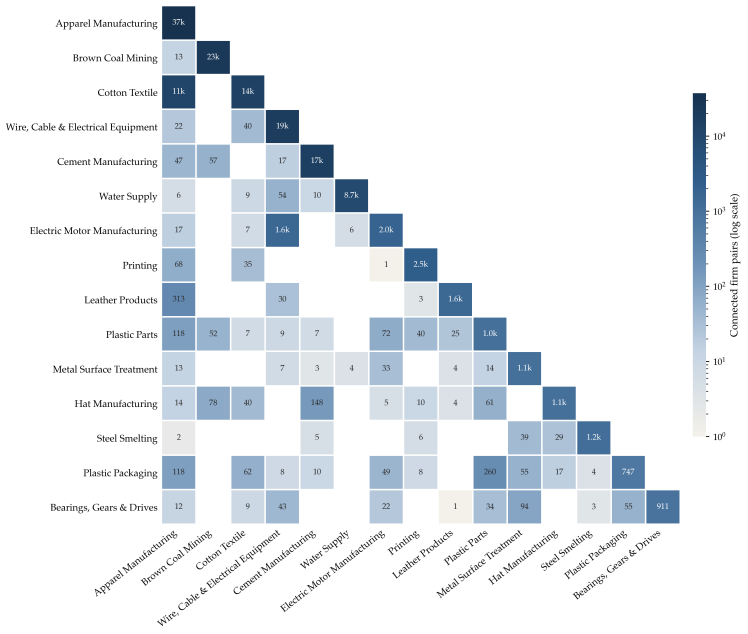
(c) Firm Network for Sichuan Province



(d) Firm Network for Hubei Province

Notes: This figure illustrates business network structures in four selected Chinese provinces/municipalities. Each node represents a different industry, with colors indicating industry categories according to the legend. The connecting lines (edges) represent business relationships between industries, with thicker lines indicating stronger connections.

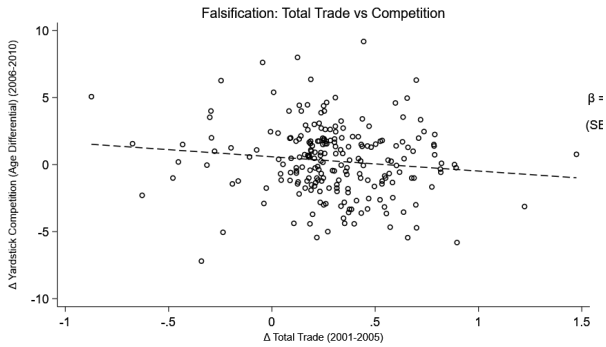
Figure G.16: Connected Sectors in Business Networks (Heatmap)



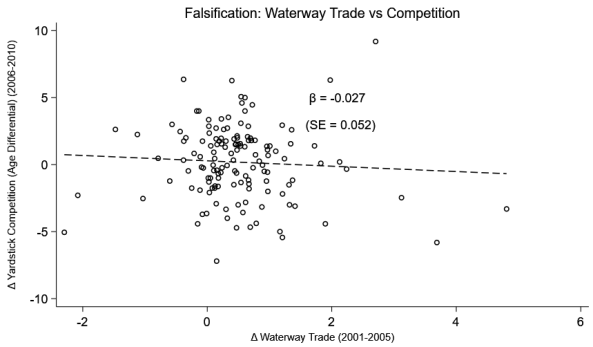
Notes: This heatmap illustrates inter-sectoral firm connections within the business network, showing connection intensity between sectors using 2000-2007 data. Lighter colors indicate higher connection strength between corresponding sectors.

Figure G.17: Falsification: Past Trade Volumes and Future Yardstick Competition

(a) Total Trade Volume



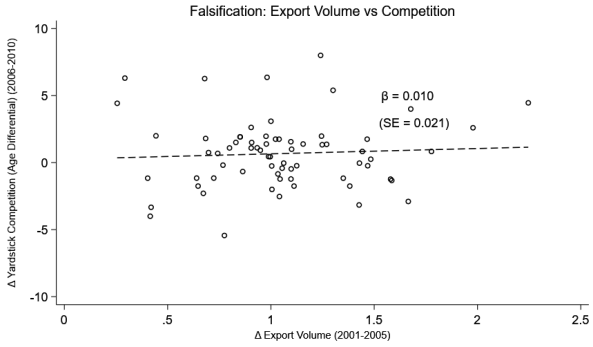
(c) Waterway Trade Volume



(b) Motorway Trade Volume

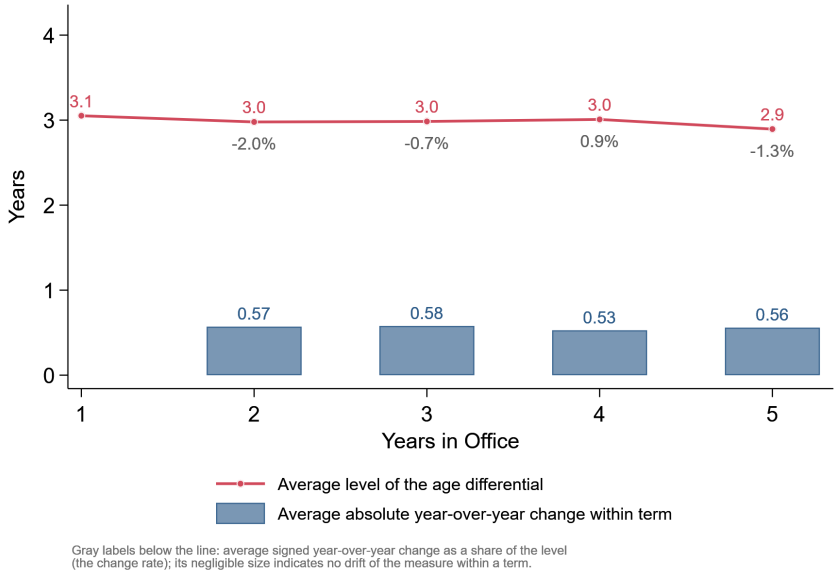


(d) Export Value (USD)



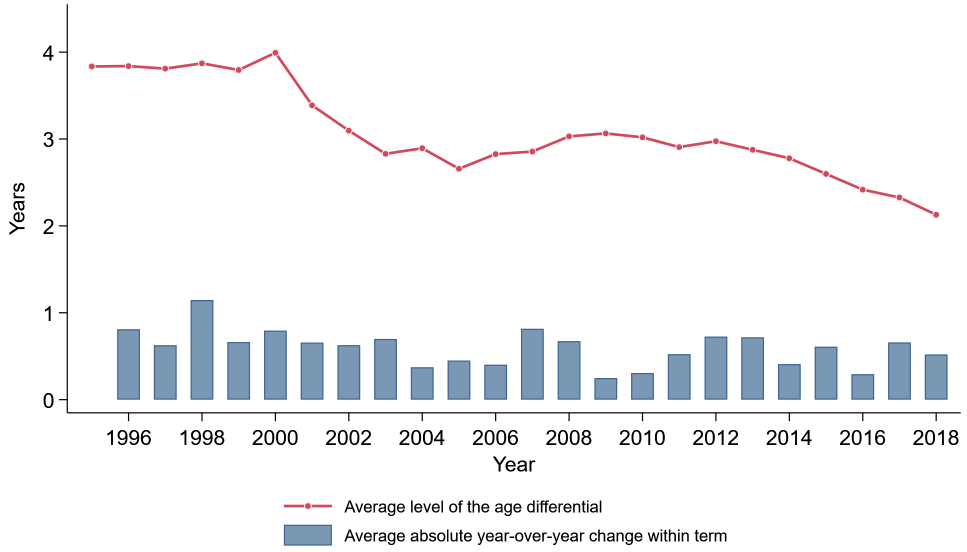
Notes: These figures test whether past trade growth predicts future political competition. The dependent variables are log changes in trade volumes during 2000–2005: total trade (panel a), motorway trade (panel b), waterway trade (panel c), and export value (panel d). The independent variable is the average Politician Age Differential during the subsequent 2006–2010 period. A significant coefficient would suggest pre-existing trends, casting doubt on the causal interpretation. The insignificant coefficients indicate no evidence of such pre-trends. All specifications include province fixed effects and baseline city controls from 2000. Standard errors are bootstrapped at the province level by 500 times.

Figure G.18: Changes in Politician Age Differential



Notes: This figure documents the within-term variation of the Politician Age Differential across tenure years (horizontal axis: years in office, 1–5). The line reports the average *level* of the differential, which is stable at about three years of age distance from provincial peers; the gray labels below the line report the average *signed* year-over-year change as a share of the level (the change rate), which never exceeds two percent in absolute value, indicating no mechanical drift of the measure over a term. The bars report the average *absolute* year-over-year change within a city-secretary term, defined from the second year in office onward. Because a secretary's own start-of-term age is fixed within a term, all of this movement – 0.53 to 0.58 years annually, about one-fifth of the level – comes from the turnover of peer secretaries in the same province, the variation that identifies the fixed-effects estimates.

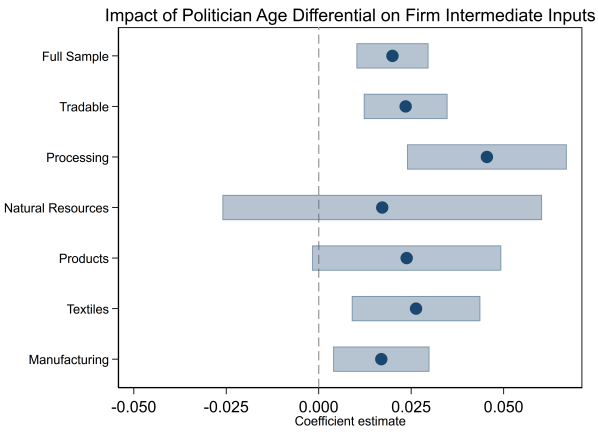
Figure G.19: Changes in Politician Age Differential over Time



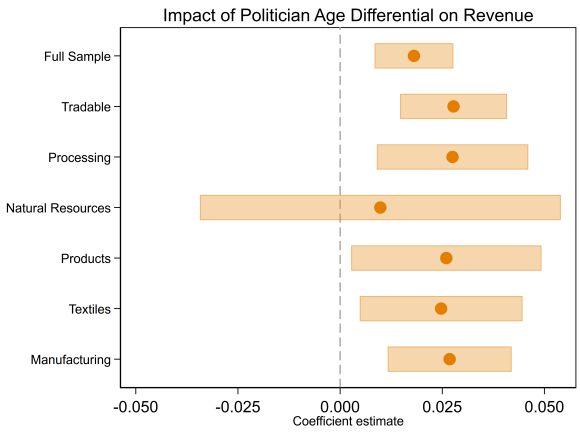
Notes: This figure plots the Politician Age Differential by calendar year. The line reports the average *level* of the differential across city-years, which declines from about 3.8 years in the late 1990s to about 2.1 years in 2018, reflecting the increasing compression of secretaries' age profiles as age-based appointment norms became standardized. The bars report the average *absolute* year-over-year change within city-secretary terms, which is sizable in every year of the sample – on average about one-fifth of the level annually – and peaks after Party Congress reshuffle waves (1998, 2007, 2012–2013), when the turnover of peer secretaries rearranges provincial age structures. The aggregate decline in the level is common to all cities and absorbed by the year fixed effects; identification relies only on the within-term variation shown by the bars.

Figure G.20: Effects of Politician Age Differential on Different Firms: Upstream vs. Downstream Sectors

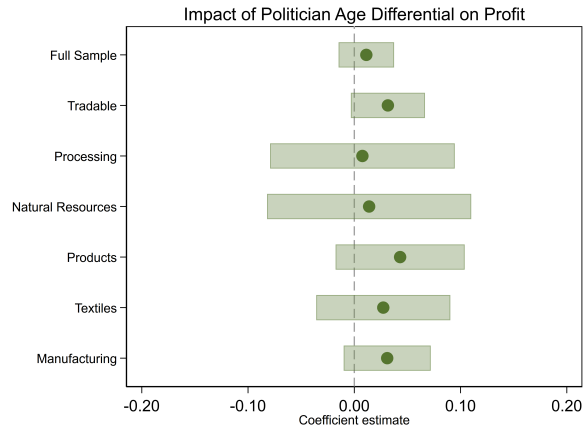
(a) Log(Intermediate Good Inputs)



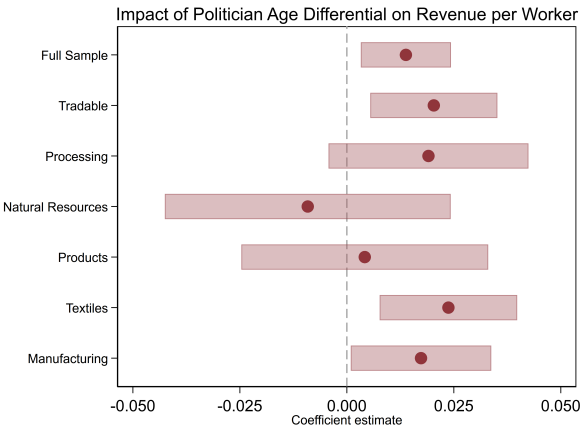
(c) Log(Revenue)



(b) Log(Profit)



(d) Log(Revenue per Worker)



Notes: These figures illustrate the effect of Politician Age Differential on various outcomes of firms operating in different sectors. Panel (a) shows the effect on firm intermediate good inputs; panel (b) shows the effect on firm profit; panel (c) shows the effect on firm revenue; and panel (d) shows the effect on firm revenue per worker. Across all specifications, I control for politician controls, firm controls, firm fixed effect, politician fixed effect, and sector-year fixed effect. Standard errors are clustered at the city level. 95% confidence intervals are shown.

H Tables

Table H.1: Promotion Statistics of City Party Secretaries (2004-2014)

Politician Category	Promotion Metric	Value
Promoted City Party Secretaries	To central gov. or other provinces	2%
	Within-province promotion	98%
	Overall promotion rate	18%
City Party Secretaries' Career Path	Previously served as city mayor	53%
	Of these, mayor post held before secretary post	96%
Promoted City Mayor	To the same city	70%

Notes: This table shows statistics of promoted city party secretaries and mayors from 2004 to 2014. The 98% within-province promotion of party secretary suggests a strong within-province competition. The career-path rows document the mobility between the two positions: a majority of secretaries previously served as city mayor, and among those who held both posts, the mayor position almost always preceded the secretary position. The mayor-to-secretary ladder, most often within the same city, means mayors' promotion prospects do not run through the within-province secretary tournament, consistent with the null results for mayors in the placebo and horse-race specifications.

Table H.2: Correlation between Trade Volumes in City i and Economic Performance Indicators in Other Cities j (Same Province)

VARIABLES	Log(Nighttime Light Intensity in City$_{j,t}$) (1)	Log(Population Size in City$_{j,t}$) (2)	Log(GDP in City$_{j,t}$) (3)
Trade Volume in City$_{i,t-1}$	0.053***	0.015***	0.050**
	(0.008)	(0.005)	(0.022)
Observations	41,554	41,554	41,554
R-squared	0.980	0.989	0.520
City Pair FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table examines the spillover effects of trade activity in one city on the economic performance of other cities within the same province. The analysis uses city-pair panel data from 1996 to 2018. The key independent variable 'Trade Volume in City$_i$' measures the total trade volume in the focal city, while the dependent variables measure different economic outcomes in connected cities (City$_j$ in the same province): nighttime light intensity, population size, and total GDP. All specifications include city-pair fixed effects and year fixed effects. Standard errors (in parentheses) are clustered at the city$_j$ level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.3: Summary Statistics

VARIABLES	N (1)	mean (2)	sd (3)	min (4)	max (5)
Panel A: Trade and Transportation Variables					
Log(Total Trade Volume)	5,281	8.609	1.149	-5.449	13.23
Log(Motorway Trade Volume)	5,157	8.450	0.951	4.078	13.23
Log(Waterway Trade Volume)	2,924	5.974	2.040	0	11.82
Log(Export Value (USD))	3,988	6.444	2.110	-4.605	12.63
Panel B: Politician Variables					
Politician Age Differential	5,281	2.934	2.208	0	10
Starting Age	5,281	50.50	3.741	36	60
Peer Average Age	5,281	50.52	1.771	40	56
Panel C: Firm Characteristics					
Log(Firm I Intermediate Goods)	26,882	9.867	1.156	2.303	16.68
Log(Firm I Revenue)	26,888	10.21	1.109	5.308	17.06
Log(Firm I Revenue per Worker)	26,883	5.012	0.972	0.408	9.249
Log(Firm I Profit)	26,889	6.498	1.913	0	14.83
Log(Firm J Intermediate Goods)	118,940	10.000	1.087	0	16.32
Log(Firm J Revenue)	118,957	10.34	1.036	5.011	16.52
Log(Firm J Revenue per Worker)	118,943	5.048	0.922	-0.0275	9.232
Log(Firm J Profit)	118,957	6.691	1.882	0	14.03
Panel D: City-sector Characteristics					
Log(City-sector Subsidies)	9,739	14.24	1.660	5.704	24.11
Log(City-sector Firm #)	9,739	0.401	0.609	0	4.078
Log(City-sector Revenue)	9,739	21.54	1.364	16.17	28.57
Log(City-sector Average Wage)	9,739	15.09	0.706	11.41	18.39
Panel E: County-sector Firm Registration					
Log(# Firm Entry)	5,765,780	2.102	1.266	0.693	9.414
1(Tradable Firms)	5,765,780	0.467	0.499	0	1
1(Real Estate Firms)	5,765,780	0.059	0.236	0	1

Notes: This table presents summary statistics for key variables used in the analysis. Panel A shows statistics for different measures of trade and transportation volume. Panel B presents statistics related to politician characteristics, where 'Politician Age Differential' measures the absolute difference between a politician's age and the average age within the region. Panel C shows statistics for firm characteristics. Panel D presents statistics related to city-sector characteristics. Panel E displays the characteristics of county-sector firm registration.

Table H.4: Correlation Between Signed Age Differential and Promotion Outcomes

	1(Promotion)			
VARIABLES	(1)	(2)	(3)	(4)
Politician's Start Age	-0.022***	-0.008	0.007	0.014
	(0.004)	(0.008)	(0.013)	(0.014)
Signed Age Differential		-0.016**	-0.031**	-0.033**
		(0.007)	(0.014)	(0.014)
Signed Age Differential × Deputy City Level				0.038*
				(0.020)
Log(Education Years)				0.018**
				(0.009)
Average Light Density				0.051
				(0.126)
Log(Jurisdictional Population)				0.038
				(0.303)
Colleague Ties (Provincial Secretary)				-0.029
				(0.060)
Colleague Ties (Governor)				0.005
				(0.051)
Hometown Ties (Provincial Secretary)				-0.248
				(0.240)
Hometown Ties (Governor)				0.044
				(0.113)
Observations	673	673	672	659
R-squared	0.050	0.056	0.408	0.424
City Characteristics	–	–	–	Yes
Politician Characteristics	–	–	–	Yes
Connections to Provincial Politicians	–	–	–	Yes
City FE	–	–	Yes	Yes
Tenure Start Year FE	–	–	Yes	Yes

Notes: This table examines the relationship between signed age differential and promotion outcomes for city party secretaries. The analysis uses politician-level data from 2000 to 2014. The dependent variable is a binary indicator for whether a politician was promoted to a higher position. The key independent variable, Signed Age Differential, is calculated as the politician's age minus the average age of party secretaries in other cities within the same province; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions use the absolute value of this gap. Control variables include politician characteristics (age, education years), jurisdictional characteristics (GDP and population), and provincial connection measures. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.5: Correlation Between Signed Age Differential and Transfer/Retirement Outcomes

	1(Transfer/Retirement)			
VARIABLES	(1)	(2)	(3)	(4)
Politician's Start Age	0.020***	0.007	-0.014	-0.023
	(0.004)	(0.008)	(0.014)	(0.015)
Signed Age Differential		0.015**	0.035**	0.039***
		(0.007)	(0.014)	(0.015)
Signed Age Differential × Deputy City Level				-0.047**
				(0.019)
Log(Education Years)				-0.013
				(0.009)
Average Light Density				-0.058
				(0.122)
Log(Jurisdictional Population)				0.618
				(0.419)
Colleague Ties (Provincial Secretary)				0.033
				(0.059)
Colleague Ties (Governor)				0.009
				(0.053)
Hometown Ties (Provincial Secretary)				0.255
				(0.241)
Hometown Ties (Governor)				-0.103
				(0.121)
Observations	673	673	672	659
R-squared	0.038	0.043	0.399	0.420
City Characteristics	–	–	–	Yes
Politician Characteristics	–	–	–	Yes
Connections to Provincial Politicians	–	–	–	Yes
City FE	–	–	Yes	Yes
Tenure Start Year FE	–	–	Yes	Yes

Notes: This table examines the relationship between signed age differential and transfer/retirement outcomes for city party secretaries. The analysis uses politician-level data from 2000 to 2014. The dependent variable is a binary indicator for whether the politician was transferred to another position or retired. The key independent variable, Signed Age Differential, is calculated as the politician's age minus the average age of party secretaries in other cities within the same province; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions use the absolute value of this gap. Control variables include politician characteristics (age, education years), jurisdictional characteristics (GDP and population), and provincial connection measures. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.6: Horse Race Specification: Party Secretary versus City Mayor Age Differential

Variables	(1) Log(Total Trade Volume)	(2) Log(Motorway Trade Volume)	(3) Log(Waterway Trade Volume)	(4) Log(Export Value)
Politician Age Differential (Secretary)	0.011**	0.011***	0.036***	0.007
	(0.005)	(0.004)	(0.009)	(0.007)
Politician Age Differential (Mayor)	0.007	0.003	-0.001	-0.003
	(0.007)	(0.004)	(0.008)	(0.005)
Observations	4,375	4,249	2,444	3,366
R-squared	0.892	0.951	0.975	0.982
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Secretary FE	Yes	Yes	Yes	Yes
City-Mayor FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a horse race between the party secretary's and the city mayor's politician age differential in the same specification. Party Secretary Age Differential measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province, while City Mayor Age Differential is defined analogously for city mayors. Including both variables jointly tests whether the trade effects documented for party secretaries are specific to party secretaries, who face within-province promotion competition of the relevant kind, rather than reflecting a general effect of local leadership age differentials. The sample is restricted to city-years with non-missing age differential measures for both the party secretary and the mayor. Column (1) reports effects on total trade volume, column (2) on motorway trade, column (3) on waterway trade, and column (4) on the value of international exports (USD). All trade volumes are logged and measured at 10,000 tons. I control for politician characteristics, city-secretary fixed effects, city-mayor fixed effects, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.7: Signed Decomposition: Politician Age Differential on City Trade

	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
VARIABLES	(1)	(2)	(3)	(4)
Years Younger than Peer Average	0.016***	0.015***	0.023**	0.012*
	(0.005)	(0.004)	(0.011)	(0.007)
Years Older than Peer Average	0.006	0.010**	0.030***	0.013*
	(0.007)	(0.004)	(0.009)	(0.008)
Younger = Older	0.240	0.212	0.491	0.869
Observations	5,297	5,113	2,858	3,913
R-squared	0.890	0.936	0.967	0.976
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table reports the signed decomposition of the absolute age differential, which tests the symmetry restriction embedded in the absolute-value specification. The two key independent variables decompose the absolute age differential into non-negative components: 'Years Younger than Peer Average' equals max(peer average age − focal age, 0), measuring how many years the focal city party secretary is below the provincial peer average (zero if focal is older); 'Years Older than Peer Average' equals max(focal age − peer average age, 0), measuring how many years the focal is above (zero if younger). By construction, their sum equals the absolute age differential used in the baseline. The third row reports the p-value of an F-test for the symmetry restriction (coefficients equal across the two variables) implicit in the absolute-value specification. Dependent variables are logged: total trade volume in column (1), motorway in (2), waterway in (3), and the value of international exports (USD) in (4). All specifications include city characteristics, politician characteristics, and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.8: Placebo Test: Cross-Provincial Age Differential and Trade Outcomes in Border Cities

VARIABLES	Log(Total Trade Volume) (1)	Log(Motorway Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Log(Export Value) (4)
Cross-Provincial Age Differential	-0.003	0.003	0.010	-0.005
(Relative to Neighboring Provinces)	(0.011)	(0.005)	(0.011)	(0.008)
Observations	3,320	3,198	1,729	2,419
R-squared	0.856	0.930	0.961	0.965
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a placebo test examining whether cross-provincial age differentials affect trade outcomes in border cities. The analysis is restricted to cities that border other provinces to test whether political competition with neighboring provinces (rather than within-province competition) drives trade patterns. The key independent variable, Cross-Provincial Age Differential, measures the absolute difference between a border city party secretary's age and the average age of party secretaries across all neighboring provinces. For cities bordering multiple provinces, this is calculated as the average across all neighboring province averages. The dependent variables are: total freight trade volume (Column 1), motorway freight volume (Column 2), waterway freight volume (Column 3), and export value (Column 4), all in logarithmic form. This serves as a placebo test because the mechanism predicts that within-province yardstick competition, not cross-provincial age differences, should affect local officials' infrastructure investment decisions and resulting trade patterns. Finding null effects supports the interpretation that age-based competition operates primarily within provincial boundaries where officials compete for promotion. The sample includes border city-year observations from 1996 to 2018. City characteristics include logged population, number of firms, and urban road length. Politician characteristics include secretary and average peer ages. All specifications include city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.9: Politician Age Differential and Railway and Airway Trade Volumes

	City Party Secretary	City Mayor
Variables	(1)	(2)
Panel A: Railway Trade Volume		
Politician Age Differential	0.011	−0.005
	(0.007)	(0.006)
Observations	3,700	3,568
R-squared	0.949	0.939
Panel B: Airway Trade Volume		
Politician Age Differential	0.023	0.001
	(0.033)	(0.004)
Observations	1,501	1,579
R-squared	0.953	0.932
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table examines how politician age differentials relate to city-level railway and airway trade, modes that mainly serve long-distance routes and are governed at higher administrative levels than city governments. The railway coefficient for city party secretaries has the same sign as, and a magnitude close to, the motorway estimate in Table 1, but it is less precisely estimated. The analysis uses city-level panel data from 1996 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's start age and the average start age of party secretaries in other cities within the same province. Column (1) reports the specification for city party secretaries, including city characteristics and politician characteristics (own age and peer average age) along with city-politician and year fixed effects. Column (2) repeats the exercise using the age differential of city *mayors*, estimated with city and politician characteristics and city-politician and year fixed effects. Trade volumes are logged and measured at 10,000 tons. Formal tests of coefficient equality across all four transport modes are reported in Table H.10. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.10: Formal Coefficient Comparison Across Transport Modes

Variables	Log(Trade Volume) (1)
Politician Age Differential × Motorway	0.013***
	(0.004)
Politician Age Differential × Waterway	0.026***
	(0.009)
Politician Age Differential × Railway	0.007
	(0.007)
Politician Age Differential × Airway	−0.002
	(0.024)
Cross-mode equality tests: F-statistic [p-value]	
All modes equal	1.05 [0.373]
Motorway = Railway	0.51 [0.474]
Waterway = Railway	2.69 [0.102]
Motorway/Waterway avg. = Railway/Airway avg.	1.52 [0.218]
Observations	13,324
R-squared	0.987
City Characteristics	Yes
Politician Characteristics	Yes
City-Politician × Mode FE	Yes
Year × Mode FE	Yes

Notes: This table compares the effect of the politician age differential across the four transport modes within a single stacked regression. The four city-level mode outcomes (motorway, waterway, railway, and airway trade volumes, logged) are stacked into a city-year-mode panel estimated on a common sample; the key independent variable 'Politician Age Differential' is interacted with mode indicators; and every fixed effect of the baseline specification is interacted with mode, so that each mode retains its own intercept structure and cross-mode differences in coefficients can be tested within one variance-covariance matrix. The middle panel reports F-tests of coefficient equality across modes with p-values in brackets. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.11: Robustness: Alternative Time Trend and Clustering

	Log(Trade Volume)			
VARIABLES	(1)	(2)	(3)	(4)
Panel A: Total Trade Volume				
Politician Age Differential	0.011***	0.013***	0.011***	0.011***
	(0.004)	(0.004)	(0.003)	(0.004)
Observations	5,281	5,281	5,281	5,281
R-squared	0.893	0.903	0.889	0.889
Panel B: Motorway Trade Volume				
Politician Age Differential	0.012***	0.014***	0.013***	0.013***
	(0.003)	(0.004)	(0.003)	(0.003)
Observations	5,101	5,101	5,101	5,101
R-squared	0.937	0.943	0.936	0.936
Panel C: Waterway Trade Volume				
Politician Age Differential	0.025***	0.021**	0.026***	0.026***
	(0.009)	(0.009)	(0.009)	(0.008)
Observations	2,858	2,858	2,858	2,858
R-squared	0.969	0.974	0.967	0.967
Panel D: Export Value (USD)				
Politician Age Differential	0.011*	0.009	0.013**	0.013**
	(0.006)	(0.006)	(0.005)	(0.006)
Observations	3,904	3,904	3,904	3,904
R-squared	0.978	0.982	0.976	0.976
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province Linear Trend	Yes	–	–	–
City Linear Trend	–	Yes	–	–
Cluster at Province-Year Level	–	–	Yes	–
Cluster at Politician Level	–	–	–	Yes

Notes: This table examines the robustness of how Politician Age Differential between party secretaries (within the same province) affects city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Panel A shows effects on total trade volume, Panel B on motorway trade, Panel C on waterway trade, and Panel D on export trade. All trade volumes are logged and measured at 10,000 tons. Column (1) includes province linear trends, Column (2) includes city linear trends, Column (3) clusters standard errors at the province-year level, and Column (4) clusters at the politician level. All specifications control for city characteristics, politician characteristics, city-politician fixed effects, and year fixed effects. Unless otherwise indicated, standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.12: Instrumental Variable Estimates: Effect of Politician Age Differential on City Trade Volume

VARIABLES	Log(Total Trade Volume) (1)	Log(Motorway Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Export Value (4)
Politician Age Differential	0.021** (0.010)	0.022** (0.010)	0.040* (0.021)	-0.002 (0.019)
Observations	4,907	4,885	2,779	3,815
First-Stage F-stat	363	362	277	304
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes

Notes: This table presents instrumental variable (IV) regression results estimating the causal effect of politician age differential on city trade volumes. The sample covers city-level panel data from 1996 to 2018. Politician Age Differential is instrumented using its one-year lagged value. The dependent variables are log-transformed trade volumes by transportation mode and trade type. All specifications include comprehensive controls for city and politician characteristics, along with city-politician and year fixed effects. The Kleibergen-Paap Wald F-statistic tests for weak identification. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.13: Robustness to Focal-City Secretary Turnover

VARIABLES	(1) Full Sample	Sample: No Focal-City Secretary Turnover (2) Drop Focal Turnover Year (t)
Panel A: Total Trade Volume		
Politician Age Differential	0.011***	0.034***
	(0.004)	(0.012)
Observations	5,281	3,710
R-squared	0.889	0.917
Panel B: Motorway Trade Volume		
Politician Age Differential	0.013***	0.019*
	(0.004)	(0.011)
Observations	5,101	3,561
R-squared	0.936	0.939
Panel C: Waterway Trade Volume		
Politician Age Differential	0.026***	0.028
	(0.009)	(0.028)
Observations	2,858	1,962
R-squared	0.967	0.972
Panel D: Export Value (USD)		
Politician Age Differential	0.013**	0.016
	(0.006)	(0.020)
Observations	3,904	2,719
R-squared	0.976	0.977
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table tests the robustness of the Politician Age Differential effect on city trade by dropping city-years close to the focal city's own secretary turnover. Column (1) uses the full sample. Column (2) drops city-years in which the focal city had its own secretary turnover in year t. Across both samples, the identifying variation in 'Politician Age Differential' comes from peer turnovers in other cities of the same province. Each panel reports a different trade outcome: total trade volume in Panel A, motorway in B, waterway in C, and the value of international exports (USD) in D. All trade volumes are logged and measured at 10,000 tons. All specifications include city characteristics and politician characteristics (politician start age and peer average age) and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.14: Robustness: Controlling for Historical Economic Conditions and Trade Volume

VARIABLES	Log(Total Trade Volume) (1)	Log(Total Trade Volume) (2)	Log(Motorway Trade Volume) (3)	Log(Motorway Trade Volume) (4)	Log(Waterway Trade Volume) (5)	Log(Waterway Trade Volume) (6)	Log(Export Value (USD)) (7)	Log(Export Value (USD)) (8)
Politician Age Differential	0.008**	0.013***	0.011***	0.013***	0.020**	0.024**	0.012**	0.009
	(0.004)	(0.004)	(0.003)	(0.004)	(0.008)	(0.010)	(0.005)	(0.006)
Log(Lagged Total Trade Volume)	0.325***							
	(0.062)							
Log(Lagged Nighttime Lighting)	0.149*		0.065		0.006		0.127	
	(0.082)		(0.068)		(0.129)		(0.113)	
Log(3-Year Avg Total Trade Volume)		0.400**						
		(0.183)						
Log(3-Year Avg Nighttime Lighting)		0.102		0.200*		-0.176		0.245
		(0.146)		(0.108)		(0.322)		(0.229)
Log(Lagged Motorway Trade Volume)			0.263***					
			(0.054)					
Log(3-Year Avg Motorway Trade Volume)				0.232***				
				(0.089)				
Log(Lagged Waterway Trade Volume)					0.269***			
					(0.046)			
Log(3-Year Avg Waterway Trade Volume)						0.249***		
						(0.068)		
Log(Lagged Export Trade Volume)							0.181**	
							(0.078)	
Log(3-Year Avg Export Trade Volume)								0.097
								(0.094)
Observations	5,200	4,516	4,942	4,250	2,667	2,278	3,640	2,998
R-squared	0.894	0.903	0.940	0.938	0.971	0.972	0.978	0.982
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the robustness of the main findings by controlling for historical economic conditions and past trade volumes. The dependent variables in Columns (1) through (8) are the log of Total Trade Volume (Columns 1-2), Motorway Trade Volume (Columns 3-4), Waterway Trade Volume (Columns 5-6), and Export Value (USD) (Columns 7-8), respectively. Odd-numbered Columns use lagged variables, while even-numbered Columns use 3-year averages (L3). The key independent variable is 'Politician Age Differential'. Specifications include controls for Log(Lagged Nighttime Lighting) or Log(3-Year Avg Nighttime Lighting), and the respective lagged or 3-year average Log(Trade Volume) for the dependent variable in each column. All specifications include city characteristics and politician characteristics (own age and peer average age) and city-politician and year fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.15: Placebo Test: Effects of Politician Age Differential on Railway Infrastructure

VARIABLES	Log(Total Railway Length) (1)	Log(Single Railway Length) (2)	Log(Railway Link Count) (3)	Log(Railway Node Count) (4)	Log(Railway Station Count) (5)	Log(Railway Betweenness) (6)
Politician Age Differential	-0.004	-0.005	-0.003	-0.002	-0.002	0.001
	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.008)
Observations	5,178	5,149	5,178	5,177	5,177	5,087
R-squared	0.986	0.982	0.988	0.995	0.995	0.978
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents placebo tests examining the effects of politician age differential on railway infrastructure development. Railway infrastructure is used as a placebo because, unlike motorways which are largely under local government control, railway construction is centrally planned by the national government and should not be affected by local politician competition dynamics. The analysis uses panel data with various measures of railway network characteristics from 1995 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Dependent variables include log-transformed measures of total railway length, single railway length, railway link count, railway node count, railway station count, and railway betweenness centrality. All specifications include politician characteristics, year fixed effects, and city-politician fixed effects. Standard errors (in parentheses) are clustered at the city level. The insignificant coefficients support the validity of the identification strategy. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.16: Effects of Similar-Age Peer Count on City Trade

	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
VARIABLES	(1)	(2)	(3)	(4)
# Similar Age Peers	-0.007***	-0.005**	-0.012**	-0.003
	(0.003)	(0.002)	(0.005)	(0.003)
Observations	5,297	5,113	2,858	3,913
R-squared	0.890	0.935	0.967	0.976
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table examines how the number of similar-age peer secretaries within the same province affects city-level trade. The analysis uses city-level panel data from 1996 to 2018. # Similar Age Peers is the count of party secretaries in other cities of the same province whose current age is within three years of the focal city secretary's start age. Columns (1) to (4) report effects on total trade volume, motorway trade, waterway trade, and the value of international exports (USD), respectively. All trade volumes are logged and measured at 10,000 tons. All specifications include city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.17: Effects of Provincial Politician Age Differential on Bilateral Trade Volume (Inter-province Trade)

	Log(Bilateral Provincial Trade Volume)				
VARIABLES	(1)	(2)	(3)	(4)	(5)
Provincial Politician Age Differential (Origin Province)	0.021**	0.022**	0.022**		
	(0.009)	(0.009)	(0.009)		
Provincial Politician Age Differential (Destination Province)	0.011	0.012		0.012	
	(0.009)	(0.009)		(0.009)	
Absolute Age Difference (Origin vs. Destination)	0.002				
	(0.003)				
# Similar Age Peers (Origin Province)					-0.013***
					(0.003)
# Similar Age Peers (Destination Province)					-0.009***
					(0.003)
Observations	16,215	16,215	16,215	16,215	16,215
R-squared	0.943	0.943	0.943	0.943	0.943
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Origin Province $\times$ Origin Politician FE	Yes	Yes	Yes	Yes	Yes
Destination Province $\times$ Destination Politician FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how politician age differentials affect bilateral trade volumes between Chinese provinces. The analysis uses province-pair panel data from the Railway Statistical Yearbook covering 2000 to 2018. The origin politician age differential measures the absolute difference between the origin province's party secretary age and the national average age of all provincial party secretaries; the destination politician age differential applies the same measure to the destination province. Column (1) includes both differentials and their absolute difference, Column (2) drops the relative-age term, Columns (3) and (4) retain only the origin and only the destination differential respectively, and Column (5) replaces the differentials with the number of similar-age provincial peers, defined as the count of other provincial party secretaries whose start age is within three years of the focal secretary, for the origin and destination provinces. All specifications include origin-destination fixed effects, year fixed effects, and origin-province-by-politician and destination-province-by-politician fixed effects. Standard errors (in parentheses) are clustered at the origin-destination pair level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.18: Motorway Underinvestment by Administrative Class

VARIABLES	Log(Provincial Motorway Length) (1)	Log(National Motorway Length) (2)	Log(Urban Motorway Length) (3)
Politician Age Differential	-0.044	-0.015	-0.004
	(0.033)	(0.042)	(0.040)
Politician Age Differential × Border Region	0.021	0.080*	0.009
	(0.041)	(0.044)	(0.039)
Observations	36,482	36,482	36,482
R-squared	0.935	0.933	0.952
County Characteristics	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes
County-Politician FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table decomposes the motorway results in Table 3 by administrative class: provincial expressways (S-prefix routes), national expressways (G-prefix routes), and urban motorways (unclassified segments). County-year lengths are computed from the annual motorway network shapefiles, classifying each main carriageway by its route number and intersecting it with county boundaries in an equal-area projection. Specifications and variable definitions follow Table 3. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.19: Effect of Motorway Development on Motorway Trade Volume

	Log(Motorway Trade Volume)				
VARIABLES	(1)	(2)	(3)	(4)	(5)
Log(Motorway Length)	0.041**				
	(0.019)				
Log(Motorway Entry/Exit)		0.093***			
		(0.021)			
Log(Motorway Link)			0.077***		
			(0.021)		
Log(Motorway Node)				0.093***	
				(0.021)	
Log(Motorway Betweenness)					0.004
					(0.006)
Observations	4,694	4,646	4,694	4,646	4,469
R-squared	0.803	0.806	0.803	0.806	0.804
City FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how different measures of motorway development affect city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variables are different measures of motorway development, including motorway length (Column 1), the number of motorway entry and exit points (Column 2), the number of motorway links connecting cities (Column 3), the number of motorway nodes or junctions (Column 4), and motorway betweenness centrality measuring a city's position in the motorway network (Column 5). The dependent variable is the logarithm of trade volume via motorway transport. All specifications include city and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.20: Politician Age Differential and Number of Firm Entries

	Log(# Firm New Entry + 1)					
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	0.0075***	0.0078***	0.0077***			
× Tradable Sector	(0.0024)	(0.0024)	(0.0024)			
Politician Age Differential				-0.0114***	-0.0105***	-0.0094***
× Real Estate Sector				(0.0034)	(0.0033)	(0.0033)
Observations	5,765,425	5,765,425	5,765,207	5,765,425	5,765,425	5,765,207
R-squared	0.548	0.558	0.574	0.548	0.558	0.574
County-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	Yes	Yes	–
Politician FE	–	Yes	Yes	–	Yes	Yes
County-Year FE	–	–	Yes	–	–	Yes

Notes: This table examines the impact of Politician Age Differential on the number of new firm entries, distinguished by sector type (tradable versus real estate). The analysis utilizes county-industry-year panel data from 1996 to 2018, with the dependent variable being the logarithm of the number of new firm entries plus one. The key independent variables are the interaction terms between Politician Age Differential and indicators for tradable sectors and real estate sectors, respectively. All specifications control for county-sector fixed effects, politician fixed effects, and county-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.21: Example Matched Firm Pair

Attribute	Firm I Details	Firm J Details
Similarity Score from LLM (96.81%)		
Firm ID	13346	1267002
Name	Shanghai Gaodong Paint Factory	Shanghai Rongsheng Coating Co., Ltd.
Industry Subcategory	Organic Chemical Products Manufacturing	Organic Chemical Products Manufacturing
Activity 1	Paint and Coating	Paint Spraying Processing
City	Pudong New District	Qingpu District

Notes: This table shows an example of firms connected via business networks based on main business activity and sectors. The similarity score also considers the financial status of firms.

Table H.22: Robustness: Effect of Politician Age Differential on Firms with High-Connectivity Input-Output Relationships (1997)

	Firm$_j$ Connected to Firm$_i$ in Provincial Input-Output Table (Above Median Connectivity) in 1997			
VARIABLES	Log(Intermediate Good Inputs) (1)	Log(Revenue) (2)	Log(Revenue per Worker) (3)	Log(Profits) (4)
Politician Age Differential in Firm$_i$ City	0.011***	0.013***	0.009**	0.001
	(0.004)	(0.003)	(0.004)	(0.003)
Observations	1,146,985	1,146,940	1,146,940	1,147,128
R-squared	0.894	0.922	0.817	0.798
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes

Notes: This table presents robustness checks examining how political age differentials affect firm intermediate good inputs and other firm outcomes through province-specific input-output linkages. The analysis focuses on firm pairs connected via provincial input-output relationships with above-median connectivity levels based on the 1997 input-output table. The key independent variable 'Politician Age Differential in Firm$_i$ City measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. All dependent variables are in logarithmic form. The sample includes manufacturing firms from the Annual Survey of Industrial Firms (ASIF) database. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.23: Effect of Politician Age Differential on Firm Performance: Heterogeneity by State Ownership

	Firm$_i$ Located in City$_i$ Direct Effect on Focal Firms		Firm$_j$ Connected to Firm$_i$ in Business Network Indirect/Spillover Effect on Connected Firms	
	Log(Revenue)	Log(Revenue per Worker)	Log(Revenue)	Log(Revenue per Worker)
VARIABLES	(1)	(2)	(3)	(4)
Politician Age Differential	0.023***	0.022***	0.007***	0.009***
	(0.007)	(0.008)	(0.002)	(0.003)
Politician Age Differential	-0.020***	-0.022***	-0.011**	-0.009**
× State-Owned Firms (SOE)	(0.006)	(0.006)	(0.005)	(0.004)
Observations	26,887	26,882	118,957	118,943
R-squared	0.909	0.863	0.905	0.861
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector×Year FE	Yes	Yes	Yes	Yes
City$_j$×Year FE	–	–	Yes	Yes

Notes: This table examines how the politician age differential between party secretaries affects firm performance, and how the effect differs for state-owned enterprises (SOEs). The analysis uses firm-level panel data, where Columns (1) and (2) report outcomes for the focal firm (Firm$_i$) located in City$_i$, and Columns (3) and (4) report outcomes for firms connected to the focal firm through business networks (Firm$_j$). The key independent variable 'Politician Age Differential in City$_i$' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. The row '× State-Owned (SOE)' interacts this differential with an indicator for whether the relevant firm is state-owned, so the total effect for an SOE is the sum of the two coefficients; the interaction offsets the main effect almost exactly (e.g., $0.023 - 0.020 \approx 0.003$ for focal-firm revenue and $0.022 - 0.022 \approx 0$ for revenue per worker), leaving a near-zero net effect for state-owned firms. All specifications include firm pair fixed effects, politician fixed effects, and sector-by-year fixed effects; Columns (3) and (4) additionally include connected city-by-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.24: Yardstick Competition and the Sectoral Composition of GDP

	Tradable VA Share of GDP (pp)	Non-Tradable VA Share of GDP (pp)
VARIABLES	(1)	(2)
Politician Age Differential	0.021	-0.010
	(0.032)	(0.029)
Observations	4,921	4,925
R-squared	0.960	0.958
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table decomposes city GDP into tradable and non-tradable components and estimates the effect of yardstick competition on the composition of value added, implementing the decomposition that underlies the reconciliation of the firm-level and GDP results. The analysis uses city-level panel data from 1996 to 2018. The dependent variables are the tradable and non-tradable (including real estate) shares of city GDP, in percentage points. The key independent variable 'Politician Age Differential' is the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. A positive coefficient in Column (1) and a negative coefficient in Column (2) indicate that weaker competition tilts the composition of GDP toward tradables, and intense competition toward non-tradables. Both specifications include city characteristics, politician characteristics, and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.25: Placebo Test: Effect of Politician Age Differential on Firm Networks in Bordering Cities

	Firm_j Connected to Firm_i in Business Network (Different Provinces) (Firms with a 95% Similarity Score)			
VARIABLES	Log(Intermediate Good Inputs) (1)	Log(Revenue) (2)	Log(Revenue per Worker) (3)	Log(Profits) (4)
Politician Age Differential in Firm_i City	-0.007	-0.012	-0.004	0.087
	(0.045)	(0.036)	(0.034)	(0.083)
Observations	1,655	1,655	1,655	1,655
R-squared	0.939	0.960	0.926	0.843
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a falsification test examining whether age differentials between party secretaries affect firms in bordering cities across different provinces. The analysis uses firm-pair panel data where firms are connected through business networks but located in different provinces. The key independent variable 'Politician Age Differential in Firm_i City measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Dependent variables measure outcomes for connected firms (Firm_j in different provinces): intermediate good inputs, revenue, revenue per worker, and profits (all log-transformed). All specifications include firm pair fixed effects, politician fixed effects, and sector-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.26: Politician Age Differential (1 SD Decrease) and Firm Value Added

Outcome	Coefficient	Loss (%)
Revenue	0.023	5.04
Intermediate inputs	0.027	5.92
Value added		2.40

Notes: This table translates the firm-level effects of a one standard deviation decrease in Politician Age Differential into a loss in value added. The coefficients are from the baseline specification of Table 7 (Panel A, Columns (3) and (1)). Loss is calculated as Coefficient $\times$ 2.192, the standard deviation of Politician Age Differential in the firm sample. The value-added loss uses $g_{VA} = (g_R - s\,g_M)/(1-s)$ with an intermediate-input share of revenue of $s = 0.75$. Data are from the Annual Survey of Industrial Firms (ASIF), 2000–2007.

Table H.27: Calculation of Secondary-Sector GDP Impact (1 SD Decrease in Politician Age Differential)

	Value (%)
Value-added loss of industrial firms	2.40
Secondary-sector share of city GDP	46.2
GDP impact	1.11

Notes: This table calculates the GDP impact of a one standard deviation decrease in Politician Age Differential, assuming that every firm in the secondary sector shares the value-added loss of the industrial firms in Appendix Table H.26. The secondary-sector share is the average share of secondary-industry value added in city GDP over 2000–2007 in the baseline city sample (Chinese City Statistical Yearbook). The GDP impact is the product of the two rows. It is a cost to the secondary sector, not an overall GDP effect, since the expansion of real estate induced by competition is excluded from this calculation.

Table H.28: Effect of Different Competition Measures

VARIABLES	Log(Total Trade Volume) (1)	Log(Motorway Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Log(Export Value (USD)) (4)
Panel A: Baseline Relative Age Competition				
Politician Age Differential	0.011***	0.013***	0.026***	0.013**
	(0.004)	(0.004)	(0.009)	(0.006)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
Panel B: Relative Age Competition ($\leq$ Age 57)				
Politician Age Differential $\leq$ Age 57	0.011**	0.013***	0.028***	0.012*
	(0.004)	(0.004)	(0.009)	(0.006)
Observations	4,835	4,677	2,659	3,572
R-squared	0.910	0.937	0.967	0.977
Panel C: Relative Current-Age Competition				
Politician Current-Age Differential	0.005	0.007**	0.019**	0.010*
	(0.005)	(0.003)	(0.009)	(0.005)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
Panel D: Relative Tenure Year Competition				
Politician Tenure Year Differential	0.001	-0.000	0.002	0.001
	(0.007)	(0.006)	(0.013)	(0.010)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
Panel E: Relative Educational Level Competition				
Politician Education Level Differential	0.006	0.003	-0.002	-0.001
	(0.008)	(0.004)	(0.013)	(0.007)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
Panel F: Relative Experience Year Competition				
Politician Working Years Differential	0.001	0.001	0.014*	0.001
	(0.002)	(0.002)	(0.007)	(0.003)
Observations	4,761	4,626	2,635	3,611
R-squared	0.911	0.937	0.968	0.976
Panel G: Similar Experience Count				
Similar Working Year (+3/-3) Counts within Province	0.000	-0.003	-0.020***	-0.000
	(0.004)	(0.003)	(0.007)	(0.005)
Observations	4,764	4,628	2,635	3,611
R-squared	0.911	0.937	0.968	0.976
Sample	Full	Full	Full	Full
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes

Notes: This table investigates the effect of different yardstick competition measures on trade volumes (log-transformed) and export values. Each panel presents results from separate regressions using different measures of political competition, all constructed as winsorized absolute differentials from the leave-one-out provincial average unless noted otherwise. Panel A shows the baseline relative age competition effect using the start-of-term politician age differential. Panel B restricts the analysis to politicians aged 57 or below to capture promotion incentives before mandatory retirement. Panel C replaces the start-of-term age with the politician's current age in the construction of the age differential. Panel D examines relative tenure competition using politician tenure differential. Panel E examines educational level competition using politician education level differential. Panel F analyzes relative experience year competition using working years differential. Panel G examines similar experience politician counts within provinces; for this count measure the expected sign is negative, since more similar peers imply more intense competition. The dependent variables are Total Trade Volume (Column 1), Motorway Trade Volume (Column 2), Waterway Trade Volume (Column 3), and Export Value in USD (Column 4). All specifications include city characteristics and politician characteristics (own age and peer average age) and city-politician and year fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.29: Effect of Politician Age Differential by City Industry Structure

	Log(Total Trade Volume)					
VARIABLES	Eastern Region (1)	Initial Trade Share (2)	City Government Firm Tax Revenue (3)	VA in Manuf Share (4)	2nd Industry Employment (5)	Industrial Firm Number (6)
Politician Age Differential	0.016***	0.033***	0.048*	0.043**	0.034***	0.056***
	(0.005)	(0.009)	(0.024)	(0.018)	(0.012)	(0.020)
Age Diff × Eastern Region	-0.012*					
	(0.007)					
Initial Trade Share		0.336***				
		(0.089)				
Age Diff × Initial Trade Share		-0.048**				
		(0.020)				
City Firm Tax Revenue			-0.027			
			(0.053)			
Age Diff × Firm Tax Revenue			-0.003			
			(0.002)			
Manufacturing Value-Added Share				0.010***		
				(0.003)		
Age Diff × Manufacturing VA				-0.001*		
				(0.000)		
City Manufacturing Employment					-0.050	
					(0.058)	
Age Diff × Manufacturing Emp					-0.007*	
					(0.004)	
Number of Industrial Firms						0.172***
						(0.063)
Age Diff × Industrial Firms						-0.006**
						(0.003)
Observations	5,281	4,426	4,151	3,933	4,892	4,103
R-squared	0.889	0.881	0.897	0.916	0.907	0.887
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how the effect of politician age differential on trade volume varies across cities with different regional characteristics, initial trade conditions, and industry structures. Column (1) examines whether a city is located in the eastern region, Column (2) examines the city's initial trade share, Column (3) examines city government firm tax revenue, Column (4) examines the manufacturing value-added share of GDP, Column (5) examines manufacturing employment, and Column (6) examines the number of industrial firms. The dependent variable is log total trade volume across all specifications. The main variable of interest is politician age differential and its interactions with city characteristics and industry structure measures. All specifications include city characteristics, politician characteristics, year fixed effects, and city-politician fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.30: Competition Effect by Tenure Proximity of Provincial Peers

	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
VARIABLES	(1)	(2)	(3)	(4)
Politician Age Differential: Near-Tenure Peers ($\leq$ 2 Yr Gap)	0.008**	0.009***	0.030***	0.007
	(0.004)	(0.003)	(0.008)	(0.005)
Politician Age Differential: Mid-Tenure Peers (3–5 Yr Gap)	0.003	0.003	0.000	0.005
	(0.004)	(0.002)	(0.005)	(0.004)
Politician Age Differential: Far-Tenure Peers ($\geq$ 6 Yr Gap)	0.003	0.006*	-0.003	0.009
	(0.004)	(0.003)	(0.007)	(0.006)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table decomposes the Politician Age Differential effect on city trade by the tenure proximity of within-province peer secretaries. For each city-year, peers are split into three disjoint bins by the absolute gap between the focal and peer secretary's tenure year: near-tenure peers (gap of 0 to 2 years), mid-tenure peers (gap of 3 to 5 years), and far-tenure peers (gap of 6 or more years). Each row reports the age differential computed over the corresponding bin, entered jointly in the same regression. Columns (1) to (4) report effects on total trade volume, motorway trade, waterway trade, and the value of international exports (USD), respectively. All trade volumes are logged and measured at 10,000 tons. All specifications include politician characteristics (politician start age and peer average age) and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table H.31: Politician Age Differential and Firm Suppliers/Competitors

	Firm$_j$ Connected to Firm$_i$ in Business Network Indirect/Spillover Effect on Firms with a 95% Similarity Score (LLM)			
VARIABLES	Log(Intermediate Good Inputs) (1)	Log(Profits) (2)	Log(Revenue) (3)	Log(Revenue per Worker) (4)
Panel A: Potential Supplier and Partner				
Politician Age Differential	0.008**	0.012	0.006***	0.007**
	(0.004)	(0.009)	(0.002)	(0.004)
Observations	99,057	99,067	99,067	99,054
R-squared	0.869	0.792	0.903	0.861
Panel B: Potential Competitor: Similar Products/Business Activities				
Politician Age Differential	-0.001	0.012	0.000	0.006
	(0.008)	(0.025)	(0.004)	(0.005)
Observations	15,395	15,400	15,400	15,399
R-squared	0.945	0.858	0.976	0.944
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes

Notes: This table examines the heterogeneous effects of yardstick political competition on connected firms (firm j), differentiating by the type of inter-firm relationship. The analysis uses firm-pair panel data from 2000 to 2007. Panel A focuses on firm pairs classified as potential suppliers or partners, while Panel B focuses on firm pairs classified as potential competitors. The key independent variable is the Politician Age Differential in the city governing firm i. Dependent variables, shown in Columns (1) through (4), are log-transformed intermediate good inputs, profits, revenue, and revenue per worker for firm j. All specifications include firm pair fixed effects, politician fixed effects, sector-by-year fixed effects, and city-by-year fixed effects. Standard errors (in parentheses) are clustered at city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.